

GR-01 : FUNCTIONS

SINGLE CORRECT

Q.(1)

Which of the following function is surjective but not injective

(A) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + x + 1$

(C) $f: \mathbb{R} \rightarrow \mathbb{R}^+ \quad f(x) = \sqrt{1+x^2}$

(D) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + 2x^2 - x + 1$

Q.(2)

Let $f(x) = \max.\{\sin t : 0 \leq t \leq x\}$

$g(x) = \min.\{\sin t : 0 \leq t \leq x\}$

and $h(x) = [f(x) - g(x)]$

where $[]$ denotes greatest integer function, then the range of $h(x)$ is-

(A) $\{0,1\}$

(B) $\{1,2\}$

(C) $\{0,1,2\}$

(D) $\{-3,-2,-1,0,1,2,3\}$

MULTIPLE CORRECT

Q.(3)

Let $f: \left(-\frac{\pi}{2}, a\right] \rightarrow (-\infty, b]$ be an invertible function with longest domain defined as

$f(x) = \tan x + \cot x - 2$ then -

(A) Ordered pair (a,b) is $\left(-\frac{\pi}{3}, -\frac{(4+2\sqrt{3})}{\sqrt{3}}\right)$

(B) Ordered pair (a,b) is $\left(-\frac{\pi}{4}, -4\right)$

(C) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2+\sqrt{x^2+4x}}{2}\right)$

(D) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2-\sqrt{x^2+4x}}{2}\right)$

Q.(4)

Which of the following statement(s) is(are) correct ?

- (A) If f is a one-one mapping from set A to A , then f is onto.
 (B) If f is an onto mapping from set A to A , then f is one-one
 (C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ f$ is injective, then f must be injective.
 (D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

Q.(5)

 If $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (where $g(x)$ is invertible function), then which of the following statement(s) is/are correct ?

- (A) $g \circ f(x) \leq g(x)$ for $x \in (-\infty, 0]$ (B) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (0, \infty)$
 (C) $g^{-1} \circ f(x) \leq g^{-1}(x)$ for $x \in (0, \infty)$ (D) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (-\infty, 0]$

Q.(6) Let $f: \mathbb{A} \rightarrow \mathbb{B}$ and $g: \mathbb{C} \rightarrow \mathbb{A}$ be functions. Then which of the following options is/are correct?

- (A) If f and g are Injective, then $f \circ g$ is also Injective.
 (B) If f and g are Surjective, then $f \circ g$ is also Surjective.
 (C) If $f \circ g$ is Injective, then g MUST be Injective.
 (D) If $f \circ g$ is Injective, then f need not be Injective.

Q.(7) Let $f: \mathbb{A} \rightarrow \mathbb{B}$ and $g: \mathbb{C} \rightarrow \mathbb{A}$ be functions. Then which of the following options is/are correct?

- (A) If $f \circ g$ is Surjective, then f MUST also be Surjective.
 (B) If $f \circ g$ is Surjective, then g need not be Surjective.
 (C) If $f \circ g$ is Injective and g is Surjective, then f MUST be Injective.
 (D) If f and g are Bijective, then $f \circ g$ need not be Bijective.

Q.(8) Consider the following sets

$$\mathbb{A} = \left\{ \frac{x^2 - 2x + 4}{x^2 + 2x + 4} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$\mathbb{B} = \left\{ \frac{x^2 - 2x - 4}{x^2 + 2x - 4} \in \mathbb{R} : x \in \mathbb{R} \right\}.$$

Then which of the following options is/are correct?

(A) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{7}{19}, \frac{19}{7}\right\} \subseteq \mathbb{A}$

(B) $\{-13, 2301, \pi, -e\} \subseteq \mathbb{B}$

(C) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{19}{39}, \frac{39}{19}\right\} \subseteq \mathbb{A}$

(D) $\{-1301, 23, -\pi, e\} \subseteq \mathbb{B}$

Q.(9) Which of the following options is/are correct, if

$$f(x) = \max(x^2, |2 - x|) \text{ and } g(x) = \min(x^3, 3x - 2)?$$

(A) If $x \in (-\infty, -2]$, then $f(g(x)) = x^6$

(B) If $x \in (-2, 0]$, then $f(g(x)) = 9x^2 - 12x + 4$

(C) If $x \in (0, 1]$, then $f(g(x)) = 4 - 3x$

(D) If $x \in (-2, 1]$, then $g(f(x)) = 4 - 3x$

Q.(10) Let $a, b, c \in \mathbb{R}$, $a + b + c = 3$ and functions f, g, h be such that

$$f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]; f(x) = \frac{x}{1 + x^2},$$

$$g(x) = ax^2 + bx + c; g(x + y) = g(x) + g(y) + xy \forall x, y \in \mathbb{R} \text{ and}$$

$$h(x) + 2h\left(\frac{1}{x}\right) = 3x \forall x \in \mathbb{R} - \{0\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then which of the following options is/are correct?

(A) f is Injective function.

(B) f is Surjective function.

(C) $g(1) + g(2) + g(3) + \dots + g(10) = 330$

(D) $|\{x \in \mathbb{R}: h(x) = h(-x)\}| = 2.$

NUMERICAL TYPE

Q.(11)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(10 - x) = f(x)$ and $f(2 - x) = f(2 + x) \forall x \in \mathbb{R}$. If $f(0) = \lambda$ (where $\lambda \in \mathbb{R}$). Then the minimum possible number of values of x satisfying $f(x) = \lambda$ in the interval $[0, 38]$, is

Q.(12)

Let $A = \{1, 2, 3, 4, 5\}$ & $B = \{a, b, c, d\}$. A function is defined from A to B . 'p' be the number of one-one functions from A to B and 'q' be the number of onto function from A to B . Then $(p + q)$ equals

Q.(13) Let

$$f: \mathbb{A} \rightarrow \left[-\frac{1}{2}, 0\right]; f^3(x) - 3x f(x) + x^3 + 1 = 0.$$

If f is an invertible function, then number of solutions of

$$(f(f(x)))^2 = \frac{3}{2}f^{-1}(x) + 1$$

is equal to

Q.(14) Let $[x] = \max\{a \in \mathbb{Z}: a \leq x\}$, $\{x\} = x - [x]$,

$$\mathbb{A} = \left\{x \in \mathbb{R} : \frac{[x+1]^2}{6} = \frac{x}{3} + \frac{\{x\}}{2}\right\},$$

$$\mathbb{B} = \left\{x \in \mathbb{R}^+ : \{x^2\} = \left\{\frac{1}{x}\right\}, x^2 \in (2, 3)\right\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then $|\mathbb{A}| + |\mathbb{B}|$ is equal to

Q.(15)

If $1 < x^2 < 9$, then number of solutions of the equation $\{[x]\} = \left\{\frac{1}{x^2}\right\}$ (where $\{.\}$ denote fractional part function) is -

Q.(16) Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{x \in \mathbb{R} : \sqrt{ax^2 + bx} \in \mathbb{R}\},$$

$$\mathbb{B} = \{\sqrt{ax^2 + bx} \in \mathbb{R} : x \in \mathbb{R}\},$$

$$\mathbb{C} = \{a \in \mathbb{R} : \mathbb{A} = \mathbb{B} \text{ for at least one } b \in \mathbb{R}^+\}.$$

Then absolute value of sum of elements of set $\mathbb{C}$ is equal to

Q.(17) Number of elements in the set

$$\{5x \in \mathbb{Z} : \log_3(\log_3|\cos x| - \log_{x^2-4x+6}|\cos x|) \in \mathbb{R}\}$$

is equal to

PASSAGES

Passage:

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \frac{e^x - e^{-x}}{2}$ and function $g : \mathbb{R} \rightarrow [1, \infty)$ is defined as

$$g(x) = \frac{e^x + e^{-x}}{2}.$$

On the basis of above information, answer the following questions :

Q.(18)

Let $h(x) = \frac{f(x)}{g(x)}$, then $\frac{2h(x)}{1+h^2(x)}$ is equal to

(A) 1

(B) 2

(C) $h(2x)$

(D) $h(x)f(x)$

Q.(19)

Number of integers contained in the range of the function $k(x) = \left[\frac{f(x)}{g(x)} \right]$ (where $[.]$ denotes G.I.F.) is

(A) 0

(B) 1

(C) 2

(D) more than 2

MATRIX MATCHING TYPE

Q.(20)

Column-I

Column-II

(A) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as
 $f(x) = e^{\operatorname{sgn} x} + e^{x^2}$, where $\operatorname{sgn} x$
denotes signum function of x , then $f(x)$ is

(P) Odd

(B) Let $f : (-1, 1) \rightarrow \mathbb{R}$ defined as

(Q) Even

$$f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$$

where $[x]$ denotes greatest integer less than
or equal to x , then $f(x)$ is

(C) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

(R) Neither odd nor even

$$f(x) = \frac{x(x+1)(x^4+1) + 2x^4 + x^2 + 2}{x^2 + x + 1} \text{ then } f(x) \text{ is}$$

(D) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

(S) One-one

$$f(x) = x + 3x^3 + 5x^5 + \dots + 101x^{101}$$

then $f(x)$ is

(T) Many-One

KEY

1. D	2. C	3. BD	4. CD	5. AB
6. ABCD	7. ABC	8. ABCD	9. ACD	10. BCD
11. 13	12. 240	13. 1	14. 4	15. 4
16. 4	17. 9	18. C	19. C	20. A-RT, B-QT, C-RT, D-PS

GR-02 : LIMITS AND CONTINUITY

SINGLE CORRECT

Q.(1)

The value of $\lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \sec(e^2x) \dots \sec(e^{50}x))}{e^2 - e^{2\cos x}}$ is equal to

- (A) $\frac{e^{100} - 1}{(e^2 - 1)}$ (B) $\frac{e^{100} - 1}{2(e^2 - 1)}$ (C) $\frac{2(e^{50} - 1)}{e^2 - 1}$ (D) $\frac{e^2(e^{100} - 1)}{2(e^2 - 1)}$

Q.(2)

$\lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4(e^{2x^4} - 1 - 2x^4)}$ is equal to

- (A) 0 (B) $-\frac{1}{6}$ (C) $\frac{1}{6}$ (D) does not exist

Q.(3)

Let $g(x) = \tan^{-1} |x| - \cot^{-1} |x|$, $f(x) = \frac{[x]}{[x+1]} \{x\}$, $h(x) = |g(f(x))|$ where $\{x\}$ denotes fractional part and

$[x]$ denotes the integral part then which of the following holds good ?

- (A) h is continuous at $x = 0$ (B) h is discontinuous at $x = 0$
(C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

Q.(4) Let f be a positive continuous function satisfying

$$f(x^3 - 4x) = \frac{2}{f(2^x + 7) - 1} \forall x \in \mathbb{R}.$$

Also, let g be a continuous function satisfying

$$g(x) = g(2x) \forall x \in \mathbb{R}, g(1) = \frac{\pi}{2},$$

then find

$$\lambda = f(15) + \lim_{x \rightarrow 0} \frac{\cos^2 g(x) + 1 - \sin^3 g(x)}{\sin^2 x}.$$

(A) 1

(B) 2

(C) 3

(D) 4

Q.(5) Which of the following options is/are correct, if

$$\mathbb{S} : \text{If } a_{n-1} = \frac{a_n - n}{n} \forall n \in \mathbb{N} - \{1\}, a_1 = 1, \text{ then } \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{1}{a_r}\right) = e.$$

$$\mathbb{T} : \lim_{n \rightarrow \infty} \left(\frac{n^2 + n + 1}{n + 1} - (an + b) \right) = 4 \Rightarrow 7a - 2b = 15.$$

 (A) $\mathbb{S}$ is True and $\mathbb{T}$ is False

 (B) $\mathbb{S}$ is False and $\mathbb{T}$ is True

 (C) Both $\mathbb{S}$ and $\mathbb{T}$ are True

 (D) Both $\mathbb{S}$ and $\mathbb{T}$ are False.

MULTIPLE CORRECT

Q.(6)

Which of the following limits are non existent ?

(A) $\lim_{x \rightarrow 0} \frac{1}{x} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$

(B) $\lim_{x \rightarrow 0} \frac{|x|}{x}$

(C) $\lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$

(D) $\lim_{x \rightarrow \pi/2} \cos^{-1} [\cot x]$ where $[\cdot]$ is greatest integer function

Q.(7)

Which of the following is/are correct :

(A) $\lim_{x \rightarrow 0} \frac{\sin 4x}{1 - \sqrt{1-x}} = 8$

(B) $\lim_{x \rightarrow \pi/2} \frac{\cos 3x + 3 \cos x}{\left(\frac{\pi}{2} - x\right)^3} = 2$

(C) $\lim_{x \rightarrow \pi/6} \frac{2 - \sqrt{3} \cos x - \sin x}{(6x - \pi)^2} = \frac{1}{36}$

(D) $\lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{\cos \frac{x}{2} \left(\cos \frac{x}{4} - \sin \frac{x}{4} \right)} = \frac{1}{2}$

Q.(8)

If $L = \lim_{x \rightarrow 1} \frac{(1-x)(1-x^2)(1-x^3) \dots (1-x^{2n})}{[(1-x)(1-x^2)(1-x^3) \dots (1-x^n)]^2}$ then L can be equal to

(A) $\prod_{r=1}^n \frac{n+r}{r}$

(B) $\frac{1}{n!} \prod_{r=1}^n (4r-2)$

(C) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(D) The coefficient of x^n in the expansion of $(1+x)^{2n}$.

Q.(9)

Let $f(x) = \frac{\cos^{-1}(\cos x)}{x}$, then which of the following is(are) correct (where $[.]$ denotes greatest integer function) ?

(A) $\lim_{x \rightarrow 0^+} [f(x)] = 1$

(B) $\lim_{x \rightarrow 0^-} [f(x)] = 1$

(C) $\lim_{x \rightarrow 0^+} [f(x)] = 0$

(D) $\lim_{x \rightarrow 0^-} [f(x)] = -1$

Q.(10)

Let $f(x) = \begin{cases} |x|, & -2 \leq x \leq 0 \\ e^x - 1, & 0 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} -x - 1, & -1 \leq x \leq 0 \\ \frac{4x-3}{3}, & 0 < x \leq 3 \end{cases}$, then -

(A) $\lim_{x \rightarrow 0} f(g(x)) = 1$

(B) $\lim_{x \rightarrow 0^-} f(g(2^x)) = 1$

(C) $\lim_{x \rightarrow 0^+} f(g(2^x)) = e^{1/3} - 1$

(D) $\lim_{x \rightarrow 0^+} f(g(2^x - 1)) = \lim_{x \rightarrow 0^-} f(g(2^x - 1))$

Q.(11) Let $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$, Then which of the following options is/are correct?

(A) If f is a continuous function such that $[f(-1)] = 2$, $[f(0)] = -2$, then $f(x) = 0$ has at least one solution in $(-1, 0)$.

(B) If f is a dis-continuous at $x = a$, and g is continuous at $x = a$, $g(a) = 0$, then $f(x)g(x)$ must be continuous at $x = a$.

(C) If $f(x) + \sin x$ is continuous and $g(x) - e^x$ is dis-continuous at $x = a$, then $f(x)g(x)$ must be dis-continuous at $x = a$.

(D) It is possible that $f(x)$ and $g(x) = \max\{t \in \mathbb{Z} : t \leq f(x)\}$ are both continuous at $x = a$, $f(a) \in \mathbb{Z}$.

Q.(12) Which of the following options is/are correct, if

$$f(x) = \lim_{n \rightarrow \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}; n \in \mathbb{N}, \text{ and}$$

$$\mathbb{A} = \{a \in \mathbb{Z} : g(x) = [x]^2 - [x^2] \text{ is continuous at } x = a.\}$$

Here $|X|$ denotes the cardinal number of set X

(A) $\lim_{x \rightarrow 1^+} f(x) = -1$

(B) $\lim_{x \rightarrow 1^-} f(x) = -1$

(C) f is dis-continuous at $x = 1$.

(D) $|\mathbb{A}| = 0$.

Q.(13) Let

$$\lambda = \lim_{x \rightarrow 0} \left(\frac{\tan x \cdot \cos \sin x - \sin \sin x}{\tan x \cdot \cos \tan x - \sin \tan x} \right), \mu = \lim_{x \rightarrow 0} \left(\frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} \right); a > 0.$$

If μ is finite, then

- (A) $a - 2\lambda = 3$ (B) $64\mu + a = 3$ (C) $a - 2\lambda = 2$ (D) $32\mu = 1$

Q.(14) Let $\mu_1 = \lim_{a \rightarrow 0^+} (p), \mu_2 = \lim_{a \rightarrow 0^+} (q)$, where p, q are roots of

$$(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0; p > -1, \text{ and}$$

$$\lambda_1 = \lim_{x \rightarrow \infty} g(x), \lambda_2 = \lim_{x \rightarrow -\infty} g(x); \text{ where } g(x) = \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})},$$

If $\mu_1 > \mu_2$, then which of the following options is/are correct?

- (A) $2\lambda_1 - \mu_2 = 2$ (B) $4\lambda_2 - 2\mu_1 = 3$ (C) $3\lambda_1 - \mu_1 = 2$ (D) $3\lambda_2 - 5\mu_2 = 3$

Q.(15) Let functions f and g are defined as

$$f(x) = \lim_{h \rightarrow 0} \frac{\sin(x+3h) - 3\sin(x+2h) + 3\sin(x+h) - \sin x}{h(\tan(x+2h) - 2\tan(x+h) + \tan x)} \text{ and}$$

$$g(x) = \frac{1 - x(1 + |1 - x|)}{|1 - x|} f\left(\frac{1}{1 - x}\right) \sin \frac{1}{1 - x} \sec^3 \frac{1}{1 - x}.$$

Then which of the following options is/are correct?

- (A) $\lim_{x \rightarrow 1^+} g(x)$ does not exist (B) $\lim_{x \rightarrow 1^-} g(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^+} g(x) = 0$ (D) $\lim_{x \rightarrow 1^-} g(x) = 0$

NUMERICAL TYPE

Q.(16)

If the value of limit $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{x^2} = \frac{Ae}{B}$, where A and B are co-prime natural numbers, then find the value of (A+B) :-

Q.(17) Find the value of $(a - 2be + c) \in \mathbb{R}$, if $b, c \in \mathbb{R} - \{0\}$,

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - \frac{1}{e}}{x^a} = b \text{ and } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{4\sqrt{2}(\sin x + \sin 3x)}{2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2} - \sqrt{2} - \sqrt{2} \cos 2x - \cos \frac{3x}{2}} \right) = c.$$

Here, e denotes the base of natural logarithm.

Q.(18) If

$$f(x) = \begin{cases} \frac{a^{\sin x} - a^{\tan x}}{\tan x - \sin x}, & x > 0 \\ k, & x = 0 \\ \frac{\ln(1+x+x^2) + \ln(1-x+x^2)}{\sec x - \cos x}, & x < 0 \end{cases}$$

is continuous at $x = 0$, and

$$g(x) = \ln \left(2 - \frac{x}{a} \right) \cot(x-a); x \neq a, a > 0,$$

is continuous at $x = a$, then $\left\lfloor g\left(\frac{1}{e}\right) + 2e \right\rfloor$ is equal to (where $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$)

Q.(19) Find the value of $\lambda + n$, where

$$\lim_{x \rightarrow 0} \left(\frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n} \right) = \lambda (\in \mathbb{R} - \{0\}); n \in \mathbb{N}.$$

Q.(20) Let

$$\mathbb{A} = \left\{ a \in \mathbb{W} : \lim_{x \rightarrow 1} \left(\frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right)^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4} \right\}, k = \lim_{x \rightarrow 0^+} \left(\frac{e^{x^x-1} - x^x}{(x^{2x} - 1)^2} \right).$$

Then $\max(\mathbb{A}) + 8k$ is equal to

PASSAGES
Passage-I

Let

$$f(x) = \begin{cases} \frac{e^{\frac{1}{x-1}} - 2}{e^{\frac{1}{x-1}} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases} \text{ and } g(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n+1] \\ b_n + \cos \pi x, & x \in (2n-1, 2n) \end{cases}; n \in \mathbb{Z}.$$

Now, based on above information, answer the following questions

Q.(21) Which of the following options is/are correct?

$$(A) \lim_{x \rightarrow 1^+} f(x) = 1 \quad (B) \lim_{x \rightarrow 1^-} f(x) = -1 \quad (C) \lim_{x \rightarrow 1} f(x) = 1 \quad (D) \lim_{x \rightarrow 1} f(x) = -1$$

Q.(22) If g is continuous for all $x \in \mathbb{R}$, then

$$(A) a_{n-1} = b_{n-1} \quad (B) a_n - b_n = 1 \quad (C) a_n - b_{n+1} = 1 \quad (D) a_{n-1} - b_n = -1$$

Passage-II

 Let f and g are continuous functions defined from $[0,1] \rightarrow \mathbb{R}$ satisfying

$$\max\{f(x): x \in [0,1]\} = \max\{g(x): x \in [0,1]\}.$$

 Also $h(x) = x \cos(\pi x + \pi [x])$, where $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$.

Now, based on above information, answer the following questions

Q.(23) Which of the following options is/are correct?

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$
 (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$

Q.(24) The function h is dis-continuous at $x =$

- (A) -1 (B) 0 (C) 2 (D) 1

MATRIX MATCHING TYPE

Q.(25)

$$\text{Let } f(x) = \begin{cases} \frac{e^{(a+2)x} - 1}{x} & , x < 0 \\ c & , x = 0 \\ \frac{(x + b(x^4))^{1/2} - x^{1/2}}{b(x^{7/2})} & , x > 0 \end{cases}$$

is continuous function $\forall x \in \mathbb{R}$

Column-I

- (A) Complete set of value(s) of a is
 (B) Complete set of value(s) of b is
 (C) Complete set of value(s) of c is
 (D) Complete set of value(s) of $2ac$ is

Column-II

- (P) $\mathbb{R}^+$
 (Q) $\left\{-\frac{3}{2}\right\}$
 (R) $\left\{\frac{1}{2}\right\}$
 (S) $\mathbb{R} - \{0\}$
 (T) $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$

KEY

1. B	2. C	3. A	4. B	5. C
6. ABCD	7. AC	8. ABCD	9. AD	10. ACD
11. AD	12. AC	13. AB	14. ABC	15. AD
16. 35	17. 10	18. 2	19. 4.5	20. 1
21. AB	22. BD	23. AD	24. ACD	25. A-Q, B-P, C-R, D-Q

GR-03 : DIFFERENTIABILITY AND DIFFERENTIATION

SINGLE CORRECT TYPE QUESTIONS

Q.(1)

$$\text{If } f(x) = \begin{cases} \ln(1+|x|) + (b-2)|x+1| + a \tan^{-1} x + 2 & , \quad -2 < x < 0 \\ b & , \quad x = 0 \\ \frac{(c+2)\sin^{-1}\sqrt{\{x\}} + e^{2x} - e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)} - 1}{x^2} & , \quad 0 < x < 2 \end{cases}$$

is derivable in $(-2, 2)$ then find the value of $(9a^2 + b^2 + c^2)$

[Note : $\{y\}$ denotes fractional part of y]

(A) 60

(B) 40

(C) 57

(D) 58

Q.(2) Let f_1 be a differentiable function satisfying

$$f_2(x) = \lim_{n \rightarrow \infty} \binom{n}{k} \left(f_1\left(\frac{x}{n}\right) - f_1(0) \right)^k ; \quad f_1'(0) = \frac{1}{3},$$

$$f_3(k) = \lim_{x \rightarrow 0} \frac{1}{x} \sum_{r=1}^k f_3\left(\frac{x}{r}\right); \quad k \in \mathbb{N}, f_3(0) = 0, f_3'(0) = 1.$$

Then which of the following option(s) is/are equal to

$$\lambda = \sum_{r=0}^{10} \left(\binom{10}{r} \cdot r! \cdot f_2(3) + \frac{f_3(r+1) - f_3(r)}{f_3(11)} \right).$$

(A) 1024

(B) 1026

(C) 1025

(D) 1203

Q.(3)

Let $\llbracket f(x) : \mathbb{A} \rrbracket$ denote the number of points on non-differentiability of any function $y = f(x)$ in the set

$\mathbb{A}$. If g is a polynomial function satisfying

$$g(x+y) = g(x) + g(y) + 2xy - 1 \forall x \in \mathbb{R}, \quad f'(0) = 3,$$

then which of the following option(s) is/are equal to

$$\lambda = \left\lfloor \lim_{n \rightarrow \infty} \sqrt[n]{\sin^{2n} x + \cos^{2n} x} : [-2\pi, 2\pi] \right\rfloor + \left\lceil |g(x) - 1| - 1 \right\rceil : \mathbb{R} \right\rceil.$$

(A) 14

(B) 15

(C) 16

(D) 17

MULTIPLE CORRECT TYPE QUESTIONS

Q.(4)

$$f(x) = \begin{cases} x+a & \text{if } x < 0 \\ |x-1| & \text{if } x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x+1 & \text{if } x < 0 \\ (x-1)^2 + b & \text{if } x \geq 0 \end{cases}. \text{ Where } a \text{ and } b \text{ are non negative real numbers.}$$

If $(g \circ f)(x)$ is continuous for all real x .

 (A) $2a + 3b = 5$

 (B) $3a - 2b = 3$

 (C) g is differentiable at 0

 (D) g is not differentiable at 0.

Q.(5)

The function f defined by $f(x) = \begin{cases} x[x], & 0 \leq x < 2 \\ (x-1)[x], & 2 \leq x \leq 3 \end{cases}$ where $[x]$ = greatest integer less than or equal to

1, is not

(A) continuous at 1

(B) continuous at 2

(C) differentiable at 1

(D) differentiable at 2.

Q.(6) The derivative of

$$f(x) = \tan^{-1} \frac{2x}{1-x^2} \text{ w.r.t. } g(x) = \sin^{-1} \frac{2x}{1+x^2}$$

is equal to

 (A) 1 at $x = -e$

 (B) -1 at $x = -e^{-1}$

 (C) 1 at $x = e^{-1}$

 (D) -1 at $x = e$
Q.(7) Which of the following options is/are correct, for

$$f(x) = \begin{cases} x \left(\frac{\frac{1}{a^x} - a^{-\frac{1}{x}}}{\frac{1}{a^x} + a^{-\frac{1}{x}}} \right), & x \neq 0 \\ 0, & x = 0 \end{cases}.$$

 (A) If $a \in (0,1)$, then f is continuous but not differentiable

- (B) If $a = 1$, then f is continuous but not differentiable
- (C) If $a > 1$, then f is continuous but not differentiable
- (D) If $a \in \mathbb{R}$, then f is continuous but not differentiable

Q.(8) Let $f: [0, \infty) \rightarrow [0, \infty); f(x) = xe^{-f(x)}$ be a bijective function. Then which of the following options is/are correct, if

$$\lambda = \max\{a \in \mathbb{Z} : a \leq f(f(e^{e+1}))\}, \quad \text{and} \quad \mu = \lim_{n \rightarrow \infty} \frac{f(x)}{\ln x}?$$

- (A) $\lambda + \mu = 2$ (B) $3\lambda + 2\mu = 4$ (C) $5\lambda - \mu = 4$ (D) $4\lambda + 5\mu = 10$

Q.(9) Let $[x] = \max\{a \in \mathbb{Z} : a \leq x\}, \{x\} = x - [x],$

$$f(x) = [x]^{\{x\}} + \{x\}^{[x]}; f'\left(\frac{5}{2}\right) = 1 + \sqrt{\lambda} \ln 2, g(x) = \begin{cases} b \sin^{-1} \frac{x+c}{2}, & -\frac{1}{2} < x < 0 \\ \frac{1}{2}, & x = 0 \\ \frac{e^{\frac{ax}{2}} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}.$$

If g is differentiable at $x = 0$ and $|c| < \frac{1}{2}$, then

- (A) $a(4 - c^2) = \lambda^6 b^2$ (B) $a(4 - c^2) = \lambda^{12} b^2$
- (C) $a(\lambda^2 - c^2) = \lambda^6 b^2$ (D) $a(4 - c^2) = \lambda^3 b^2$

Q.(10) Let $f_n(\alpha)$ denote the n^{th} derivative of function f at $x = \alpha$. Also

$$f(x) = -\frac{x^3}{3} + x^2 \sin \frac{3a}{2} - x \sin a \sin 2a - 5 \sin^{-1}(a^2 - 8a + 17),$$

$$g(x) = h\left(x - \sqrt{1 - x^2}\right), h'(x) = 1 - x^2 \text{ and } p(x) = \frac{3x - 1}{x^2 - 1}.$$

Then, which of the following options is/are correct?

- (A) $f'(\sin 8) < 0$ (B) $25 g'\left(\frac{3}{5}\right) = 42$ (C) $p_{50}(0) = 50!$ (D) $f'(\sin 8) > 0$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(11)

If $\sqrt{4 \cos x - 3 \sin x + 5 + y \sin x} = y \cos 2x$, then $y'(\pi)$ is equal to

Q.(12)

If $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $g(f(x)) = h(x)$, then $\lim_{x \rightarrow 1} \frac{h^{-1}(x) - h^{-1}(1)}{x - 1}$ is equal to -

Q.(13)

Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$ then $265 \left[\lim_{h \rightarrow 0} \frac{h^4 + 3h^2}{(f(1-h) - f(1)) \sin(5h)} \right]$ equals

Q.(14)

Let f be a function that is differentiable every where and that has the following properties:

(i) $f(x+h) = f(x) \cdot f(h)$

(ii) $f(x) > 0$ for all real x .

(iii) $f'(0) = -1$

Then find $\sum_{r=1}^{10} \ln \left(\frac{1}{f(r)} \right)$

Q.(15) Find

$$\lambda = \max\{a \in \mathbb{Z} : a \leq \max(A)\}, \text{ where}$$

$$A = \left\{ |12a + 4b + c| \mid |ax^3 + bx^2 + cx + 5| \leq |e^x - e^2| \forall x \geq 0 \right\}.$$

PASSAGES

Passage-I:

Let f be a differentiable function satisfying

$$\lim_{y \rightarrow 0} \left(\frac{f(x+y)}{f(x)} \right)^{\frac{1}{y}} = e^{f(-x) \tan x}.$$

Also right hand derivative at $x = 0$ exists for h , satisfying

$$g(x - y) = g(x)g(y) + h(x)h(y) \forall x, y \in \mathbb{R}.$$

and

$$h(x - y) = h(x)g(y) - g(x)h(y) \forall x, y \in \mathbb{R}.$$

Now, based on above information, answer the following questions

Q.(16) The value of $f'(0)$ is equal to

Q.(17) The value of $g'(0)$ is equal to

Passage-II:

The function f satisfies the following conditions:

$$f(x) = \begin{cases} ax(x-1) + b, & x < 1 \\ x-1, & 1 \leq x \leq 3; \\ p(x)^2 + qx + 2, & x > 3 \end{cases} \quad \begin{matrix} f \text{ is continuous for all } x \in \mathbb{R}, \\ f'(1) \text{ d.n.e.}, \\ f'(x) \text{ is continuous at } x = 3. \end{matrix}$$

Further given that function g is differentiable at $x = 0$,

$$g(x) = (\cos x)^{\frac{1}{x}} \quad \text{and} \quad \lambda = \lim_{n \rightarrow \infty} \left(\frac{2}{\pi} (n+1) \cos^{-1} \frac{1}{n} - \left(n + \frac{2}{\pi} \right) \right).$$

Now, based on above information, answer the following questions.

Q.(18) The value of $b + p + |q|$ is equal to

Q.(19) The value of $\lambda - 2g'(0)$ is equal to

MATRIX MATCHING

Q.(20) Consider the following

$$f(x) = g(g(x)); \quad \text{where} \quad g(x) = \begin{cases} 1-x, & x \in [0,1] \\ x+2, & x \in (1,2). \\ 4-x, & x \in [2,4] \end{cases}$$

Now, Match the following columns for function f

Column-I

Column-II

(A) $x = 0$

(P) continuous

 (B) $x = 1$

(Q) dis-continuous

 (C) $x = 2$

(R) differentiable

 (D) $x = 3$

(S) non-differentiable

KEY

1. C	2. C	3. A	4. BD	5. ABD
6. CD	7. ABC	8. AC	9. AC	10. ABC
11. 1	12. 4	13. 3	14. 45	15. 7
16. 0	17. 0	18. 1.33	19. 2	20. A-PR, B-PS, C-QS, D-QS

GR-04 : TANGENTS AND NORMALS, MONOTONICITY

SINGLE CORRECT TYPE QUESTIONS

Q.(1) Consider the following statements:

S-1: A closed vessel tapers to a point both at its top E and its bottom F and is fixed with EF vertical.

When the depth of the liquid in it is x cm, the volume of the liquid in it is $x^2(15 - x)$ cu. cm.

S-2: A horse runs along a circle with a speed of 20 km/hr. A lantern is at the center of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $\left(\frac{1}{8}\right)^{th}$ of the circle is v km/hr.

Then the numerical value of $v + EF$ is equal to

- (A) 30 (B) 40 (C) 50 (D) 60

Q.(2) Let θ be the angle of intersection of curves $\mathbb{C}$ and $\mathbb{D}$. Let ϕ be acute the angle of intersection of curves $\mathbb{E}$ and $\mathbb{F}$. If

$$\mathbb{C}: \frac{x^2}{a^2} + \frac{y^2}{4} = 1, \quad \mathbb{D}: y^2 = 16x, \quad \mathbb{E}: y = |x^2 - 1|, \quad \mathbb{F}: y = |x^2 - 3|,$$

and $\theta = \frac{\pi}{2}$, then value of $a^2 + 7 \tan \theta$ is equal to

- (A) $2 + 4\sqrt{2}$ (B) $4\sqrt{2} - 2$ (C) $2 + 3\sqrt{2}$ (D) $3 + 4\sqrt{2}$

Q.(3) Let

$$\mathbb{A} = \left\{ k \in \mathbb{R} \mid \alpha, \beta, \gamma \text{ are roots of } \frac{x^3}{a^2} + \left(a^2 + 4 + \frac{1}{a^2}\right)x = \frac{x^2}{a^2} + 1 \right\}$$

$$\mathbb{B} = \left\{ \left| \frac{a}{2} - \frac{b}{2} \right| \mid |a - 1| + |b - 1| = |a| + |b| = |a + 1| + |b + 1|, a, b \in \mathbb{R} \right\},$$

$$\lambda = \min(\mathbb{A}), \quad \mu = \min(\mathbb{B}), \quad k = \sum_{\alpha, \beta, \gamma} \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right).$$

Then which of the following options is/are equal to $\lambda + \mu$?

- (A) 5 (B) 2 (C) 3 (D) 4

Q.(4)

Let $f : \mathbb{R} - \{a, b\} \rightarrow \mathbb{R}$, $f(x) = \frac{(x-1)(x-3)}{(x-a)(x-b)}$, then which of the following is / are correct :-

- (A) $f(x) = K$ has 3 distinct solution $\forall K \in (0, \infty)$ and $\forall a, b \in (3, \infty)$ and $(a < b)$
 (B) $f(x) = K$ has two solutions distinct $\forall K \in (0, \infty)$ if $a \in (1, 3)$ and $b \in (3, \infty)$
 (C) range of $f(x)$ can-not be $\mathbb{R}$ if $a, b \in (3, \infty)$ and $(a < b)$
 (D) $f(x)$ is strictly increasing in the interval $(-\infty, a)$ if $a, b \in (3, \infty)$ and $(a < b)$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(5) Consider the following statements:

S-1: A point $P(x, y)$ moves on curve $\mathbb{C}: x^{\frac{2}{3}} + y^{\frac{2}{3}} = k^{\frac{2}{3}}; k > 0$, such that perpendiculars drawn from origin on tangent and normal at P, are of length $p_1(x)$ and $p_2(x)$ respectively. Let

$$\lambda = \frac{dp_1}{dx} \cdot \frac{dp_2}{dx},$$

S-2: Sides of ΔABC vary in such a way that its circumradius remains constant. Let

$$\mu = \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C},$$

where a, b, c represent the sides BA, CA, AB of ΔABC (dx represents small increment in variable x).

Then which of the following options is/are correct?

- (A) $\lambda + \mu \geq 0$ (B) $\lambda + \mu \leq 0$ (C) $\lambda + \mu = 0$ (D) $\lambda \leq 0$

Q.(6) Let L denote a tangent of $y^2 - 2x^3 - 4y + 8 = 0$ passing through the point (1,2). Also let P be the point of intersection of tangents drawn to $x^2y = 1 - y$ at points where it is intersected by the curve $xy = 1 - y$. Then which of the following options is/are correct?

(A) $L: 2\sqrt{3}x - y = 2(\sqrt{3} - 1)$

(B) $P = \left(0, \frac{1}{2}\right)$

(C) $2\sqrt{3}x + y = 2(\sqrt{3} + 1)$

(D) $P = (0, 1)$

Q.(7) Which of the following options is/are correct?

(A) 1 is the length of x-intercept of tangent to both branches of function

$$f(x) = \begin{cases} -x^2, & x < 0 \\ x^2 + 8, & x \geq 0 \end{cases}$$

(B) If the area of triangle formed by tangent to $xy^n = a^{n+1}$ and coordinate axes is a constant, then $n=1$,

(C) The segment of normal to curve $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ contained between coordinate-axes is equal to $2a$.

(D) The area of triangle formed by tangent to $xy^n = a^{n+1}$ and coordinate axes is constant, for infinitely many real values of n .

Q.(8) Consider the following curves, where t is a parameter

$$\mathbb{C}: x = \sqrt{a} \sec t, y = \sqrt{b} \csc t, \quad \mathbb{D}: x = \sec^2 t, y = \cot t.$$

If λ is the x-intercept of the tangent of $\mathbb{C}$ at any arbitrary point $P(\alpha, \beta)$, and tangent to $\mathbb{D}$ at $A\left(t = \frac{\pi}{4}\right)$ meets it again at B, then

(A) λ is proportional to α^3

(B) $2AB = 3\sqrt{5}$

(C) λ is proportional to $\sqrt[3]{\alpha}$

(D) $3AB = 2\sqrt{5}$

Q.(9) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying

$$f'(x) > 2f(x) \forall x \in \mathbb{R}; f(0) = 1.$$

Then which of the following options is/are correct for all $x \in (0, \infty)$?

(A) $f(x) > e^{2x}$

(B) $f(x) < -e^{2x}$

(C) $|f(x)| > e^{2x}$

(D) $f'(x) > 0$

Q.(10) If $ax + by + c = 0$ is equation of a line which is tangent at one point and normal at another point of the curve $27y^2 = 4x^3$, then which of the following option(s) is/are possible?

- (A) $c = 2a$ (B) $c + 2a = 0$ (C) $a = b\sqrt{2}$ (D) $a = -b\sqrt{2}$

Q.(11) Let tangent to $\mathbb{C}$ at $P(1,7)$ touches the circle $\mathbb{D}$ at (α, β) , where

$$\mathbb{C}: y = x^2 + 6, \quad \mathbb{D}: x^2 + y^2 + 16x + 12y + c = 0, \quad \mathbb{E}: y = e^x.$$

If $A(c, e^c)$, $B(c-1, e^{c-1})$, $C(c+1, e^{c+1})$, then which of the following options is/are correct?

- (A) Tangent of $\mathbb{E}$ at A intersects AB on LHS of A (B) $2\alpha - 3\beta = 9$
 (C) Tangent of $\mathbb{E}$ at A intersects AB on RHS of A (D) $3\alpha - 2\beta = 4$

Q.(12) Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{x \in \mathbb{R} \mid x^4 - 4x^3 + 12x^2 + x - 1 = 0\},$$

$$\mathbb{B} = \{x \in \mathbb{R} \mid x^5 - 5x + a = 0\}.$$

Then, which of the following options is/are correct?

- (A) $a > 2|\mathbb{A}| \Rightarrow |\mathbb{B}| = 1$ (B) $|\mathbb{B}| = 1 \Rightarrow |a| > 2|\mathbb{A}|$
 (C) $|a| < 2|\mathbb{A}| \Rightarrow |\mathbb{B}| = 3$ (D) $|\mathbb{B}| = 3 \Rightarrow |a| \leq 2|\mathbb{A}|$

Q.(13) Let function f be defined as

$$f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}; f(x) = (\ln(\sec x + \tan x))^3.$$

Then which of the following options is/are correct?

- (A) f is an even function (B) f is an odd function
 (C) f is an injective function (D) f is a surjective function

Q.(14) Which of the following options is/are correct?

- (A) $x \in (0,1) \Rightarrow \sin \pi x \leq 4x(1-x)$
 (B) $f(x) = -\sin^3 x + 3\sin^2 x + 5$ is increasing in $x \in \left(0, \frac{\pi}{2}\right)$

(C) $f(x) = -\sin^3 x + 3 \sin^2 x + 5$ is decreasing in $x \in \left(-\frac{\pi}{2}, 0\right)$

(D) $f(x) = \sec x - \csc x$ is increasing in $x \in \left(0, \frac{\pi}{2}\right)$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(15) Let $P(x_1, y_1)$ be a point on curve

$$\mathbb{C}: (x^2 - 11)(y + 1) + 4 = 0,$$

where $x_1, y_1 \in \mathbb{N}$. If the area of the triangle formed by normal drawn to $\mathbb{C}$ at P and the coordinate axes is equal to

$$\frac{a}{b}; a, b \in \mathbb{N} \quad \text{and} \quad \lambda = \min(a - 6b),$$

then find the value of λ .

Q.(16) Find λ , if

$$\left\{a \in \mathbb{R} \mid \{x \in \mathbb{R}^- \mid 3 - x^2 > |x - a|\} \text{ is NOT a NULL set} \right\} = \left(-\frac{13}{4}, \lambda\right).$$

Q.(17) Let function f be defined as

$$f: \mathbb{R} \rightarrow \mathbb{R}; f(x) = ax + \cos 2x + \sin x + \cos x.$$

Further $m, n \in \mathbb{Z}$ such that

$$\left\{a \in \mathbb{R} \mid f \text{ is a strictly increasing function.} \right\} = \left[\frac{m}{n}, \infty\right); \gcd(m, n) = 1.$$

Then the value of $(m - n)$ is equal to

Q.(18)

Let p be a 6-degree polynomial satisfying

$$p(x) = p(2 - x) \forall x \in \mathbb{R}, \quad \text{and} \quad |\{x \in \mathbb{R} \mid p(x) = 0\}| = 5.$$

If S denotes the sum of roots of polynomial p , and a is a positive constant satisfying

$$\{x \in \mathbb{R} \mid f''(x) > 0\} = \left(-\frac{\lambda}{a}, \frac{a}{\mu}\right), \quad f(x) = \begin{cases} xe^{ax}, & x \leq 0 \\ x + ax^2 - x^3, & x > 0 \end{cases}$$

then $\lambda + \mu + S$ is equal to (Here $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$)

Q.(19) Let r_f denote the number of distinct real roots of a polynomial f . Suppose

$$\mathbb{A} = \{(x^2 - 1)^2(ax^3 + bx^2 + cx + d) \mid a, b, c, d \in \mathbb{R}\}.$$

If $f \in \mathbb{A}$, then minimum possible value of $r_{f'} + r_{f''}$ is equal to

Q.(20) Let θ be acute angle between $y = x^x \ln x$ and $y \ln 2 = 2^x - 2$ at their point of intersection on line $y = 0$. If

$$\left\{a \in \mathbb{R} \mid f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax; f'(x) \geq 0 \forall x \in \mathbb{R}\right\} = [\lambda, \infty),$$

then $7\lambda + 9 \tan \theta$ is equal to

PASSAGES

Passage-I:

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following properties:

$$f'(x) > 0 \forall x \in (-\infty, -2) \cup (3, \infty), \quad f'(x) \leq 0 \forall x \in (-2, 3), \quad \lim_{x \rightarrow \infty} f(x) = 0,$$

$$f \text{ is continuous } \forall x \in \mathbb{R} - \{3\}, \quad f \text{ is differentiable } \forall x \in \mathbb{R} - \{-2, 3\}, \quad \lim_{x \rightarrow 3} f(x) \rightarrow -\infty,$$

$$f''(x) > 0 \forall x \in (-\infty, 0) - \{-2\}, \quad f''(x) < 0 \forall x \in (0, \infty) - \{3\}, \quad \lim_{x \rightarrow -\infty} f(x) = 3,$$

and $f(0) = 0$. Now, answer the following questions:

Q.(21) Let λ be the maximum number of solutions of $f(x) = |x|$. If

$$f'(0) > -3 \text{ and } f(-2) > 6,$$

then number of solutions of $f(x) + 3x = 0$ is equal to

(A) $\lambda + 1$

(B) $\lambda - 1$

(C) $2\lambda - 1$

(D) $3\lambda - 4$

Q.(22) The function $y = f(-|x|)$ is

- (A) differentiable $\forall x \in \mathbb{R}$, if $f'(0) = 0$.
 (B) continuous but NOT differentiable at two points, if $f'(0) = 0$.
 (C) continuous but NOT differentiable at one point, if $f'(0) = 0$.
 (D) dis-continuous at two points if $f'(0) = 0$.

Passage-II:

Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{27\lambda \in \mathbb{R} \mid |\mathbb{B}| = 3\}, \mathbb{B} = \{x \in \mathbb{R} \mid x^3 + x^2 - x + \lambda = 0\} \text{ and } f(x) = 2 - \sqrt{4 - (x - 1)^2}.$$

Now, answer the following questions:

Q.(23) Which of the following options lie in set $\mathbb{A}$?

- (A) 5 (B) -16 (C) -27 (D) 4

Q.(24) Which of the following options is/are correct?

- (A) f is monotonic in interval $(-1, 1)$. (B) f is monotonic in interval $(1, 3)$.
 (C) $y = 2x - 2\sqrt{5}$ is tangent to f (D) $y = -3x + 5 + 2\sqrt{10}$ is tangent to f

MATRIX MATCHING

Q.(25) Match the curves in Column-I with curves in Column-II, if they touch each other.

Column-I

Column-II

(A) $y^2 = 4x$

(P) $x^2 + 676y^2 = 676$.

(B) $xy = 4$

(Q) $5x - 4y = 1$.

(C) $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 9$

(R) $x^2 + y^2 = 8$.

(D) $x = 2t - t^2, y = t + t^2$

(S) $x^2 + y^2 = 6x - 1$.

KEY

1. C	2. A	3. D	4. C	5. BD
6. ABD	7. ABC	8. AB	9. ACD	10. BCD
11. AB	12. ACD	13. BCD	14. ABCD	15. 3
16. 3	17. 9	18. 11	19. 5	20. 10
21. A	22. B	23. BD	24. ABC	25. A-S, B-R, C-P, D-Q

GR05-MAXIMA-MINIMA, MVT

SINGLE CORRECT TYPE QUESTIONS

Q.(1) Which of the following options is/are correct, if $m_{\mathbb{X}}$ denotes the smallest element in a set $\mathbb{X}$,

$$\mathbb{A} = \{e^{x^2} + e^{-x^2} | x \in [0,1]\}, \quad \mathbb{B} = \{xe^{x^2} + e^{-x^2} | x \in [0,1]\}, \quad \mathbb{C} = \{x^2e^{x^2} + e^{-x^2} | x \in [0,1]\}?$$

- (A) $m_{\mathbb{A}} = m_{\mathbb{B}} \neq m_{\mathbb{C}}$ (B) $m_{\mathbb{B}} \neq m_{\mathbb{A}} = m_{\mathbb{C}}$ (C) $m_{\mathbb{A}} \neq m_{\mathbb{B}} \neq m_{\mathbb{C}}$ (D) $m_{\mathbb{A}} = m_{\mathbb{B}} = m_{\mathbb{C}}$

Q.(2)

$$\text{Let } f(x) = \begin{cases} 5, & x = 0 \\ x^2 + 5x + \alpha, & 0 < x < 2 \\ \beta x + \gamma, & 2 \leq x \leq 3 \end{cases}$$

If $f(x)$ satisfies all the conditions of Lagrange's mean value theorem in $[0,3]$ and $P(c, f(c))$ is the point on the curve $f(x)$ in $c \in [0,3]$ where the tangent is parallel to the chord joining the end points, then

find the value of $\frac{1}{17}(\alpha + 2\beta + 3\gamma + 6c)$

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(3)

If $f(x)$ is a differentiable function and $0 < a_1 < a_2 < \dots < a_{100} < 100$ and

$f(a_r) = (-1)^r \cdot r$, $1 \leq r \leq 99$ and $f(a_{100}) = 0$ and roots of equation $f(x) = 0$ are rational

then the minimum number of zeroes of $f(x) \cdot \cos x + f'(x) \cdot \sin x$ for $x \in [0, 100]$ is

- (A) 129 (B) 130 (C) 131 (D) 128

Q.(4)

Let $f(x)$ satisfy the requirement of Lagrange's mean value theorem in $[0, 2]$. If $f(0) = 0$ and $|f'(x)| \leq \frac{1}{2}$ for all $x \in [0, 2]$, then which of the following is not true ?

- (A) $f(x) \leq 2$ (B) $|f(x)| \leq 2x$
 (C) $f(x) = 3$ for at least one $x \in [0, 2]$ (D) $|f(x)| \leq 1$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(5)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 2x^2 \ln |x| - 5x^2, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Which of the following statement(s)

is/are correct?

- (A) $f(x)$ has exactly one local maximum and two local minimum points
 (B) $f(x)$ is strictly increasing in $(10, \infty)$
 (C) Absolute minimum value of $f(x)$ exist but absolute maximum value of $f(x)$ does not exist
 (D) In the interval $x \in (0, \infty)$, $f(x) = k$ has two distinct roots

Q.(6)

Which of the following statement(s) is/are correct ?

- (A) If $f(x) = x^2 + 10 \sin x$, $-\infty < x < \infty$ then $\exists$ some $c \in \mathbb{R}$ such that $f(c) = 2015$.
 (B) Suppose f is continuous function on $[1, 5]$ and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, then $f(3) > 6$.
 (C) The functions $f(x) = \operatorname{sgn}(\tan^{-1} e^{-x})$ and $g(x) = \cos(2[\cot^{-1} x]\pi)$ are identical functions, where $\operatorname{sgn}(k)$ and $[k]$ denote signum function of k and greatest integer less than or equal to k respectively.
 (D) If the function $g(x) = \operatorname{sgn}(kx)$ is continuous for every $x \in \mathbb{R}$, then number of real values of k is one. (where $\operatorname{sgn}(x)$ denotes signum function of x).

Q.(7)

Let $f(x) = \frac{(x-1)^2 \cdot e^x}{(1+x^2)^2}$, then which of the following statement(s) is(are) correct?

- (A) $f(x)$ is strictly increasing in $(-\infty, -1)$
- (B) $f(x)$ is strictly decreasing in $(1, \infty)$
- (C) $f(x)$ has two points of local extremum
- (D) $f(x)$ has a point of local minimum at some $x \in (-1, 0)$

Q.(8)

If $f(x)$ is double differentiable function such that $|f''(x)| \leq 5$ for each x in the interval $[0, 4]$ and f takes its largest value at an interior point of this interval, then value of $|f'(0)| + |f'(4)|$ cannot be :-

- (A) 10
- (B) 15
- (C) 21
- (D) 28

Q.(9) Which of the following options is/are correct?

(A) If f and g are continuous functions in $[a, b]$ and differentiable in (a, b) then there exists at least

one $c \in (a, b)$ such that $\left| \frac{f(a)}{g(a)} - \frac{f(b)}{g(b)} \right| = (b-a) \left| \frac{f'(c)}{g'(c)} \right|$.

(B) If functions f, g, h are continuous in $[a, b]$ and differentiable in (a, b) , then there exists at least

one $c \in (a, b)$ such that $\left| \frac{f(a)}{g(a)} - \frac{f(b)}{g(b)} \right| = 0$.

(C) Let $a > 0$ and f be continuous in $[-a, a]$ and $f'(x) \leq 1 \forall x \in (-a, a)$. If $f(a) = a, f(-a) = -a$, then $f(0) = 0$.

(D) Let f is continuous in $[a, b]$ and differentiable in (a, b) . If $f(a) = a$ and $f(b) = b$, then there exists $(c_1, c_2) \subseteq (a, b)$ such that $f'(c_1) + f'(c_2) = 2$.

Q.(10) Let $x = a$ be a point where $f(x) = 2|x| + |x+2| - |2|x| - |x+2||$ has a local extremum, and

$$\beta = \min \left\{ k \in \mathbb{R} \mid 4kx^2 + \frac{1}{x} \geq 1 \forall x > 0 \right\}.$$

Then which of the following option(s) is/are possible?

- (A) $54\beta = -a$ (B) $81\beta = -a$ (C) $54\beta = -3a$ (D) $27\beta = -2a$

Q.(11) Let $f: \mathbb{R} \rightarrow \mathbb{R}^+$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $(f'(2), g(2)) = (0, 0) \neq (f''(2), g'(2))$,

$$\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1 \text{ and } h(x) = \begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}.$$

Then which of the following options is/are INCORRECT?

- (A) f has local minimum at $x = 2$, and h' has exactly three solutions in $(-\pi, \pi)$
 (B) f has local maximum at $x = 2$, and h attains maximum at $x = 0$.
 (C) $f''(2) > f(2)$, and h attains minimum at $x = 0$.
 (D) $f(x) = f''(x)$ has at least one real root, and h' has more than 3 roots in $(-\pi, \pi)$

Q.(12) Which of the following options is/are correct?

- (A) There exists $a < c < b$ satisfying

$$\frac{\sin b - \sin a}{\cos a - \cos b} = \cot c \quad \forall (a, b) \in \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid \cos x \neq \cos y\}.$$

- (B) For all $0 < a < b$, we have

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.$$

- (C) Let f is continuous in interval $[a, b]$ and differentiable in interval (a, b) . Then $\forall (a, b) \subseteq \mathbb{R} - \{0\}$, there exists at least one $c \in (a, b)$ for which

$$\frac{f(b) - f(a)}{\frac{1}{b} - \frac{1}{a}} = -c^2 f(c).$$

- (D) If f is differentiable in interval $[1, 5]$, then there exists $(a, b) \subseteq [1, 5]$ satisfying

$$\frac{f^2(5) - f^2(1)}{f'(a)f(b)} = 8.$$

Q.(13) For every twice differentiable function $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 10$, which of the following options is/are correct?

- (A) There exists interval $I_1 \subseteq \mathbb{R}$, and $I_2 \subseteq \mathbb{R}$ such that f is one-one in I_1 and many-one in I_2 .
- (B) There exists $\alpha \in (-4, 0)$ and $\beta \in (0, 4)$ such that $|f'(\alpha)| \leq 1$ and $|f'(\beta)| \leq 1$.
- (C) f must be a many-one function satisfying $\lim_{n \rightarrow \infty} f(n) = 1$.
- (D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

Q.(14) Which of the following options is/are correct for

$$f(x) = \begin{cases} \sqrt{x} \ln x, & x > 0 \\ 0, & x = 0 \end{cases} ?$$

- (A) f is continuous at $x = 0$ and has no absolute maxima.
- (B) Smallest possible value of f is $-\frac{1}{e}$.
- (C) $f'(0) = d.n.e$ and $\lim_{x \rightarrow 0} f'(x) = d.n.e$.
- (D) f has an inflection point at $x = 1$.

NUMERICAL ANSWER TYPE QUESTIONS

Q.(15) A conical vessel is to be prepared out of a circular sheet of gold of unit radius. If $\lambda\pi$ is the sectorial area which is used to form the sheet so that the vessel has maximum volume, then the integer nearest to value of $\sqrt{300\lambda}$ is equal to

Q.(16) Find $13(\max \mathbb{A}) + 2(\min \mathbb{B})$, if

$$\mathbb{A} = \left\{ \frac{1}{|x-4|+1} + \frac{1}{|x-8|+1} \mid x \in \mathbb{R} \right\}, \text{ and } \mathbb{B} = \left\{ \frac{b+1}{a+b-2} \in \mathbb{R} \mid a^2 + b^2 = 1 \right\}.$$

Q.(17)

Let A (1, -1), B (4, -2) and C (9, 3) be the vertices of the triangle ABC. A parallelogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.

Q.(18) From a fixed point A on circumference of a circle of radius $\sqrt{2\sqrt{3}}$, let the perpendicular AB fall on the tangent at C on it. Then greatest possible area of ΔABC is equal to

Q.(19) Consider all the rectangles lying in the region

$$\{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq \frac{\pi}{2} \text{ and } 0 \leq y \leq 2 \sin 2x\}$$

and having one side on X-axis. The area of the rectangle which has maximum perimeter among all such rectangles, is $k\pi$. Then value of $k\sqrt{3}$ is equal to

Q.(20) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow (0, \infty)$ are functions satisfying

$$f(x) = \ln g(x), \quad f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4.$$

Then number of points of local minima of function g is/are

PASSAGES

Passage-I:

Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ and $g: \mathbb{R}^+ \rightarrow \mathbb{R}$ be functions defined by

$$f(x) = \int_0^x \prod_{r=1}^{21} (t-r)^r dt; x > 0 \quad g(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450; x > 0.$$

Let m_p and M_p denote the number of points of local minimum and local maximum of a certain function p . Then, answer the following questions:

Q.(21) The value of $2m_f + 3M_f + m_f M_f$ is equal to

Q.(22) The value of $6m_g + 4M_g + 8m_g M_g$ is equal to

Passage-II:

If $f(x)$ is defined in $[a, b]$ and (i) $f(x)$ is continuous on $[a, b]$
 (ii) $f(x)$ is derivable on (a, b)
 (iii) $f(a) = f(b)$. Then has atleast one point $c \in (a, b)$ such that $f'(c) = 0$. This is known as Rolle's theorem. This means $f(x)$ is a polynomial then between any two roots of $f(x) = 0$ there is always lies a root of $f'(x) = 0$

Q.(23)

A twice differentiable function f such that $f(a) = f(b) = 0$ and $f(c) > 0$, $a < c < b$. Then there is atleast one value ' ϵ ' between a and b for which

- (A) $f''(\epsilon) = 0$ (B) $f''(\epsilon) > 0$ (C) $f''(\epsilon) < 0$ (D) $f''(\epsilon) \geq 0$

Q.(24) If f is continuous in $[a, b]$ and differentiable in (a, b) , then

$$\frac{f'(c)}{3c^2} = \frac{f(b) - f(a)}{b^3 - a^3}$$

holds for

- (A) at least one $c \in (a, b)$ (B) at most one $c \in (a, b)$
 (C) exactly one $c \in (a, b)$ (D) no $c \in (a, b)$

MATRIX MATCHING
Q.(25)

Let $f(x)$ be a real valued function defined by $f(x) = x^2 - 2|x|$ and

$$g(x) = \begin{cases} \text{minimum} \{f(t); -2 \leq t \leq x\}, & x \in [-2, 0) \\ \text{maximum} \{f(t); 0 \leq t \leq x\}, & x \in [0, 3] \end{cases}$$

Now, match the following columns:

Column-I

- (A) g is NOT continuous at $x =$
 (B) g is NOT differentiable at $x =$

Column-II

- (P) -2
 (Q) 0

(C) Number of points of strict local extremum of g is

(R) 1

 (D) Maximum value of g is equal to

(S) 3

KEY

1. D	2. B	3. B	4. C	5. ABC
6. ABCD	7. AC	8. CD	9. ABCD	10. AC
11. AC	12. ABD	13. ABD	14. ACD	15. 3
16. 32	17. 5	18. 2.25	19. 0.5	20. 1
21. 57	22. 6	23. C	24. A	25. A-Q, B-Q, C-Q, D-S

GR-06 : INTEGRATION

SINGLE CORRECT TYPE QUESTIONS

Q.(1)

Let $f(t)$ be a continuous function for $t \in \mathbb{R}$. Let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} xf(x(2-x))dx$ and $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f(x(2-x))dx$, then :-

- (A) $\frac{I_1}{I_2} = 2$ (B) $\frac{I_1}{I_2} = 1$ (C) $\frac{I_1}{I_2}$ is not defined (D) $\frac{I_1}{I_2} = \frac{1}{2}$

Q.(2)

The absolute value of $\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx}$ is equal to -

- (A) e (B) πe (C) $e/2$ (D) π/e

Q.(3)

The value of definite integral $\int_{-\pi}^{\pi} \frac{x^2}{1 + \sin x + \sqrt{1 + \sin^2 x}} dx$, is :

- (A) $\frac{3x^2}{2}$ (B) $\frac{\pi^3}{3}$ (C) $\frac{2\pi^3}{3}$ (D) $\frac{\pi^3}{6}$

Q.(4) Which of the following option(s) is/are equal to

$$I = \int \frac{\cos 2x \sin 4x}{\cos^4 x (1 + \cos^2 2x)} dx.$$

Here C denotes the constant of integration.

- (A) $\sec^2 x - \ln(1 + \cos^2 2x) + 2 \ln(1 + \cos 2x) + C$
- (B) $\sec^2 x + \ln(1 + \cos^2 2x) + 2 \ln(1 + \cos 2x) + C$
- (C) $\sec^2 x - \ln(1 + \cos^2 2x) - 2 \ln(1 + \cos 2x) + C$
- (D) $\sec^2 x + \ln(1 + \cos^2 2x) - 2 \ln(1 + \cos 2x) + C$

Q.(5) Let

$$I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx, \quad J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx,$$

$$K = \frac{1}{2} \ln \frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \quad \text{and} \quad L = \frac{1}{2} \ln \frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1}.$$

Then which of the following options is/are correct? (Here C denotes the constant of integration.)

- (A) $I - J = K + C$ (B) $I - J = C - L$ (C) $I + J = C - K$ (D) $I + J = L + C$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(6)

Let $I_1 = \int_0^{\pi/2} \cos(\pi \sin^2 x) dx$, $I_2 = \int_0^{\pi/2} \cos(2\pi \sin^2 x) dx$ and $I_3 = \int_0^{\pi/2} \cos(\pi \sin x) dx$ then -

- (A) $I_1 = 0$ (B) $I_2 + I_3 = 0$ (C) $I_1 + I_2 + I_3 = 0$ (D) $I_2 + I_3 \neq 0$

Q.(7)

Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'(1/4) = 0$. Then

- (A) $f''(x)$ vanishes at least twice on $[0, 1]$ (B) $f'(1/2) = 0$

(C) $\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x \, dx = 0$

(D) $\int_0^{1/2} f(t) e^{\sin \pi t} dt = \int_{1/2}^1 f(1-t) e^{\sin \pi t} dt$

Q.(8) Suppose a and b are positive real numbers such that $ab = 1$. Let for any real parameter t , the distance from the origin to the line $(ae^t)x + (be^{-t})y = 1$ be denoted by $D(t)$. If

$$I = \int_0^1 \frac{dx}{(D(x))^2},$$

then which of the following options is/are correct?

- (A) The value of I is $\frac{e^2-1}{2} \left(b^2 + \frac{a^2}{e^2}\right)$. (B) I is minimum when $b = \sqrt{e}$
- (C) Minimum value of I is $e - \frac{1}{e}$ (D) The value of I is $\frac{e^2+1}{2} \left(b^2 + \frac{a^2}{e^2}\right)$

Q.(9)

If $\int \cos(2016x) \cdot (\sin^{2014} x) dx = f(x) + c, f(0) = 0$ then -

- (A) $f(x)$ is odd (B) $\int_{\pi}^{2\pi} f(x) dx = 0$ (C) $\int_0^{\pi} f(x) dx = 0$ (D) $\lim_{x \rightarrow 0} \frac{2015f(x)}{x^{2015}} = 1$

Q.(10) Which of the following option(s) satisfy the equation

$$\int_{-\infty}^{\infty} \frac{\ln|t|}{t^2 + x^4} dx = \frac{\pi}{2} \ln 2; x \in \mathbb{R}.$$

- (A) Number of values of x satisfying the equation is 5
- (B) Roots of $x^2 - (2 + \sqrt{2})x + 2\sqrt{2} = 0$, satisfy the above equation
- (C) Product of values of x satisfying above equation is 8
- (D) Roots of $x^3 + 31x^2 - 2x - 21 = 0$ satisfy the above equation

Q.(11) Choose the correct option(s), if

$$\int_{\frac{3}{2}}^x \frac{x^2 + 3}{x^4 + x^3 - x^2 - 3x + 9} dx = f(x).$$

- (A) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{2}{\sqrt{19}}$ (B) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{4}{\sqrt{19}}$
- (C) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{5}{\sqrt{19}}$ (D) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{3}{\sqrt{19}}$

Q.(12) Let

$$f(x) = \int_1^x \frac{x^2 - 1}{x^3 \cdot \sqrt{2x^4 - 2x^2 + 1}} dx.$$

Then which of the following options is/are correct?

(A) $f(2) = \frac{1}{4}$

(B) $f(2) = \frac{1}{8}$

(C) $f(\sqrt{2}) = \sin \frac{\pi}{10}$

(D) $f(\sqrt{2}) = \sin \frac{\pi}{10} - \frac{1}{2} \sin \frac{\pi}{6}$

Q.(13) Which of the following options is/are correct?

A) If $f(x) = \sqrt{x}$, and $g(x) = x^{\frac{1}{3}}$, then

$$\int_0^2 (f(1+x^3) + g(x^2+2x)) dx = 6$$

B) Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a twice differentiable function such that $2f'(x) + xf''(x) = 1 \quad \forall x$, then

$$\int_{-1}^1 x^3 f(x) dx = \frac{1}{5}, \text{ if } f'\left(\frac{1}{2}\right) = \frac{1}{2}$$

C) If f is a function satisfying

$$f(x) = x + \int_0^1 (x+t)f(t) dt, \text{ then } \int_0^1 f(x) dx = -6$$

D) If $f(x) = \ln x$ then

$$\int_1^{e^2} \frac{f(xe^{x+1})}{(x+1)^2 + f^2(x^x)} dx > \frac{\pi}{4}$$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(14) Find the value of I, where

$$I = \int_0^{\frac{\pi}{2}} (\cos(2\pi \sin^2 x) - 2 \cos(\pi \sin^2 x) + \cos(\pi \sin x)) dx.$$

Q.(15) Evaluate the integral I, where (take $\pi = 3.14$)

$$I = \int_0^{\pi} e^{\cos x} \cdot \cos(\sin x) dx.$$

Q.(16) Let us define the integral I, as

$$I = \lim_{n \rightarrow \infty} \int_1^n \frac{x^4 + 2x^2 + 1}{\left(x^{\frac{113}{25}} + \frac{11}{3}x^{\frac{63}{25}} + 11x^{\frac{13}{25}}\right)^5} dx = \frac{p}{q}; p, q \in \mathbb{N}; HCF(p, q) = 1.$$

Number of divisors of 'p' is equal to

Q.(17) Find the value of 3I, where I is defined as

$$I = \sum_{r=0}^{10} \int_r^{r+4} |x-1-r| \cdot |x-2-r| \cdot |x-3-r| dx$$

Q.(18) Find the area bounded by following

$$f(x) = x^3 - 6x^2 + 11x - 6; y = 0; x = -1; x = 4.$$

Q.(19) Given that

$$\int_{-1}^1 \tan^{-1} \frac{1}{\sqrt{1-x^2}} \times \cot^{-1} \frac{x}{\sqrt{1-x^2|x|}} dx = k\pi^2, \text{ where } k \in \mathbb{R}.$$

Then $(1-k)^2$ is equal to

Q.(20) Consider the following two integrals

$$A = \int_0^1 \sqrt{x^7(1-x)^9} dx \quad \text{and} \quad B = \int_0^1 \sqrt{\frac{x^7(1-x)^9}{(x+5)^{20}}} dx.$$

If $|\ln A - \ln B| = a \ln p_1 + b \ln p_2 + c \ln p_3$, where $p_1 < p_2 < p_3$ are prime and $a, b, c \in \mathbb{Q}$, then the value of $a - 2b + c$ is equal to

PASSAGES

Passage-I:

Consider the following integrals

$$I = \int_1^2 \frac{1}{x-1+\sqrt{x^2-3x+3}} dx \quad \text{and} \quad J = \int_1^2 \frac{x^2-x+1}{x(2x-1)^2} dx$$

Now, answer the following questions:

Q.(21) The value of $I - 2J$ is equal to

Q.(22) The value of $\text{sgn}(4(I - J) - 1) + 8 + \text{sgn}(2 - 4(I - J))$ is equal to

Passage-II:

Let

$$\int \frac{x^2}{(x^2-1)\sin 2x + 2x \cos 2x} dx = f(x) + C; f\left(\frac{\pi}{2}\right) = \frac{1}{2} \ln \frac{\pi}{2}.$$

Now, answer the following questions:

Here C denotes the constant of integration.

Q.(23) The integer nearest to value of $e^{-2f(\pi)}$ is equal to

Q.(24) The integer nearest to value of $e^{2f(\frac{\pi}{4})}$ is equal to

MATRIX MATCHING

Q.(25) Let

$$a = \int_0^1 \frac{dx}{1 + (1-x)^{\frac{\pi}{2}}}, \quad b = \int_0^1 \frac{dx}{1 + x^4}$$

$$c = \int_0^1 \frac{dx}{\sqrt{4-x^2+x^5}}, \quad d = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^{100}}}$$

Match the truth values of entries in Column-I with those in Column-II.

Column-I

Column-II

(A) $a > \ln 2$

(P) $c < \frac{1}{2}$

(B) $a > \frac{\pi}{4}$

(Q) $c < \frac{\pi}{6}$

(C) $b < \frac{3}{4}$

(R) $d > 0.5$

(D) $b < \frac{9}{10}$

(S) $d > 0.524$

KEY

1. B	2. A	3. B	4. A	5. B
6. ABC	7. ABCD	8. BC	9. ABCD	10. BC
11. AC	12. BD	13. ABD	14. 0	15. 3.14
16. 10	17. 165	18. 18.75	19. 0.5	20. 1
21. 0	22. 10	23. 3	24. 8	25. A-QR, B-PS, C-PS, D-QR

GR-07 : AREA AND DIFFERENTIAL EQUATIONS**SINGLE CORRECT TYPE QUESTIONS****Q.(1)**

Let $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function

such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of $\int_{1/2}^1 f(x)dx$ lies in the interval

- (A) $(2e - 1, 2e)$ (B) $(e - 1, 2e - 1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

Q.(2)

Let the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = e^{x-1} - e^{-|x-1|} \text{ and } g(x) = \frac{1}{2}(e^{x-1} + e^{1-x}).$$

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is

- (A) $(2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$ (B) $(2 + \sqrt{3}) + \frac{1}{2}(e - e^{-1})$
(C) $(2 - \sqrt{3}) + \frac{1}{2}(e + e^{-1})$ (D) $(2 + \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

Q.(3)

The area of the region $\left\{ (x, y) : 0 \leq x \leq \frac{9}{4}, 0 \leq y \leq 1, x \geq 3y, x + y \geq 2 \right\}$ is

(A) $\frac{11}{32}$

(B) $\frac{35}{96}$

(C) $\frac{37}{96}$

(D) $\frac{13}{32}$

Q.(4)

A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be

$\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is

(A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(B) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$

(C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$

(D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

Q.(5)

The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ in $(-1, 1)$ satisfying

$f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$ is

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$

(B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

(D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Q.(6) A tank contains 1000 ℓ of water. Brine that contains 0.05 kg/ ℓ of salt enters the tank at the rate of 5 ℓ /min. Brine that contains 0.04 kg/ ℓ of salt enters the tank at the rate of 10 ℓ /min. The solution is kept thoroughly mixed and drains from the tank at the rate of 15 ℓ /min. If the amount of salt left in the tank after 200 minutes, is A, then 3A is equal to

- (A) $130(1 - e^{-1})$ (B) $130(1 - e^{-2})$ (C) $130(1 - e^{-3})$ (D) $130(1 - 3e^{-1})$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(7)

Let Γ denote a curve $y = y(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to Γ at a point P intersect the y-axis at Y_p . If PY_p has length 1 for each point P on Γ , then which of the following options is/are correct ?

(1) $y = \log_e \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) - \sqrt{1 - x^2}$ (2) $xy' - \sqrt{1 - x^2} = 0$

(3) $y = -\log_e \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) + \sqrt{1 - x^2}$ (4) $xy' + \sqrt{1 - x^2} = 0$

Q.(8)

Let b be a nonzero real number. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 1$.

If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$

for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE ?

- (A) If $b > 0$, then f is an increasing function
 (B) If $b < 0$, then f is a decreasing function
 (C) $f(x) f(-x) = 1$ for all $x \in \mathbb{R}$
 (D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

Q.(9)

For any real numbers α and β , let $y_{\alpha, \beta}(x)$, $x \in \mathbb{R}$, be the solution of the differential equation

$$\frac{dy}{dx} + \alpha y = x e^{\beta x}, y(1) = 1$$

Let $S = \{y_{\alpha, \beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S ?

(A) $f(x) = \frac{x^2}{2} e^{-x} + \left(e - \frac{1}{2}\right) e^{-x}$

(B) $f(x) = -\frac{x^2}{2} e^{-x} + \left(e + \frac{1}{2}\right) e^{-x}$

(C) $f(x) = \frac{e^x}{2} \left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right) e^{-x}$

(D) $f(x) = \frac{e^x}{2} \left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right) e^{-x}$

Q.(10)

Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true ?

(A) $y(-4) = 0$

(B) $y(-2) = 0$

(C) $y(x)$ has a critical point in the interval $(-1, 0)$

(D) $y(x)$ has no critical point in the interval $(-1, 0)$

Q.(11)

Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here

$y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true ?

(A) $P = y + x$

(B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$

(D) $P - Q = x + y - y' - (y')^2$

Q.(12)

Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ?

- (A) The curve $y = f(x)$ passes through the point (1, 2)
- (B) The curve $y = f(x)$ passes through the point (2, -1)
- (C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$
- (D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

Q.(13) Let $y = f(x)$ and $y = g(x)$ be functions satisfying

$$x^3 - 3x^2 f(x) + (f(x))^3 = 0, \quad g'(x) = \frac{f(x)}{x}, \quad g(0) = 1.$$

Then, $g(x)$ is equal to which of the following option(s)

- (A) $1 + f(x)$ (B) $x f'(x) + 1$ (C) $f'(x) + 1$ (D) $x f(x) + 1$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(14)

A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n (n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR , then the value of n is _____

Q.(15)

Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$ for all $x \in \mathbb{R}$ and $f : \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is

Q.(16)

If $y(x)$ is the solution of the differential equation

$$x dy - (y^2 - 4y) dx = 0 \text{ for } x > 0, y(1) = 2,$$

and the slope of the curve $y = y(x)$ is never zero, then the value of $10y(\sqrt{2})$ is _____.

Q.(17)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx} = (2 + 5y)(5y - 2), \text{ then the value of } \lim_{x \rightarrow -\infty} f(x) \text{ is } \underline{\hspace{2cm}}.$$

Q.(18) Let A_n represents the area formed by the elements of set A , defined below.

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in \mathbb{N}\}$$

If $[.]$ denotes the greatest integer function and $\{.\}$ denotes the fractional part function, then find the value of $A_3 + A_6$.

Q.(19) Let $f(x)$ and $g(x)$ be two differentiable functions satisfying the following condition:

$$f(0) = 2, g(0) = 1, f(x) + g'(x) = 1 = f'(x) + g(x), x \geq 0.$$

The value of $2f(\ln 2) + 3g(\ln 3)$ is equal to

Q.(20) The functions $f(x)$ and $g(x)$ are defined as

$$f(x) = \begin{cases} -2, & -3 \leq x \leq 0 \\ x - 2, & 0 < x \leq 3 \end{cases}, \text{ and}$$

$$g(x) = \min \left(f(|x|) + |f(x)|, f(|x|) - |f(x)| \right).$$

The area bounded by $y = g(x)$, $y = 0$, $x = 3$, $x = -3$, is equal to

PASSAGES

Passage-I:

Let $f : [0, 1] \rightarrow \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$.

Q.(21)

If the function $e^{-x}f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = \frac{1}{4}$, which of the following is true ?

- (A) $f'(x) < f(x)$, $\frac{1}{4} < x < \frac{3}{4}$ (B) $f'(x) > f(x)$, $0 < x < \frac{1}{4}$
- (C) $f'(x) < f(x)$, $0 < x < \frac{1}{4}$ (D) $f'(x) < f(x)$, $\frac{3}{4} < x < 1$

Q.(22)

Which of the following is true for $0 < x < 1$?

- (A) $0 < f(x) < \infty$ (B) $-\frac{1}{2} < f(x) < \frac{1}{2}$ (C) $-\frac{1}{4} < f(x) < 1$ (D) $-\infty < f(x) < 0$

Passage-II:

Consider one side AB of a square ABCD, (read in order) on the line $y = 2x - 17$, and the other two vertices C, D on the parabola $y = x^2$.

Q.(23)

Find the minimum intercept of the line CD on y-axis.

Q.(24)

Find the maximum possible area of the square ABCD.

Q.(25)

 Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$.

KEY

1. D	2. A	3. A	4. A	5. B
6. C	7. 14	8. AC	9. AC	10. AC
11. BC	12. BC	13. AB	14. 4	15. 3
16. 8	17. 0.4	18. 24	19. 3.5	20. 11.5
21. C	22. D	23. 3	24. 1280	25. 10.66-10.67

GR-08 : STRAIGHT LINES**SINGLE CORRECT****Q.(1)**

A variable line passes through $P(2,3)$ and cuts the co-ordinates axes at A and B. If the parallelogram OACB (where O is the origin) is completed then find number of ordered pairs (x,y) of integers which lie on the locus of point C.

(A) 11

(B) 7

(C) 9

(D) 5

Q.(2)

Triangle ABC lies in the Cartesian plane and has an area of 70 sq. units. The coordinates of B and C are $(12, 19)$ and $(23, 20)$ respectively and the coordinates of A are (p, q) . The line containing the median to the side BC has slope -5 . Find the largest possible value of $(p + q)$.

(A) 44

(B) 45

(C) 46

(D) 47

MULTIPLE CORRECT

Q.(3) Let $x + y = 2$ and $2x + y = 5$ are internal angle bisectors of angles B and C of $\triangle ABC$ respectively. If $A(3,4)$, then which of the following options is/are correct?

(A) Image of A about internal bisector of angle B is $(-2, -1)$.(B) If r denotes inradius of $\triangle ABC$, then $2r = 3\sqrt{10}$.(C) Equation of BC is $y = 3x + 5$ (D) Image of A about internal bisector of angle C is $(-1, -2)$.**Q.(4)** Let

$$L \equiv ax + by + c = 0,$$

$$M \equiv bx + cy + a = 0,$$

$$N \equiv cx + ay + b = 0.$$

If $\lambda = a + b + c, \mu = a^2 + b^2 + c^2, \nu = ab + bc + ca; a, b, c \in \mathbb{R}$, the which of the following options is/are correct?

- (A) If $\lambda = 0, \mu \neq \nu$, then the lines L, M, N always pass through a fixed point.
- (B) If $\lambda = 0, \mu = \nu$, then the each of L, M, N are true for all points in xy -plane.
- (C) If $\lambda \neq 0, \mu \neq \nu$, then then lines L, M, N form a triangle.
- (D) If $\lambda \neq 0, \mu = \nu$, then graph of L, M, N are same.

Q.(5) Let M and N be line mirrors. A ray of light along L_1 strikes M and reflects along line L_2 . After second reflection from N , the ray moves along line L_3 and after third reflection from M , it moves along line L_4 . If

$$L_1 \equiv x - 3y = 1, M \equiv y = 0 \text{ and } N \equiv x = 4$$

then which of the following options is/are correct?

- (A) Equation of line L_4 is $x + 3y - 5 = 0$
- (B) Area of the figure so formed is 12 square units.
- (C) Equation of line L_3 is $x - 3y = 7$
- (D) Equation of line L_2 is $x + 3y = 1$.

Q.(6) Let $L_i; i = 1, 2, 3 \dots 7$ be lines such that

$$\begin{aligned} L_1 &\equiv x + y = 1, & L_2 &\equiv x + y = 3, \\ L_3 &\equiv 2x - y = 5, & L_4 &\equiv 3x + y = 5. \end{aligned}$$

Line L_5 passes through origin, and meets L_1 and L_2 at P and Q respectively. Through P and Q , lines L_6 and L_7 are drawn parallel to L_3 and L_4 respectively intersecting at point R . Which of the following option(s) can be point R .

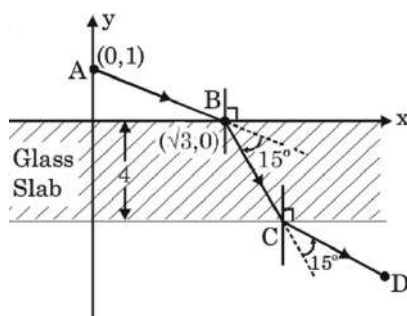
- (A) (4,3) (B) (10,4) (C) (19,7) (D) (25,10)

Q.(7)

Consider two isosceles triangles ΔAB_1C_1 and ΔAB_2C_2 such that $AB_1 = AC_1$, $AB_2 = AC_2$ and $P(1, -2)$ is point of intersection of lines B_1C_1 & B_2C_2 . If A, B_1, C_1 lies on $7x - y + 3 = 0$ and A, B_2, C_1 lies on $x + y - 3 = 0$, then which of the following is/are correct ?

- (A) point A lies on y-axis.
(B) product of x-coordinates of points B_1, C_1, B_2, C_2 is negative
(C) product of y-coordinates of points B_1, C_1, B_2, C_2 is positive
(D) sum of slope of lines B_1C_1 and B_2C_2 is $\frac{8}{3}$.

Q.(8) A ray of light from A to glass slab of thickness 4 units is incident at point B(as shown in figure). After refraction, it turns by 15° towards the normal and moves along BC. After second refraction it turns by 15° away from the normal, and moves along CD.



If $CD = AB$, then which of the following options is/are correct?

- (A) Equation of BC is $x + y = \sqrt{3}$ (B) Coordinates of C are $(2 + \sqrt{3}, -4)$
(C) Equation of DC is $\sqrt{3}y + x = 2 - 3\sqrt{3}$ (D) Coordinates of D are $(2\sqrt{3} + 4, -5)$

Q.(9)

Point A lies on x-axis and point B lies on $y = x$. If PA.PB is minimum where P divides segment AB internally, then (given P is $(1, 2 - \sqrt{3})$ and 'O' is origin), -

- (A) $OA = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$ (B) $OB = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$
(C) $\angle OAB = 45^\circ$ (D) $\angle OBA = 22.5^\circ$

NUMERICAL TYPE

Q.(10) If $|x| = \max\{a \in \mathbb{Z} : a \leq x\}$, then find the minimum value of

$$\left| \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 4x + 29} \right|.$$

Q.(11) Find the value of $4x$, if λ is minimum, where

$$\lambda = AB + BC + CA; A(5,3), B(\alpha, \alpha), C(x, 0).$$

Q.(12)

ΔPOR has vertices $P(0,12)$, $R(5,0)$ and $O(0,0)$. There exist a line ' ℓ ' cutting PR and OP at A and B respectively, such that circles can be inscribed in ΔPAB and quadrilateral $ORAB$. Also these circles are tangent to the line ' ℓ ' at the same point. If the area of quadrilateral $ORAB$ is A , then find the value of $\left\lceil \frac{A}{5} \right\rceil$?

Q.(13) Find the area of pentagon formed by points lying in $\mathbb{A} \cup \mathbb{B}$, where

$$\mathbb{A} = \left\{ (x, y) \in \mathbb{Z}^2 \mid \det \begin{pmatrix} x^2 & 4-y \\ y-4 & 1 \end{pmatrix} \begin{pmatrix} x-3 & y \\ y & 3-x \end{pmatrix} \begin{pmatrix} x^2+y^2+37 & 2 \\ 6x+y & 1 \end{pmatrix} = 0 \right\},$$

$$\mathbb{B} = \left\{ (x, y) \in \mathbb{Z}^2 \mid \det \begin{pmatrix} |x-4| & -1 \\ |y-9| & 1 \end{pmatrix} \begin{pmatrix} x-6 & y-1 \\ y-1 & 6-x \end{pmatrix} \begin{pmatrix} y-5 & 7-x \\ 7-x & 5-y \end{pmatrix} = 0 \right\}.$$

Here $\det(A)$ denotes the determinant of matrix A .

Q.(14)

If value of the expression

$$\left(\sqrt{(x-2)^2 + (y-4)^2} - \sqrt{(x+2)^2 + (y-2)^2} \right)^4 + \left(\sqrt{(x-2)^2 + (y-4)^2} + \sqrt{(x+2)^2 + (y-2)^2} \right)^4$$

is minimum for $x = x_0$ and $y = y_0$, then $|x_0 + y_0|$ is equal to

Q.(15)

Let PQRS be a square such that point P lies on positive y-axis and Q lies on positive x-axis. If S is (7, 10) and the co-ordinate of point R is (a, b), then value of b is

PASSAGES

Passage-I

Consider a triangle ABC.

Let $x + y - 3 = 0$, $x - 4y + 7 = 0$ and $2x - y = 0$ respectively the equations of the altitudes BE, CF and AD on the sides AC, AB and BC.

Q.(16)

If one of the vertex of triangle is of the form $(\alpha, 2\alpha)$, then the locus of centroid of triangle ABC as α varies is -

(A) $9x - 63y + 129 = 0$

(B) $9x + 63y - 129 = 0$

(C) $9x + 45y - 89 = 0$

(D) $9x + 45y - 99 = 0$

Q.(17)

Let Q be any point on the line $L : x - k = 0$, where k is abscissa of orthocenter of $\triangle ABC$, P is a point (k, 0) other than point Q, QR is the acute angle bisector of angle OQP, (O is origin) meets the x-axis at point R, then the locus of foot of perpendicular from R on the OQ is -

(A) $(x^2 + y^2)(x + 2) + x = 0$

(B) $(x^2 + y^2)(x - 1) + x = 0$

(C) $(x^2 + y^2)(x - 2) + x = 0$

(D) $(x^2 + y^2)(x + 1) + x = 0$

Passage-II

Given four parallel lines L_1, L_2, L_3 and L_4 . Let the distances between them be d_{12}, d_{23}, d_{34} respectively (here d_{ij} is distance between lines L_i and L_j). Let P be a point, sum of whose distances from four lines is k ($d_{12} < d_{23} < d_{34}$).

Q.(18)

 If $k = d_{12} + 2d_{23} + d_{34}$, then locus of 'P' is -

- (A) entire region between lines L_2 and L_3 . (B) entire region between lines L_1 and L_2 .
 (C) entire region between lines L_3 and L_4 . (D) not possible.

Q.(19)

 If $k = d_{12} + 2d_{23} + d_{34} + 2\alpha$, $0 < \alpha < d_{34}$, then locus of 'P' is -

- (A) entire region between lines L_2 and L_3 . (B) entire region between lines L_1 and L_4 .
 (C) entire region between lines L_3 and L_4 . (D) not possible.

MATRIX MATCHING TYPE

Q.(20)
Column-I

- A. In the triangle ABC, equation of the perpendicular bisector of the sides AB & AC are $x + y = 0$ & $y - x = 0$ respectively. If $A \equiv (5, 7)$ then equation of side BC is
- B. The straight line of the family $(x + y) + \lambda(2x - y + 1) = 0$ that is farthest from $(1, -3)$ is
- C. In triangle ABC, $A(1, 1)$, $B(4, -2)$, $C(5, 5)$. Equation of the internal angle bisector of angle A is
- D. The diagonals AC and BD of a rhombus intersect at $(5, 6)$ If $A = (3, 2)$ then equation of diagonal BD is

Column-II

- (P) $7y - 5x = 0$
- (Q) $y = 1$
- (R) $15y - 6x = 7$
- (S) $2y + x = 17$

KEY

1. B	2. D	3. ABC	4. ABCD	5. CD
6. AD	7. ABC	8. AD	9. AB	10. 6
11. 17	12. 5	13. 36.5	14. 3	15. 3

16. Ç	17. C	18. A	19. C	20. A-P,B-R,C-Q,D-S
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GR-09 : CIRCLES**SINGLE CORRECT TYPE QUESTIONS****Q.(1)**

Circles are drawn by taking sides of a triangle ABC as diameters. If equations of the sides AB, BC, CA are $x - y = 0$, $3x - 4y - 17 = 0$ and $x + y - 2 = 0$, respectively, then co-ordinates of the radical centre is-

- (A) $(-1, 2)$ (B) $(3, -1)$ (C) $(1, 1)$ (D) $(1, 5)$

Q.(2)

Let $A \equiv (0, 4)$ and $B \equiv (3, 8)$ be two points such that $\angle AXB$ is maximum where X is a point on x-axis. If x-co-ordinate of X is $a\sqrt{2} - b$, then $(a^4 + b^4)$ is

- (A) 106 (B) -602 (C) 17 (D) 706

Q.(3)

Consider the circles $x^2 + y^2 + 3\sqrt{2}(x + y) = 0$ and $x^2 + y^2 + 5\sqrt{2}(x + y) = 0$. A third circle is drawn internally tangent to larger circle and externally tangent to smaller circle and tangent to their common

diameter. If radius of such circle is $\frac{p}{q}$ (p & q are relatively prime), then value of $p - q$ is

- (A) 7 (B) 8 (C) 10 (D) -7

MULTIPLE CORRECT TYPE QUESTIONS

Q.(4)

An insect can go from point $A(-3, 2)$ to $B(3\sqrt{3}, 2)$ by two paths such that centre of circle $x^2 + y^2 - 10y + 16 = 0$ lies in area enclosed by the paths but insect cant go inside the circle. If both paths are of smallest possible length, then which of the following option(s) is/are true -

- (A) Area enclosed by both paths is $\frac{15\pi}{4} + 9 + 9\sqrt{3}$ sq. units.
- (B) Area enclosed by both paths is $\frac{15\pi}{2} + 9 + 3\sqrt{3}$ sq. units.
- (C) Absolute difference between the lengths of two paths is $\frac{5\pi}{2}$ units.
- (D) Absolute difference between the lengths of two paths is $\frac{3\pi}{2}$ units.

Q.(5)

Given a circle of radius 3 and a point P outside the circle such that $\angle APB = \frac{2\pi}{3}$, where A and B are points of tangency from P to the circle. Line joining P and center O of the circle cut the circle at C and D (where C is near to point P), which of the following statement(s) is/are correct -

- (A) $|\text{Area of triangle ABC} - \text{Area of triangle ABD}|$ is $\frac{9\sqrt{3}}{2}$
- (B) $|\text{Area of triangle ABC} - \text{Area of triangle ABD}|$ is $\frac{3\sqrt{3}}{2}$
- (C) Circum radius of triangle PAB is $\sqrt{3}$
- (D) Circum radius of triangle PAB is $2\sqrt{3}$

Q.(6)

If locus of middle points of all chords of $S_1 \equiv x^2 + y^2 - 4x - 4y - 36 = 0$ which subtends 90° at point 'P'(6,8) is a circle 'S' then

- (A) Equation of circle 'S' is $x^2 + y^2 - 8x - 10y + 32 = 0$
- (B) Equation of circle 'S' is $x^2 + y^2 + 8x + 10y - 32 = 0$
- (C) Equation of circle of family $S_1 + \lambda S = 0$ and passes through (0, 0) is $17(x^2 + y^2) = 104x + 112y$
- (D) Equation of circle with diametric ends P(6, 8) & centre of 'S' is $x^2 + y^2 - 8x - 10y + 28 = 0$

Q.(7) Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two fixed points, and $R(\alpha, \beta)$ be a point satisfying

$$\frac{PR}{QR} = k; k > 0, k \neq 1.$$

Let locus of R for different values of k be curves $S_1, S_2, S_3, \dots$ and $S'_1, S'_2, S'_3, \dots$ be circles such that S_i is orthogonal to $S'_j \forall i, j$.

Then, which of the following options is/are correct?

- (A) It is possible that S_i and $S_j (i \neq j)$ have four common tangents.
 (B) It is possible that S'_i and $S'_j (i \neq j)$ have two common tangents
 (C) If S'_k has minimum possible radius equal to that of S_i , then distance between their centers is $\frac{PQ}{\sqrt{2}}$.
 (D) It is possible that S_i and $S_j (i \neq j)$ have two common tangents.

Q.(8) Circle $x^2 + y^2 - 8x - 6y + 16 = 0$ and line $y = mx + c$ intersect at two distinct points A and B, such that circle with AB as diameter always passes through origin. If $x^2 + y^2 + 2gx + 2fy + k = 0$ represents the locus of center of circle with AB as diameter, then

- (A) $c^2 + (4m - 3)c + 8(m^2 + 1) = 0$ (B) $k + g = 6$
 (C) $4c^2 + (4m - 3)c + 8(m^2 + 1) = 0$ (D) $k - 2f = 11$

Q.(9) Let $x^2 + y^2 + 2gx + 2fy + c = 0$ represent the equation of circle passing through points lying in $\mathbb{A} \cup \mathbb{B}$, given that it exists and $|\mathbb{A}| = 2 = |\mathbb{B}|$, where $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 4, x + 2y = a\},$$

$$\mathbb{B} = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 4x - 2y + 1 = 0, 12x - 6y = 41\}.$$

Then which of the following options is/are correct?

- (A) $a + 2 = 0$ (B) $5g - 5f = 4$ (C) $c = -36$ (D) $a - 5c = 34$

Q.(10) Circle C_1 touches X-axis at a distance of 3 units from origin, and has an intercept of length $2\sqrt{7}$ on Y-axis. Circle C_2 passes through point (0,1) and is orthogonal to

$$C_3 \equiv (x - 1)^2 + y^2 = 16 \text{ and } C_4 \equiv x^2 + y^2 = 1.$$

Then which of the following options is/are correct?

- (A) Possible equation of C_1 is $x^2 + y^2 - 6x + 8y + 9 = 0$
 (B) Possible equation of C_1 is $x^2 + y^2 - 6x - 8y + 9 = 0$
 (C) Radius of C_2 is 7
 (D) Center of C_2 is $(-8, 1)$

Q.(11) Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$ and 'n' denote the number of possible values of p, such that $|\mathbb{A} \cap \mathbb{B}| = 3$, where

$$\mathbb{A} = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 + 2x + 4y = p\}, \quad \mathbb{B} = \{(x, y) \in \mathbb{R}^2 | xy = 0\}.$$

Let $R(-1,0), S(1,0)$ be ends of diameter of circle $C : x^2 + y^2 = 1$, and P be a variable point (other than R and S) be points on circle C , such that tangents to circle C at S and P meet at point Q . The normal to circle C at P intersects a line through Q parallel to RS at point E . Then the locus of E passes through

- (A) $\left(\frac{1}{2n-1}, \frac{1}{\sqrt{n+1}}\right)$ (B) $\left(\frac{1}{2n}, \frac{1}{n}\right)$ (C) $\left(\frac{1}{2n-1}, \frac{-1}{\sqrt{1+n}}\right)$ (D) $\left(\frac{1}{2n}, \frac{-1}{n}\right)$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(12)

Let area bounded by the locus of P is A sq. units such that $\angle APB = 30^\circ$ and A, B are $(0,0)$ and $(2,0)$ respectively, then greatest integer less than $\left(\frac{A}{4}\right)$ is

Q.(13)

A ray of light incident at the point $(-3,-2)$ gets reflected from the tangent at $(0,-2)$ to the circle $x^2 + y^2 = 4$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is $Jx - Ky + 46 = 0$.

The value of $J + K$ is

Q.(14)

The number of integral values of λ for which the variable line $3x + 4y - \lambda = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ without having any common points with any circle is equal to $2k$, then value of k is

Q.(15)

Tangents drawn from $P(1,8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touches the circle at the points A and B respectively. If the radius of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circumcircle of the $\triangle PAB$ orthogonally is equal to $\frac{\sqrt{p}}{a}$ where $p, q \in \mathbb{N}$, then find the minimum value of $(p + q)$.

PASSAGES

Passage-I

Let points $A_i; i = 1, 2, 3, 4, 5, 6$ lying on circle $S \equiv x^2 + y^2 = 4$, satisfy the following conditions.

Chords A_1A_2 and A_3A_4 of S pass through $P_0(1,1)$ and are parallel to X-axis and Y-axis respectively.

A_5A_6 is chord of S passing through P_0 and having slope -1 .

Tangents to S at A_1, A_2 intersect at A_7 , the tangents of S at A_3, A_4 meet at A_8 and the tangents of S at A_5, A_6 meet at A_9

P is a point on circle S in first quadrant and tangent of S at P intersects coordinate axes at M and N .

Now, based on above information, answer the following questions

Q.(16) The points A_7, A_8, A_9 lie on curve

(A) $(x - 4)^2 + (y - 4)^2 = 16$

(B) $x + y = 4$

(C) $(x - 4)(y - 4) = 4$

(D) $xy = 4$

Q.(17) Mid-point of MN lies on the curve

(A) $(x + y)^2 = 3xy$

(B) $x^{2/3} + y^{2/3} = 2^{4/3}$

(C) $x^2 + y^2 = 2xy$

(D) $x^2 + y^2 = x^2y^2$

Passage-II

Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x-3)^2 + (y-4)^2 = 16$, intersect at the points X and Y. Suppose that another circle $C_3 : (x-h)^2 + (y-k)^2 = r^2$ satisfies the following conditions :

- (i) centre of C_3 is collinear with the centres of C_1 and C_2
- (ii) C_1 and C_2 both lie inside C_3 , and
- (iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

There are some expression given in the List-I whose values are given in List-II below :

List-I	List-II
(I) $2h + k$	(P) 6
(II) $\frac{\text{Length of ZW}}{\text{Length of XY}}$	(Q) $\sqrt{6}$
(III) $\frac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}}$	(R) $\frac{5}{4}$
(IV) α	(S) $\frac{21}{5}$
	(T) $2\sqrt{6}$
	(U) $\frac{10}{3}$

Q.(18)

Which of the following is the only INCORRECT combination ?

- (1) (IV), (S) (2) (IV), (U) (3) (III), (R) (4) (I), (P)

Q.(19)

Which of the following is the only CORRECT combination ?

- (1) (II), (T) (2) (I), (S) (3) (I), (U) (4) (II), (Q)

MATRIX MATCHING
Q.(20) Match the following columns

Column-I

Column-II

(A) Let $2x^2 + y^2 = 3xy$ be the equation of pair of tangents drawn

(P) 2

from origin to circle of radius 3 with center in first quadrant. If O is origin

- and A is one of the points of contact, then $OA - 3\sqrt{10}$ is equal to
- (B) Let $2x + 3y = 1$ is tangent to a circle at $(1, -1)$. This circle is (Q) 3
 orthogonal to a circle which is drawn having diameter as a line segment
 with end points $(0, -1)$ and $(-2, 3)$. If $x^2 + y^2 + 2gx + 2fy + c = 0$ is the
 equation of circle, then $4f - 4g$ is equal to
- (C) If S and T are two concentric circles, such that from any point P on S (R) 5
 pair of tangents drawn to T, touch it at A, B and centroid of $\triangle ABP$ lies
 on T. Then ratio of ratio of radii of S to T is
- (D) If A, B lie on $x^2 + y^2 = 5$ such that tangents at A and B intersect (S) 9
 at $P(5, -5)$, and (α, β) is ortho-center of $\triangle ABP$, then $\alpha + \beta + 3$ is equal to

KEY

1. C	2. D	3. A	4. AC	5. AC
6. AD	7. ABC	8. ABD	9. ABD	10. ABC
11. AC	12. 6	13. 7	14. 4	15. 77
16. B	17. D	18. A	19. 4	20. A-S, B-R, C-P, D-Q

GR10-PARABOLA

SINGLE CORRECT TYPE QUESTIONS

Q.(1)

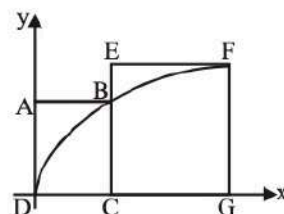
The axis of parabola is along the line $y = x$ and the distance of vertex from origin is $\sqrt{2}$ and that of origin from its focus is $2\sqrt{2}$. If vertex and focus both lie in the 1st quadrant, then the equation of the parabola is

- (A) $(x + y)^2 = (x - y - 2)$ (B) $(x - y)^2 = (x + y - 2)$
 (C) $(x - y)^2 = 4(x + y - 2)$ (D) $(x - y)^2 = 8(x + y - 2)$

Q.(2)

ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is -

- (A) $\frac{\sqrt{5} + 1}{2}$ (B) $\frac{\sqrt{3} + 1}{2}$
 (C) $\frac{\sqrt{5} + 1}{4}$ (D) $\frac{\sqrt{3} + 1}{4}$



MULTIPLE CORRECT TYPE QUESTIONS

Q.(3) Which of the following options is/are correct?

- (A) $105xy = 64$ is the locus of vertex of family of parabolas $6y = 2a^3x^2 + 3a^2x - 12a$.
 (B) $\sqrt{3}y = x + 3$ is the common-tangent to $(x - 3)^2 + y^2 = 9$ and $y^2 = 4x$.
 (C) $64xy = 105$ is the locus of vertex of family of parabolas $6y = 2a^3x^2 + 3a^2x - 12a$.
 (D) $y = \sqrt{3}x + 3$ is the common-tangent to $(x - 3)^2 + y^2 = 9$ and $y^2 = 4x$.

Q.(4) Which of the following options is/are correct?

- (A) $x = a$ is the directrix of locus of mid-point of segment joining focus of $y^2 = 4ax$ to any point on it.
 (B) Acute angle between tangents from $(1,4)$ to $y^2 = 4x$ is 45° .
 (C) $x = 0$ is the directrix of locus of mid-point of segment joining focus of $y^2 = 4ax$ to any point on it.
 (D) Acute angle between tangents from $(1,4)$ to $y^2 = 4x$ is 60° .

Q.(5) P, Q are points on parabola $y^2 = 8x$ with focus S, such that tangents at P, Q intersect at T and normals at P, Q intersect at N on same parabola. Then, which of the following options is/are correct?

- (A) Vertex of locus of mid-points of chord PQ is $(-2,0)$.
 (B) If chord PQ always passes through a fixed-point F, then $SF = 6$.
 (C) Focus of locus of mid-points of chord PQ is $(-3,0)$.
 (D) If chord PQ always passes through a fixed-point F, then $SF = 4$.

Q.(6) Let $A_i(t_i^2, 2t_i); t_i \neq 0; i = 1, 2, 3, 4$; be four points on parabola $y^2 = 4x$. Let (x_1, y_1) is the mid-point of focal chord A_1A_2 , and lines A_2A_3, A_1E are parallel, where $E(2,0)$. If tangent at A_4 and normal at A_1 intersect at (x_2, y_2) , and $t_1t_4 = 1$, then

- (A) $t_3 = y_1$ (B) $t_3 = 2y_1$ (C) $x_2 = 1 + x_1 - t_2^2$ (D) $x_2 = 1 - x_1 + t_2^2$

Q.(7) Let points P and Q lie on parabola $y^2 = 4ax; a > 0$, such that chord PQ is a focal chord. The tangents at P and Q intersect on a line $y = 2x + a$. If chord PQ subtends angle θ at the vertex of parabola, then

- (A) $3 \tan \theta = -2\sqrt{5}$ (B) $3 \tan \theta = -2\sqrt{7}$ (C) $PQ = 5a$ (D) $PQ = 7a$

Q.(8) The common tangents to $\mathbb{C}_1: x^2 + y^2 = 2$ and $\mathbb{C}_2: y^2 = 8x$ touch the circle $\mathbb{C}_1$ at P, Q and parabola $\mathbb{C}_2$ at R, S. Also, normal of parabola $\mathbb{C}_3: y^2 = 4x$ drawn at ends of its latus-rectum are tangents to a circle $\mathbb{C}_4$ with center $(3, -2)$ and radius r. Then area of quadrilateral PQRS is equal to

- (A) $4r^2 + 7$ (B) $5r^2 + 5$ (C) $3r^2 + 7$ (D) $7r^2 - 2$

Q.(9)

The circle $C_1: x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3 , respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centres Q_2 and Q_3 , respectively. If Q_2 and Q_3 lie on the y-axis, then -

(A) $Q_2Q_3 = 12$

(B) $R_2R_3 = 4\sqrt{6}$

(C) area of the triangle OR_2R_3 is $6\sqrt{2}$

(D) area of the triangle PQ_2Q_3 is $4\sqrt{2}$

Q.(10)

The focus of the parabola is (1,1) and the tangent at the vertex has the equation $x + y = 1$. Then :

(A) equation of the parabola is $(x - y)^2 = 2(x + y - 1)$

(B) equation of the parabola is $(x - y)^2 = 4(x + y - 1)$

(C) the co-ordinates of the vertex are $\left(\frac{1}{2}, \frac{1}{2}\right)$

(D) length of the latus rectum is $2\sqrt{2}$

Q.(11)

Let E denote the parabola $y^2 = 8x$. Let $P = (-2, 4)$, and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E. Let F be the focus of E. Then which of the following statements is (are) TRUE?

(A) The triangle PFQ is a right-angled triangle

(B) The triangle QPQ' is a right-angled triangle

 (C) The distance between P and F is $5\sqrt{2}$

(D) F lies on the line joining Q and Q'

NUMERICAL ANSWER TYPE QUESTIONS

Q.(12)

$P(x, y)$ is a variable point moves in xy plane such that $x = -2t^2$, $y = t^2 - 2t + 2$, where $t \in \mathbb{R}$, then locus of P is a conic section whose latus rectum is L . Then $\frac{5\sqrt{5}}{4}L$ is equal to

Q.(13)

If (h, k) be a point on the axis of parabola $3y^2 + 4y - 6x + 8 = 0$ from which three distinct real normals could be drawn to the parabola, then $[h]_{\text{least}}$ is equal to (where $[.]$ denotes greatest integer function)

Q.(14)

If the radius of the circle touching the parabola $y^2 = 4x$ at an end of latus rectum and passing through the focus is $\sqrt{N}$, then the value of N is

Q.(15)

Square of the area of the triangle formed by end points of a focal chord PQ of length 32 units of the parabola $y^2 = 8x$ and its vertex, is

PASSAGES

Passage-I:

Let $a, r, s, t \in \mathbb{R} - \{0\}$, and

$$P(at^2, 2at), Q, R(ar^2, 2ar), S(as^2, 2as)$$

are distinct points on parabola $y^2 = 4ax$. Suppose that PQ is focal chord, and lines QR, PK are parallel where $K(2a, 0)$. Let tangent at P and normal at S to parabola meet at $T(\alpha, \beta)$. If curve C is the mirror image of parabola w.r.t. line L_1 , and A, B are the points of intersection of C with line L_2 , where

$$L_1: x + y + 4 = 0 \quad \text{and} \quad L_2: y + 5 = 0.$$

Now, based on above information, answer the following questions

Q.(16) If $t = 4$, then $(r^2) \cdot AB$ is equal to

Q.(17) If $a = 5t = 125s = 25$, then $\sqrt{2.5\beta}$ is equal to

Passage-II:

Let us consider the following parabolas:

$$\Pi_1: y^2 = 2x \quad \text{and} \quad \Pi_2: y^2 = 4x.$$

Let A and B are distinct points on parabola Π_1 , such that circle with AB as diameter passes through its vertex O, such that A lies in first quadrant and area of ΔOAB is $3\sqrt{2}$ square units.

Let S be focus of parabola Π_2 , and pair of tangents drawn from $P(-2,1)$ to Π_2 , touch it at P_1 and P_2 . Let Q_1, Q_2 be points on lines SP_1, SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2 .

Now, based on above information, answer the following questions

Q.(18) Which of the following is/are coordinates of A?

- (A) $(4, 2\sqrt{2})$ (B) $(9, 3\sqrt{2})$ (C) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (D) $(1, \sqrt{2})$

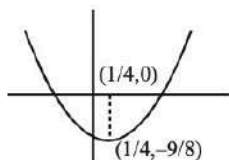
Q.(19) Which of the following options is/are correct?

- (A) $SQ_1 = 2$ (B) $5Q_1Q_2 = 3\sqrt{10}$ (C) $PQ_1 = 3$ (D) $SQ_2 = 1$

MATRIX MATCHING

Q.(20)

Suppose a parabola has vertex $\left(\frac{1}{4}, -\frac{9}{8}\right)$ and equation $y = ax^2 + bx + c$ (where $a > 0$ and $a+b+c$ is integer). Let 'a' takes its minimum possible value under given condition, then



Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- | | |
|---|--------|
| (P) 2(latus rectum of parabola) is | (1) 9 |
| (Q) If equation of directrix is $4y + \lambda = 0$, then λ^2 is | (2) 81 |
| (R) If the value of a is $\frac{p}{q}$ (where p & q are relatively prime) then p + q is | (3) 11 |
| (S) If $(\alpha, 0)$ lies on parabola, then $16\left(\alpha - \frac{1}{4}\right)^2$ is | (4) 36 |

KEY

1. D	2. A	3. BC	4. CD	5. BC
6. AC	7. AC	8. AB	9. ABC	10. BCD
11. ABD	12. 4	13. 2	14. 2	15. 256
16. 15	17. 13	18. AD	19. BCD	20. P-1, Q-2, R-3, S-2

GR-11 : ELLIPSE AND HYPERBOLA

SINGLE CORRECT TYPE QUESTIONS

Q.(1) If a circle of radius 5 is concentric with ellipse $3x^2 + 7y^2 = 105$, then m is slope of the common tangent to circle and ellipse. If normal at $P(\theta)$ to ellipse $5x^2 + 14y^2 = 70$ intersects it again at $Q(2\theta)$, then $m^2 - 3 \cos \theta$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(2) Ellipse $\mathbb{E}: 4x^2 + 9y^2 = 36$ is inscribed in a rectangle $\mathbb{R}$, whose sides are parallel to coordinate axes. Ellipse $\mathbb{E}'$ passing through $(0,4)$ circumscribes the rectangle $\mathbb{R}$. If e denotes the eccentricity of ellipse $\mathbb{E}'$, then $4e$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(3)

If circle whose diameter is major axis of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) meets minor axis at point P & orthocentre of ΔPF_1F_2 lies on ellipse where F_1 & F_2 are focii of ellipse, then square of eccentricity of ellipse is-

- (A) $2 \sin 18^\circ$ (B) $2 \sin 15^\circ$ (C) $\sin 45^\circ$ (D) $\sin 60^\circ$

Q.(4) If (α, β) is the point on ellipse $x^2 + 2y^2 = 6$, which is closest to the line $x + y = 7$, then $2\alpha + 3\beta$ is equal to

- (A) 7 (B) 6 (C) 3 (D) 4

Q.(5)

If $x = 9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangents, is -

- (A) $9x^2 - 8y^2 + 18x - 9 = 0$ (B) $9x^2 - 8y^2 - 18x + 9 = 0$
 (C) $9x^2 - 8y^2 - 18x - 9 = 0$ (C) $9x^2 - 8y^2 + 18x + 9 = 0$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(6) Let $A_1(x_1, y_1)$ and $A_2(x_2, y_2)$ be foci of ellipse

$$E \equiv \frac{x^2}{9} + \frac{y^2}{8} = 1; x_1 < x_2.$$

Let P be a parabola having vertex at origin, focus at A_2 which intersects ellipse E at point M in first quadrant, and point N in fourth quadrant. Let tangents to ellipse E at M and N intersect at R , normal to parabola P at M intersects X -axis at Q . Which of the following options is/are correct if (x_3, y_3) are coordinates of ortho-center of ΔA_1MN and

$$\lambda = \frac{\text{Area of } \Delta MQR}{\text{Area of quadrilateral } MA_1NA_2}.$$

- (A) $10x_3 + 8y_3 + 9 = 0$ (B) $8\lambda = 5$
 (C) $9x_3 + 8y_3 + 10 = 0$ (D) $5\lambda = 4$

Q.(7)

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are

- (A) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$ (B) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$
 (C) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$ (D) $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

Q.(8)

C_1 is a circle described on the line joining the positive ends of the major axis and minor axis of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, as diameter and C_2 is the circle described on the line joining the negative ends of major axis and minor axis of the same ellipse, as diameter. If $y = mx + c$ is a common tangent to C_1 and C_2 , then

- (A) $5m - 4 = 10c - 41 = 0$ (B) $4m + 5 = c = 0$
 (C) $5m - 4 = 10c + 41 = 0$ (D) $5m - 4 = c = 0$

Q.(9)

Consider an ellipse E : $\frac{x^2}{16} + \frac{y^2}{4} = 1$

- (A) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $4x^2 + y^2 = 9$ (where O is the origin)
 (B) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $x^2 + 9y^2 = 4$ (where O is the origin)
 (C) Area of quadrilateral formed by foci and end points of minor axes of the given ellipse is $8\sqrt{3}$ sq. units
 (D) If a tangent is drawn at P on E to intersect the x-axis and y-axis at L and M respectively then minimum value of LM is 6 units

Q.(10)

An ellipse and a hyperbola have their principal axes along the coordinate axes and have a common foci separated by a distance $2\sqrt{13}$, the difference of their focal semi axes is equal to 4. If the ratio of their eccentricities is $3/7$. Find the equation of these curves.

- (A) $36x^2 + 49y^2 = 1764$ (B) $4x^2 - 9y^2 = 36$
 (C) $49x^2 + 36y^2 = 1764$ (D) $9x^2 - 4y^2 = 36$

Q.(11) Let A and A' be ends of major axis of ellipse $E \equiv x^2 + 2y^2 = 2$. Let P be any point on ellipse E . Let C be the locus of point of intersection of perpendiculars drawn from A and A' to $A'P$ and AP respectively. Let $(x_1, y_1), (x_2, y_2)$ be foci of C , (x_3, y_3) be a point of intersection of C and $E, ax + y + b = 0$ and ℓ be the equation of directrix and length of latus-rectum of C . Then which of the following options is/are correct?

(A) $x_1^2 + 2x_2^2 + 3x_3^2 = 4$

(B) $\ell = 2$

(C) $3x_1^2 + 2x_2^2 + x_3^2 = 2$

(D) $b^2 = 8$

Q.(12) Let $\mathbb{E}$ be an ellipse with major and minor axis of length 4 and $2\sqrt{3}$ respectively. Let P be any point on $\mathbb{E}$, and S, S' are foci of $\mathbb{E}$. Let C be the center of $\mathbb{E}$, foot of perpendicular from P on major axis is P' , normal at P cuts major axis at N and λ denote the length of portion of tangent at P intercepted between principal axes. Then which of the following options is/are correct?

(A) $PS \cdot PS' \leq 4$

(B) $PS^2 + PS'^2 \geq 8$

(C) $\lambda \geq 2 + \sqrt{3}$

(D) $4CN = CP'$

Q.(13) Let $m_1 < m_2 < m_3$ be the slopes of distinct normal of ellipse $\mathbb{E}$ passing through point A , where

$$\mathbb{E} \equiv \frac{x^2}{16} + \frac{y^2}{12} = 1, A = \left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{2}}{2\sqrt{3}} \right).$$

Then which of the following options is/are correct?

(A) $m_1 + m_2 + m_3 = 2\sqrt{3}$

(B) $3\sqrt{3}m_1m_2m_3 = -8$

(C) $\sqrt{3}(m_1 + m_3 - m_2) = 4\sqrt{3} + 2$

(D) $\sqrt{3}(m_3 - m_2 - m_1) = 4\sqrt{3} - 2$

Q.(14) An ellipse has focus at origin, and passes through $P(4, 3)$ and $Q(-4, -3)$. Length of major axis is $2a$, and that of minor axis is $2b$. (x_1, y_1) is its centre and (x_2, y_2) is the other focus. If eccentricity of ellipse is $\frac{1}{\sqrt{2}}$ and $y_i > 0$, then

(A) $x_1 + y_1 = \sqrt{2}$

(B) $x_2 + y_2 = 3\sqrt{2}$

(C) $b = 5\sqrt{2}$

(D) $a = 20$

Q.(15) Which of the following options is/are correct for ellipse

$$\mathbb{E} \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

(A) Foot of perpendicular from focus of $\mathbb{E}$ on any tangent lies on its auxiliary circle.

(B) Perpendicular from focus of $\mathbb{E}$ on any tangent, and line from center to corresponding point of contact of tangent intersect on corresponding directrix.

(C) If tangent at any point of ellipse $\mathbb{E}$ makes angle α with major axis and β with focal radius, then eccentricity of $\mathbb{E}$ is

$$e = \left| \frac{\cos \beta}{\cos \alpha} \right|.$$

(D) If $\mathbb{E}$ is the ellipse having least possible area, and also containing the circle $(x - 1)^2 + y^2 = 1$, then eccentricity of $\mathbb{E}$ is

$$e = \sqrt{\frac{2}{3}}.$$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(16)

The graphs of $x^2 + y^2 + 6x - 24y + 72 = 0$ & $x^2 - y^2 + 6x + 16y - 46 = 0$ intersect at four points. Compute the sum of the distances of these four points from the point $(-3, 2)$.

Q.(17)

An equilateral triangle is inscribed in ellipse whose equation is $x^2 + 4y^2 = 4$. One vertex of triangle is $(0, 1)$ and one altitude is contained in y-axis & length of each side is $\sqrt{\frac{m}{n}}$ (where m & n are relatively prime), then $m + n$ is-

Q.(18)

Let C be the centre of ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Let P be any point on the ellipse such that angle between tangent to ellipse at P and CP is minimum. If length of CP is $\sqrt{N}$, then $\frac{N}{2}$ is

Q.(19) Let M_{AB} denote the mid-point of line-segment AB. Let P, Q, R be distinct points on ellipse

$$E \equiv \frac{x^2}{16} + \frac{y^2}{9} = 1 \quad \text{and} \quad d = (\text{distance between } M_{PQ} \text{ and } M_{PR}).$$

Then maximum possible value of d is equal to

Q.(20) Consider the hyperbola

$$H \equiv \frac{x^2}{100} - \frac{y^2}{64} = 1,$$

with foci S and S_1 , where S lies on positive X-axis. Let P be a point on hyperbola H, in first quadrant and $\angle SPS_1 = \alpha$; $2\alpha < \pi$. The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight lines PS_1 at P_1 . Let δ be the distance of P from the line SP_1 and $\beta = S_1P$. Then find the value of

$$\lambda = \max \left\{ a \in \mathbb{Z} : a \leq \frac{\beta\delta}{9} \sin \frac{\alpha}{2} \right\}.$$

Q.(21) If Δ is the area of parallelogram formed by common tangents of ellipse E and hyperbola H, and λ is the smallest distance of any point on ellipse E' from its center, then find $(b - a)\Delta$, where

$$E \equiv \frac{x^2}{4} + y^2 = 1, \quad H \equiv \frac{x^2}{5} - \frac{y^2}{3} = 1, \quad E' \equiv 5x^2 + 4y^2 + xy = 2, \quad \lambda = \frac{a}{\sqrt{b - \sqrt{2}}}.$$

PASSAGES

Passage-I:

Let circle $\mathbb{C}$ and hyperbola $\mathbb{H}$ intersect at points A and B. Let S be focus, L be the corresponding

directrix of ellipse $\mathbb{E}$. $\mathbb{P}$ is a parabola with focus at S and line L as directrix, where

$$\mathbb{C}: x^2 + y^2 = 8x, \quad \mathbb{E}: \frac{x^2}{4} + \frac{y^2}{3} = 1, \quad \mathbb{H}: \frac{x^2}{9} - \frac{y^2}{4} = 1.$$

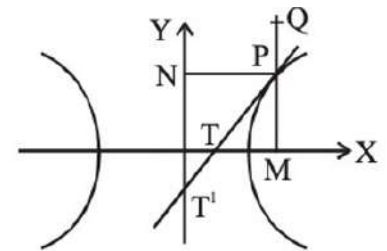
Now, based on above information, answer the following questions

Q.(22) If number of common tangents between $\mathbb{P}$ and $\mathbb{E}$ is n , and $ax + by + 4 = 0$ is equation of common tangents with positive slope to $\mathbb{C}$ and $\mathbb{H}$, then $n + a + b^2$ is equal to

Q.(23) If m denotes the number of normals between $\mathbb{P}$ and $\mathbb{E}$, and $x^2 + y^2 + 2gx + 2fy + c = 0$ is the equation of circle with AB as diameter, then $2g + 2f + 2c + m$ is equal to

Passage-II:

Figure shows a hyperbola whose centre is C . Tangent is drawn at a variable point ' P ' in Ist quadrant meeting transverse and conjugate axes at T and T^1 respectively. M and N are feet of perpendiculars from P on transverse and conjugate axes as shown. Q is a point on ordinate MP produced such that $MQ = SP$ where S is the focus. $CM \cdot CT = 4$ and $CN \cdot CT^1 = 5$.



On the basis of above information, answer the following questions :

Q.(24)

Eccentricity of hyperbola is -

- (A) $\sqrt{2}$ (B) $\frac{3}{2}$ (C) 3 (D) 2

Q.(25)

If locus of Q is straight line, then its

- (A) x-intercept is $\frac{3}{4}$ (B) y-intercept is -2
(C) slope is $\frac{3}{2}$ (D) slope is $-\frac{3}{2}$

MATRIX MATCHING

Q.(26)

Column-I	Column-II
(A) For an ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with vertices A and A', tangent drawn at the point P in the first quadrant meets the y-axis in Q and the chord A'P meets the y-axis in M. If 'O' is the origin then $OQ^2 - MQ^2$ equals to	(P) 2
(B) If the product of the perpendicular distances from any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ of eccentricity $e = \sqrt{3}$ from its asymptotes is equal to 6, then the length of the transverse axis of the hyperbola is	(Q) 3
(C) The locus of the point of intersection of the lines $\sqrt{3}x - y - 4\sqrt{3}t = 0$ and $\sqrt{3}tx + ty - 4\sqrt{3} = 0$ (where t is a parameter) is a hyperbola whose eccentricity is	(R) 4
(D) If F_1 & F_2 are the feet of the perpendiculars from the foci S_1 & S_2 of an ellipse $\frac{x^2}{6} + \frac{y^2}{3} = 1$ on the tangent at any point P on the ellipse, then $(S_1F_1) \cdot (S_2F_2)$ is equal to	(S) 6

Q.(27)

Column-I	Column-II
(A) If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, the locus of P is	(P) Straight line
(B) A variable circle C has the equation $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, where t is a parameter. The locus of the centre of the circle is	(Q) Circle
(C) The locus of point of intersection of tangents to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at two points the sum of whose eccentric angles is constant is	(R) Parabola
(D) An ellipse slides between two perpendicular straight lines. Then the locus of its centre is	(S) Hyperbola

Q.(28) Let

$$E \equiv \frac{x^2}{4} + \frac{y^2}{3} = 1, \quad H(\alpha, 0), \quad 0 < \alpha < 2.$$

A straight line L drawn through H and parallel to Y -axis crosses the ellipse E and its auxiliary circle at points E_E and E_A respectively in the first quadrant. The tangent to ellipse E at E_E intersects the positive X -axis at point G and the line joining E_A and origin makes an angle θ with positive X -axis.

Now, match the following columns

Column-I(Value of θ)

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{12}$

Column-II(Area of $\Delta E_A GH$)

(P) $\frac{(\sqrt{3}-1)^4}{8}$

(Q) 1

(R) $\frac{3}{4}$

(S) $\frac{1}{2\sqrt{3}}$

(T) $\frac{3\sqrt{3}}{2}$

KEY

1. C	2. B	3. A	4. A	5. B
6. AB	7. BC	8. ABC	9. ACD	10. AB
11. BCD	12. ABCD	13. AB	14. AC	15. ABCD
16. 40	17. 937	18. 5	19. 4	20. 7
21. 119	22. 9	23. 38	24. B	25. BC
26. A-R,B-S, C-P,D-Q	27. A-S,B-R, C-P,D-Q	28. A-Q,B-T, C-S,D-P	29.	30.

GR-12 : QUADRATIC EQUATIONS AND POLYNOMIALS**SINGLE CORRECT**

Q.(1) If $|A|$ denotes the cardinal number of set A ,

$$A = \{x \in \mathbb{Z} : ax^2 + 2x(1 - a) - 4 < 0, a \in \mathbb{R}\} \text{ and}$$

$$B = \{a \in \mathbb{R} : |A| = 3\},$$

then B is equal to

(A) $(0, 2)$

(B) $(0, 1]$

(C) $[1, 2)$

(D) $[2, \infty)$

Q.(2) Let $a, b \in \mathbb{R}$. If the polynomials

$$ax^2 + bx + a = 0 \text{ and } x^3 - 2x^2 + 2x - 1 = 0$$

have two roots in common, then the smallest value of k for which both the roots of the equation

$$x^2 - 8kx + 16(k^2 - k + 1) = a + b$$

are real, distinct and have values at least 4, is equal to

(A) 1

(B) 2

(C) 3

(D) 4

MULTIPLE CORRECT

Q.(3) Let set A be defined as

$$A = \{x \in \mathbb{R} : x^2 + (2 - m)x - 4 = 0, m \in \mathbb{R}\}.$$

Then which of the following options is/are correct?

(A) If $m < 5$, then $x < -1$ is true for some $x \in A$.

(B) If $m > 5$, then $x > -1$ is true for all $x \in A$.

(C) $A = \emptyset$ for atleast one $m \in \mathbb{R}$.

(D) $A \neq \emptyset$ for all $m \in \mathbb{R}$

Q.(4) Graph of

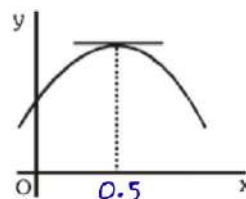
Graph of $f(x) = px^2 - qx - r$ is shown in the adjacent figure then -

(A) $r(p - q - r) < 0$

(B) $q < 0$

(C) $\sqrt{q^2 + 4pr} > |q|$

(D) $p < 0$



Q.(5)

If the quadratic equation $x^2 - \frac{(\alpha^2 + 11)x}{9} + \frac{15}{4}(\alpha + \beta) + 16 = 0$ has two integral roots α & β , then

(A) $\alpha + \beta = 20$

(B) $|\alpha - \beta| = 120$

(C) $|\alpha - \beta| = 6$

(D) $\alpha + \beta = 128$

Q.(6)

Graph of $y = ax^2 + bx + c$ is shown in the figure and length of PQ & OR are 9 units & 5 units respectively (where Q is the vertex of the parabola)

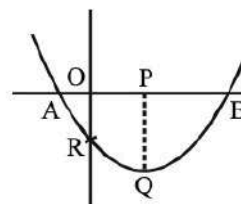
and area of $\triangle OBQ$ is $\frac{45}{4}$ (unit)², then

(A) length of AB is 3

(B) length of AB is 4

(C) $\left|\frac{b}{a}\right| > 1$

(D) $\left|\frac{b}{a}\right| < 1$



Q.(7)

If the equation $x^4 + 2ax^3 + x^2 + 2ax + 1 = 0$ have at least two distinct negative roots, then the possible values of a is/are-

(A) 1

(B) 2

(C) 3

(D) 4

Q.(8) If $|A|$ denotes the cardinal number of set A , then which of the following options is/are correct?

(A) $|\{n \in \mathbb{N} : \sqrt{n^2 + 96} \in \mathbb{N}\}| = 4$.

(B) $|\{n \in \mathbb{Z} : \sqrt{n^2 + 19n + 92} \in \mathbb{N}\}| = 2$

(C) $|\{n \in \mathbb{A} : x^2 + x - n = 0, x \in \mathbb{Z}\}| = 9; \mathbb{A} = \{k \in \mathbb{N} : k < 100\}$

(D) $|\{n \in \mathbb{Z} : (n^2 - n - 1)^{n+2} = 1\}| = 4$

Q.(9) Let

$$\{x \in \mathbb{C} : x^3 + 5x^2 + \ell x + m = 0, \ell, m \in \mathbb{R}\} = \{p, q, r_1\},$$

$$\{x \in \mathbb{C} : x^3 + 7x^2 + \ell x + n = 0, \ell, n \in \mathbb{R}\} = \{p, q, r_2\}.$$

If all the roots of $x^4 - 8x^3 + ax^2 - bx + 16 = 0$ are positive real numbers, then which of the following options is/are correct?

(A) $r_1 + a = 19$

(B) $r_2 + b = 25$

(C) $r_1 + b = 27$

(D) $r_2 + a = 17$

NUMERICAL TYPE

Q.(10)

The number of solutions of equation $|2ax - 3| + |ax + 1| + |5 - ax| = \frac{1}{2}$ is/are -

Q.(11) If

$$\lambda \in \left\{ \sqrt[4]{5x^4 - x^8} : x + \sqrt[4]{5 - x^4} = 2 \right\},$$

then $2\lambda + \sqrt{42}$ is equal to

Q.(12) The value of

$$\lambda = \frac{(7^4 + 64)(15^4 + 64)(23^4 + 64)(31^4 + 64)(39^4 + 64)}{(3^4 + 64)(11^4 + 64)(19^4 + 64)(27^4 + 64)(35^4 + 64)}$$

is equal to

Q.(13) Let

$$\mathbb{A} = \{x < 0 : x^4 + (a - 1)x^3 + x^2 + (a - 1)x + 1 = 0\}$$

$$\mathbb{B} = \{a \in \mathbb{R} : |\mathbb{A}| \geq 2\}.$$

If $|A|$ denotes the cardinal number of set A , then number of elements in $\mathbb{B} \cap \{x \in \mathbb{N} : x < 20\}$ is equal to

Q.(14) Let λ denote the number of possible values of α for which the expression

$$\frac{\alpha x_1^2 + 7x_1 - 2}{\alpha + 7x_1 - 2x_1^2} = \frac{\alpha x_2^2 + 7x_2 - 2}{\alpha + 7x_2 - 2x_2^2} \Rightarrow x_1 = x_2 \forall x_1, x_2 \in \{x \in \mathbb{R} : \alpha + 7x - 2x^2 \neq 0\}.$$

Q.(15) Let a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2024 = 0$, and c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2024 = 0$. Then the sum of digits of the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is equal to

Q.(16) Let p denote the number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$. If there exists only one real value of λ for which the quadratic equation $\lambda x^2 - 2px + 9 = 0$ has two positive integral solutions, then the sum of these two solutions is equal to

PASSAGES

Passage-III

Consider the following sets

$$A = \{x \in \mathbb{Z} : a^3 + a^2|x + a| + |a^2x + 1| = 1, a \in \mathbb{R}\},$$

$$B = \{x \in \mathbb{Z} : |x + a^3| + |x + 1| \leq 1 - a^3, a \in \mathbb{R} - \{0\}\}.$$

Based on above information, answer the following questions

Q.(17) Number of elements in $\{a \in \mathbb{Z} : |A| \geq 4\} \cap \{x \in \mathbb{Z} : x > -10\}$ is equal to

Q.(18) Number of elements in $\{a \in \mathbb{Z} : |B| \geq 5\} \cap \{x \in \mathbb{Z} : x > -10\}$ is equal to

MATRIX MATCHING TYPE

Q.(19) Let A denote the set of possible values of $a (\in \mathbb{R})$ for which

$$x^2 + (a - b)x + (1 - a - b) = 0$$

has unequal real roots for all $b \in \mathbb{R}$. If

$$f(x) = \frac{x^2 - 11x + 18}{x^2 - 7x + 12},$$

then match the following columns

Column-I

Column-II

(A) $\{x \in \mathbb{R} : (f(x))^{-1} \in \mathbb{A}\}$

(P) (3,4)

(B) $\{x \in \mathbb{R} : f(x) \in \mathbb{A}\}$

(Q) $\left(\frac{3}{2}, 2\right) \cup (9, \infty)$

(C) $\{x \in \mathbb{R} : f(x) \in \mathbb{R} - \mathbb{A} - [0,1]\}$

(R) $\left(-\infty, \frac{3}{2}\right) \cup (3,4)$

(D) $\{x \in \mathbb{R} : f(x) \in \mathbb{A} - (1,2]\}$

(S) $(2,3) \cup (4,9)$

Q.(20) Match the following

Column-I

Column-II

(A) The set of values of 'a' for which the $ax^2 + 4x + 10 \leq 0$

(P) [0,1]

has at least one real solution and every solution of $x^2 - x - 2 < 0$

is larger than any solution of $ax^2 + 4x + 10 \leq 0$ contains

(B) The set of values of k for which the solution set of

(Q) [-1,0]

$x^2 + 3k^2 - 1 \geq 2k(2x - 1)$ is a subset of solution set of

$x^2 - (2x - 1)k + k^2 \geq 0$ contains

(C) The set of values of k for which there is at least one common

(R) [0,0.4]

solution of the inequalities $x^2 + 4kx + 3k^2 > 1 + 2k$ and

$x^2 + 2kx \leq 3k^2 - 8k + 4$ contains

(D) The set of values of k for which any real x is a solution of at least

(S) [-3, -2]

one of the inequalities $x^2 + 5k^2 + 8k > 2(3kx + 2)$ and

$x^2 + 4k^2 \geq k(4x + 1)$ contains

KEY

1. C	2. B	3. ABD	4. ABCD	5. AC
6. AC	7. ABCD	8. ABCD	9. ABCD	10. 0
11. 8	12. 337	13. 17	14. 2	15. 7
16. 12	17. 8	18. 8	19. $A \rightarrow Q, B \rightarrow R,$ $C \rightarrow S, D \rightarrow P$	20. A-R, B-PQR, C-QRS, D-QS

GR-13 : PROGRESSIONS AND SERIES

SINGLE CORRECT

Q.(1)

The sum of the series, $\frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \dots$ upto infinite terms. If $|x| < 1$, is

- (A) $\frac{x}{1-x}$ (B) $\frac{1}{1-x}$ (C) $\frac{1+x}{1-x}$ (D) 1

Q.(2) Let

$$A = \left\{ \sum_{k=1}^{4n} \left((-1)^{\frac{k(k+1)}{2}} \cdot k^2 \right) : n \in \mathbb{N} \right\}.$$

Which of the following number(s) lie in A?

- (A) 1088 (B) 1336 (C) 1056 (D) 1176

MULTIPLE CORRECT

Q.(3)

Consider a sequence $\langle t_n \rangle$ such that $t_n^2 + t_n = t_{n+1}$, $\forall n \in \mathbb{N}$ and $t_1 = 1$, then-

- (A) $\prod_{m=1}^{99} (t_m + 1) = t_{99}$ (B) $\prod_{m=1}^{99} (t_m + 1) = t_{100}$ (C) $1 + \sum_{m=1}^{99} t_m^2 = t_{100}$ (D) $1 + \sum_{m=1}^{99} t_m^2 = t_{99}$

Q.(4)

If a, b, c are positive real numbers, such that minimum value of $E = a^4 + 2b^4 + 4c^4 - 8abc$ is 'm' and a_0, b_0, c_0 are respective value of a, b, c for which E is minimum, then -

- (A) $m = 2$ (B) $m = -2$ (C) $a_0 b_0 c_0 = 1$ (D) $a_0 b_0 c_0 = 4$

Q.(5) Let

$$a = \frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \cdots \infty,$$

$$b = \sum_{r=0}^{\infty} (-1)^{r+1} \left(\frac{r}{5^r} \right),$$

$$\frac{c}{d} = \sum_{k=1}^{360} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}.$$

 If $c, d \in \mathbb{N}$ and $\gcd(c, d) = 1$, then which of the following options is/are correct?

- (A) $a + 36b = 8$ (B) $5a + 72b = 20$ (C) $c + d = 37$ (D) $c + d = 39$

Q.(6) Let $|A|$ denote the cardinal number of set A ,

$$\mathbb{A} = \{5n - 4 : n \in \mathbb{N}, n < 2019\},$$

$$\mathbb{B} = \{7n + 2 : n \in \mathbb{N}, n < 2019\}.$$

 Suppose $\ln b_1, \ln b_2, \ln b_3, \dots, \ln b_{101}$ are in A.P. (Arithmetic Progression) with common difference

$$d = \ln(|\mathbb{A} \cup \mathbb{B}| - 3746).$$

 Also $a_1, a_2, a_3, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If

$$s = \sum_{r=1}^{101} (a_r - b_r) \text{ and } t = (a_{101} - b_{101}),$$

then which of the following options is/are correct?

- (A) $s > 0$ (B) $s < 0$ (C) $t > 0$ (D) $t < 0$

Q.(7) Let

$$\lambda = \sum_{k=1}^{100} \frac{k}{k^4 + k^2 + 1}, \mu = \sum_{n=0}^{\infty} \frac{n}{n^4 + 4}.$$

Then which of the following options is/are correct?

- (A) $10101\lambda + 80\mu = 5080$ (B) $10101\lambda - 8\mu = 5058$
 (C) $10101\lambda - 1600\mu = 4000$ (D) $10101\lambda + 400\mu = 5200.$

Q.(8) Let p, q, r are non-zero distinct real numbers such that sum of roots of equation

$$p x^2 + qx + r = 0$$

is equal to the sum of squares of their reciprocals. If

$$2a^2, b^2, 2c^2$$

are three distinct real numbers in Harmonic Progression, then which of the following options is/are correct?

(A) $4a^2 - b^2, b^2, 4c^2 - b^2$ are in G.P. (Geometric Progression)

(B) $a^2b^2, 2a^2c^2, b^2c^2$ are in A.P. (Arithmetic Progression)

(C) pq^2, p^2r, qr^2 are in A.P. (Arithmetic Progression)

(D) p^2q, q^2r, qr^2 are in A.P. (Arithmetic Progression)

Q.(9) Let $|A|$ denotes the cardinal number of set A , $[x]$ denotes the greatest integer $\leq x$,

$$A = \{3n - 2 : n \in \mathbb{N}, n < 2005\},$$

$$B = \{7m + 2 : m \in \mathbb{N}, m < 2005\}.$$

Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with

$$a_1 = [\log_6 |A \cap B|], S_p = \sum_{r=1}^p (a_r).$$

For any $n \in \{x \in \mathbb{N} : x \leq 20\}$, let $m = 5n$. If $S_m : S_n$ is independent of n , then a_2 can be

(A) 12

(B) 6

(C) 3

(D) 9

Q.(10)

Let $-\frac{\pi}{2} < \alpha_1 < \alpha_2 < \alpha_3 < \dots < \alpha_n < 2\pi$ and $0 < \beta_1 < \beta_2 < \dots < \beta_m < 2\pi$ are values of α and β respectively such that $2\cos 2\alpha - 3 = 2\cos(2\alpha + \beta) - 2\cos \beta$, then which of the following is(are) true ?

(A) $2n > 3m$

(B) $n > m$

(C) $\sum_{i=1}^m \beta_i \leq 2\pi$

(D) $\sum_{i=1}^m \beta_i \leq \sum_{i=1}^m \alpha_i$

NUMERICAL TYPE

Q.(11) Sum of the series

$$\frac{5}{1 \times 2 \times 3} + \frac{7}{2 \times 3 \times 3^2} + \frac{9}{3 \times 4 \times 3^3} + \frac{11}{4 \times 5 \times 3^4} + \dots \infty$$

is equal to

Q.(12)

Let $l_1, l_2, \dots, l_{100}$ be consecutive terms of an arithmetic progression with common difference d_1 , and let $w_1, w_2, \dots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2 , where $d_1 d_2 = 10$. For each $i = 1, 2, \dots, 100$, let R_i be a rectangle with length l_i , width w_i and area A_i . If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is

Q.(13)

Let $a_1, a_2, a_3, \dots$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, \dots$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c , for which the equality

$$2(a_1 + a_2 + \dots + a_n) = b_1 + b_2 + \dots + b_n$$

holds for some positive integer n , is —

Q.(14) The value of

$$\sum_{r=2}^{\infty} \frac{16r}{1 + r^2(r^2 - 2)}$$

is equal to

Q.(15)

If a_i is positive or negative according as 'i' is even or odd and equation

$x^4 + a_1x^3 + a_2x^2 + a_3x + 5 = 0$ has four positive real roots then if the minimum value of a_1a_3 is N, then the sum of the digits in 'N' is-

PASSAGES

Passage-I

Let

$$\frac{T_n}{\mu} = \frac{(n^2 + n + 2)2^n}{(n+1)(\mu - 2n)(2n + \lambda)},$$

$$f(x) = x^4 - 4x^3 + \lambda x^2 + \mu x + 1.$$

If $f(x) = 0$ has four real positive real roots, then answer the following two questions

Q.(16) Which of the following options is/are correct?

- (A) $|\lambda| + |\mu| = 2$ (B) $\lambda - \mu = 10$ (C) $|\lambda| + |\mu| = 10$ (D) $|\lambda| - |\mu| = 2$

Q.(17) Which of the following options is/are correct?

(A) $\sum_{r=1}^5 T_r = \frac{6.2^6}{7.8} - \frac{2}{2.3}$

(B) $\sum_{r=1}^4 T_r = \frac{5.2^5}{6.7} - \frac{2}{2.3}$

(C) $\sum_{r=1}^{\infty} T_r = -\frac{2}{2.3}$

(D) $\sum_{r=1}^{\infty} T_r$ does not exist finitely

Passage-2

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence such that

$$a_1 = 1, a_2 = 2017, \frac{a_{n-1} + na_{n-2}}{n} = \frac{a_{n-1}^2}{a_n} \forall n \geq 2,$$

and

$$S_n = n + \frac{1}{2} \left(n-1 + \frac{1}{2} \left(n-2 + \frac{1}{2} \left(n-3 + \dots + \frac{1}{2} (2) \right) \right) \dots \right)$$

Now, based on above information, answer the following questions

Q.(18) Which of the following options is/are correct?

- (A) $S_{2005} = 4008$ (B) $S_{2005} = 4010$ (C) $S_{2004} = 4006$ (D) $S_{2004} = 2008$

Q.(19) Which of the following options is/are correct?

- (A) a_5 is the greatest term of the sequence
 (B) a_6 is the greatest term of the sequence
 (C) $a_m < a_{m-1}$ holds true for $m = 6, 7, 8, \dots$
 (D) $a_m < a_{m-1}$ holds true for $m = 7, 8, 9, \dots$

MATRIX MATCHING TYPE

Q.(20) Let

$$\frac{1}{2} + \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{5}{32} + \frac{8}{64} + \frac{13}{128} + \frac{21}{256} + \frac{32}{512} + \dots \infty = \frac{p}{q}; p, q \in \mathbb{N}, \gcd(p, q) = 1,$$

$$T_r = \frac{rx}{(1-x)(1-2x)(1-3x)\dots(1-rx)}, x \neq 0 \text{ \& } V_r = \frac{1}{(r^2 + 7r + 12)},$$

$$f(x) = \left(\frac{1 + T_p + T_{p+1} + T_{p+2} + \dots \infty}{T_p + T_{p+1} + T_{p+2} + \dots \infty} \right)^2.$$

Then match the following columns

Column-1

Column-2

(A) $f(\sqrt{2})$ is equal to

(P) 4

(B) $4(V_q + V_{q+1} + V_{q+2} + \dots \infty)$ is equal to

(Q) 3

(C) $f(\sqrt{3})$ is equal to

(R) 2

 (D) $100(V_1 + V_2 + V_3 + \dots V_{21}) - 17$ is equal to

(S) 1

KEY

1. C	2. C	3. BC	4. BC	5. BC
6. BD	7. AD	8. ABC	9. CD	10. ABC
11. 1	12. 18900	13. 1	14. 5	15. 8
16. BCD	17. ABD	18. AC	19. BD	20. A-R, B-S, C-Q, R-P

GR-14 : PERMUTATIONS AND COMBINATIONS**SINGLE CORRECT TYPE QUESTIONS**

Q.(1) Consider a determinant of order 3 all whose entries are either 0 or 1. Five of these entries are 1 and four of them are '0'. Also $a_{ij} = a_{ji} \forall i, j \in \{1, 2, 3\}$. Find the number of such determinants.

- (A) 12 (B) 21 (C) 13 (D) 22

Q.(2) 10 identical balls are distributed in 5 different boxes kept in a row and labeled A, B, C, D and E. Find the number of ways in which the ball can be distributed in the boxes if no two adjacent boxes remain empty.

- (A) 770 (B) 771 (C) 772 (D) 773

Q.(3) Let x & y be 4 digit numbers in each of which digits do not repeat. Then the number of pair of such numbers (x, y) , if no digit in numbers x & y are common is equal to

- (A) $2 \cdot 9!$ (B) $4 \cdot 9!$ (C) $3 \cdot 9!$ (D) $5 \cdot 9!$

MULTIPLE CORRECT TYPE QUESTIONS

Q.(4) Which of the following options is/are correct?

- (A) The number of 4 digit natural numbers which contain exactly 2 distinct digits is 567.
(B) The number of 4 digit natural numbers which contain at least 2 distinct digits is 8090.
(C) The number of divisors of the form $(4k + 2); k \in \mathbb{N}$ of the number $2^8 5^{10} 7^3 13^2$ is 132.
(D) The number of divisors of the form $(4k + 1); k \in \mathbb{N}$ of the number $2^8 5^{10} 7^3 13^2$ is 66.

Q.(5) Consider the following statements

S-1: Let m be the total number of unordered pairs of disjoint subsets of set $S = \{1, 2, 3, 4, 5\}$.

S-2: The total number of ways in which 7 balls of different colors can be distributed among 5 persons so that each person gets at least one ball is $100n$.

Then which of the following options is/are correct?

- (A) $m + n = 300$ (B) $n - m = 46$ (C) $3n - 4m = 16$ (D) $5m - 3n = 100$.

Q.(6) Let a_n denote the number of all n – digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n – digit integers in which no three consecutive digits are 0, and c_n = the number of such n -digit integers in which a maximum of two consecutive digits can be 0. Then

- (A) $a_{2024} = 3a_{2021} + 2a_{2020}$ (B) $b_{2024} = b_{2022} + 3b_{2021} + 2b_{2020} + b_{2019}$
 (C) $c_{2024} = c_{2023} + c_{2022} + c_{2021}$ (D) $a_{2024} + c_{2024} = b_{2024}$

Q.(7) Let $n_1 > n_2 > n_3 > n_4 > n_5$ be positive integers such that

$$n_1 + n_2 + n_3 + n_4 + n_5 = 21.$$

Let $p^n(x)$ denote the n^{th} derivative of $p(x)$ w.r.t. x , i.e.

$$\frac{d^n}{dx^n}(p(x)) = p^n(x), \quad f(x) = \prod_{n=1}^5 \frac{x^n}{1-x^n}, \quad g(x) = \frac{1}{5!} \left(\frac{x}{1-x} \right)^5.$$

Then, the number of solutions of the given equation is equal to

Here $\binom{n}{r}$ denotes nC_r .

- (A) $21! \cdot f^{21}(0)$ (B) $21! \cdot g^{21}(0)$ (C) $\binom{6}{2}$ (D) $\binom{5}{2}$

Q.(8) Let $O_{n,m}$ denote the number of onto and non-decreasing functions

$$f: \{1, 2, 3, \dots, n\} \rightarrow \{1, 2, 3, \dots, m\}.$$

Then, which of the following options is/are correct?

- (A) $(O_{10,5})^2 + (O_{10,6})^2 + (O_{10,7})^2 = 125$ (B) $O_{8,5} = 35$
 (C) $O_{20,10} + 2O_{20,11} + O_{20,12} = O_{22,12}$ (D) $3O_{30,10} = 7 \cdot O_{30,21}$

Q.(9) Number of obtuse angled triangles which can be formed by joining vertices of an n – sided regular polygon can be ($n < 10$)

- (A) 24 (B) 54 (C) 21 (D) 6

Q.(10) Let $f(m, n, k)$ be the number of ways in which m boys and n girls can stand in a queue in such a way that a maximum of k girls can stand consecutively in the queue. Then, which of the following options is/are correct?

- (A) $f(5, 5, 0) = 2 \cdot 5! \cdot 5!$ (B) $f(5, 7, 2) = 146 \cdot 5! \cdot 7!$
(C) $f(7, 5, 2) = 504 \cdot 5! \cdot 7!$ (D) $f(6, 6, 3) = 628 \cdot 6! \cdot 6!$

Q.(11) Which of the following options is/are correct about

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right); n \in N, n > 1$$

- (A) $D_{20} = 19(D_{19} + D_{18})$ (B) $D_{30} = \left\lfloor \frac{30!}{e} \right\rfloor$
(C) $D_{50} = \left\lfloor \frac{50!}{e} + \frac{1}{2} \right\rfloor$ (D) $D_{70} = 70(D_{69} + D_{68})$

Here $[.]$ denotes the greatest integer function and $\lfloor . \rfloor$ denotes the nearest integer function

Q.(12) Which of the following options is/are correct about

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right); n \in N, n > 1, \text{ and } E_n = \int_0^\infty e^{-t} (t-1)^{n-1} dt.$$

- (A) $D_{20} = 20D_{19} + 1$ (B) $D_{35} = 35D_{34} - 1$
(C) $D_{50} = E_{50}$ (D) $D_{71} = E_{70}$

Q.(13) Each of the n persons have a gun, with 1 bullet which they fire randomly at any other person. If all the shots are fired simultaneously, and no one misses the target at which they aim and shoot, then P_n denotes the number of ways in which exactly 1 person is alive after shooting. Which of the

following options is/are correct?

(A) $P_2 = 0$

(B) $P_3 = 6$

(C) $P_4 = 48$

(D) $P_6 = 3840$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(14)

A committee of four members is to be formed from 5 men and 5 women. If atleast one of Miss X and Miss Y is in the committee then Mr. A & Mr. B must serve together and if both the above ladies are not in the committee then both the above mentioned men will not serve the committee. If committee spend ₹300 per women and ₹200 per men, then number of committies with minimum expenditure is

Q.(15)

Let A is set of first 7 natural numbers and B is set of first 6 even natural numbers. Consider function $f : A \rightarrow B$ such that $f(x) \neq 2x \forall x \in A$ and $f(x)$ is an onto function. If total number of such functions is $181 \times N$, then sum of digits of N is

Q.(16)

Let P_n denotes the number of ways of choosing three points out of n distinct points in a line such that no two of them are consecutive. If Q_n denotes the number of ways of choosing similar points on circle

then $\frac{Q_6}{P_6} + \frac{Q_7}{P_7} + \frac{Q_8}{P_8}$ is

Q.(17)

If L.C.M. of A, B, C is $r^2 t^2 s^2$, where r, s, t are distinct prime numbers and A, B, C are positive integers, the number of ordered triplets (A, B, C) is K, then value of $\left[\frac{K}{1000} \right]$ is (where $[.]$ denotes greatest integer function)

Q.(18) Let m_n denote the number of ways of seating n-married couples, around a circle, such that no husband sits next to his wife. Then, the remainder obtained on dividing m_5 by 5 is equal to

Q.(19) The set S , is defined as

$$S = \left\{ (p, q, r) \in \mathbb{N}^3 : LCM(p, q, r) = 2^5 3^6 4^8 5^9 \text{ and } HCF(p, q, r) = 2^2 3^3 4^4 5^5 \right\}.$$

Then, the digit in units place of $n(S)$, is equal to

Q.(20) Let A_n denote the number of ways of seating n -married couples, around a circle, such that men and women sit alternately, and no husband sits next to his wife. Then, A_5 is equal to

PASSAGES

Passage-II:

Consider seven different points $P_1, P_2, P_3, P_4, Q_1, Q_2, Q_3$ in a plane such that P_1, P_2, P_3, P_4 are on straight line ' ℓ '; Q_1, Q_2, Q_3 are non-collinear points and none of them lies on straight line ' ℓ ' (Q_1, Q_2, Q_3 are on same side of ℓ).

Q.(21)

The maximum number of possible points of intersection of straight lines formed by joining any two of the given seven points (Excluding the given points) -

Q.(22)

From each of the three points Q_1, Q_2, Q_3 perpendicular lines are drawn to the straight lines formed by joining any two of the given points (excluding the point from which perpendicular line is drawn). The maximum possible number of points of intersection of perpendicular lines (excluding points Q_1, Q_2, Q_3) is-

Passage-II:

Let set A be defined as

$$A = \{1, 2, 3, \dots, n\}.$$

Let a_n denote the number of functions $f: A \rightarrow A$ which satisfy

$$f(f(x)) = x.$$

Let b_n denote the number of functions $f: A \rightarrow A$, such that

$$f(f(x)) = f(x).$$

Now, answer the following questions:

Q.(23) The value of $a_6 - a_4$ is equal to

Q.(24) The value of b_5 is equal to

MATRIX MATCHING

Q.(25)

In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .

- (i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boy and 2 girls.
- (ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.
- (iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.
- (iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are **NOT** in the committee together.

LIST-I

- P.** The value of α_1 is
- Q.** The value of α_2 is
- R.** The value of α_3 is
- S.** The value of α_4 is

LIST-II

- 1.** 136
- 2.** 189
- 3.** 192
- 4.** 200
- 5.** 381
- 6.** 461

KEY

1. A	2. B	3. B	4. ACD	5. BC
6. ABC	7. AD	8. BC	9. ABCD	10. ABC
11. ABC	12. ABC	13. ABCD	14. 9	15. 3
16. 2	17. 6	18. 3	19. 2	20. 312
21. 51	22. 283	23. 66	24. 196	25. P-4, Q-6, R-5, S-2

GR-15 : PROBABILITY

SINGLE CORRECT TYPE QUESTIONS

Q.(1)

There are three different Urns, Urn-I, Urn-II and Urn-III containing 1 Blue, 2 Green ; 2 Blue, 1 Green; 3 Blue, 3 Green balls respectively. If two Urns are randomly selected and a ball is drawn from each Urn and if the drawn balls are of different colours then the probability that chosen Urn was Urn-I and Urn-II is-

- (A) $\frac{1}{7}$ (B) $\frac{5}{13}$ (C) $\frac{5}{14}$ (D) $\frac{5}{7}$

Q.(2)

A tennis match of best of 5 sets, where a player winning 3 sets first is declared as winner of the match, is played between two players A & B. The probability that a set is won by A is $\frac{1}{2}$ and if he loses the set then probability of his winning the next set is $\frac{1}{4}$, otherwise it remains same, then probability that A wins is -

- (A) $\frac{5}{16}$ (B) $\frac{7}{16}$ (C) $\frac{3}{32}$ (D) $\frac{1}{8}$

Q.(3)

From a pack of 52 well shuffled cards, cards are drawn one by one without replacement. If 4th drawn card is found to be ace, then what is the probability, that there are no more aces left in the pack, is-

- (A) $\frac{1}{{}^{48}C_3 + 3 \cdot {}^{49}C_2 + 1}$ (B) $\frac{1}{{}^{48}C_3 + {}^{49}C_2 + 1}$ (C) $\frac{1}{3 \cdot {}^{48}C_3 + {}^{49}C_2 + 1}$ (D) $\frac{1}{{}^{52}C_4 + 1}$

Q.(4) Family A has 6 members of which 4 are males and 2 are females & family B has 5 members consisting at least one male and one female. (Assume it is equiprobable for a member of unknown gender to be a male or female). 2 members are selected randomly either from family A or family B. If both the members are female, then the probability that they belong to family A is equal to p . The value of $9p$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(5)

A fair coin is tossed 10 times and the outcomes are listed. Let E_i be the event that the i^{th} outcome is a head and E_m be the event that the list contains exactly m heads. If E_i and E_m are independent, then 'm' is equal to

- (A) 6 (B) 5 (C) 3 (D) 4

MULTIPLE CORRECT TYPE QUESTIONS

Q.(6) Let X and Y be two events such that

$$P(X | \bar{Y}) = \frac{1}{2}, \quad P(Y | \bar{X}) = \frac{1}{3} \quad \text{and} \quad P(X \cap Y) = \frac{1}{6}.$$

Then which of the following options is/are correct?

- (A) $P(X \cup \bar{Y} | \bar{X} \cup Y) = \frac{3}{4}$ (B) $P(X \cup Y | \bar{X} \cup \bar{Y}) = \frac{3}{5}$
 (C) $P(Y \cup \bar{X} | \bar{Y} \cup X) = \frac{4}{5}$ (D) $P(\bar{X} \cup \bar{Y} | X \cup Y) = \frac{3}{4}$

Q.(7) Let ω be an imaginary cube root of unity. A fair die is thrown three times. If r_1, r_2 and r_3 are the numbers obtained on the die, then the reciprocal of probability that

- (A) $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is 4.5. (B) $1 + \omega^{r_1+r_2+r_3} + (\omega^2)^{r_1+r_2+r_3} = 0$ is 1.5.
 (C) $\begin{vmatrix} \omega^{r_1} & \omega^{r_2} & \omega^{r_3} \\ \omega^{r_3} & \omega^{r_1} & \omega^{r_2} \\ \omega^{r_2} & \omega^{r_3} & \omega^{r_1} \end{vmatrix} = 0$ is 4. (D) $\omega^{r_1+r_2} + \omega^{r_2+r_3} + \omega^{r_3+r_1} = 0$ is 4.5.

Q.(8) A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$

Then which of the following options is/are correct?

- (A) If the signal received at B is green, then the probability that the original signal was green is $\frac{20}{23}$.
 (B) If the signal received at B is red, then the probability that the original signal was green is $\frac{12}{17}$.
 (C) If the original signal is red, then the probability that signal received at B is green is $\frac{3}{8}$.
 (D) If the original signal is green, then the probability that signal received at B is red is $\frac{5}{8}$.

Q.(9) Let U and W be two urns such that

U contains 3 white and 2 red balls,
 and
 W contains only 1 white ball.

A fair coin is tossed. If head appears then 1 ball is drawn at random from U and put into W. However, if tail appears then 2 balls are drawn at random from U and put into W. Now 1 ball is drawn at random from W.

- (A) The probability that ball drawn from W is white is $\frac{43}{60}$.
- (B) Given that ball drawn from W is white, the probability that coin's toss was a head, is $\frac{12}{23}$.
- (C) Given that ball drawn from W is red, the probability that coin's toss was a tail, is $\frac{12}{23}$.
- (D) Given that ball drawn from W is white, the probability that W had at least 2 white balls is $\frac{33}{43}$.

Q.(10) Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T and $P(E) > P(F)$, then

- (A) $23 \cdot P\left(\frac{E}{E \cup F}\right) - P\left(\frac{F}{E \cap F}\right) = 17$
- (B) $11 \cdot P\left(\frac{E}{E \cup F}\right) + 17 \cdot P\left(\frac{F}{E \cup F}\right) = 25$
- (C) $11 \cdot P\left(\frac{E \cap \bar{F}}{E \cup \bar{F}}\right) - 17 \cdot P\left(\frac{\bar{E} \cap F}{\bar{E} \cup F}\right) = 1$
- (D) $11 \cdot P\left(\frac{\bar{E} \cup F}{E \cup \bar{F}}\right) + 17 \cdot P\left(\frac{E \cup \bar{F}}{E \cup F}\right) = 21$

Q.(11) Which of the following options is/are correct for any three non-null events A, B and C?

- (A) If events A, B, C are independent, then events A, $B \cup C$ are also independent.
- (B) If events A, B are independent, events A, C are independent, & events B, C are mutually exclusive, then events A, $B \cup C$ are also independent.
- (C) If events A, C are independent, events B, C are independent, & events A, $B \cap C$ are independent, then events B, $A \cap C$ are also independent.
- (D) If events A, B are independent and events B, C are mutually exclusive, then events C, A are also independent

Q.(12)

There are three bags B_1 , B_2 and B_3 . The bag B_1 contains 5 red and 5 green balls, B_2 contains 3 red and 5 green balls, and B_3 contains 5 red and 3 green balls, Bags B_1 , B_2 and B_3 have probabilities $\frac{3}{10}$, $\frac{3}{10}$ and $\frac{4}{10}$ respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct ?

- (1) Probability that the selected bag is B_3 and the chosen ball is green equals $\frac{3}{10}$
- (2) Probability that the chosen ball is green equals $\frac{39}{80}$
- (3) Probability that the chosen ball is green, given that the selected bag is B_3 , equals $\frac{3}{8}$
- (4) Probability that the selected bag is B_3 , given that the chosen balls is green, equals $\frac{5}{13}$

Q.(13) In an election where candidate A receives a votes and candidate B receives b votes, p denotes the probability that candidate A remains strictly ahead of candidate B throughout the count. Then, which of the following options is/are correct?

- | | |
|--|---|
| (A) If $a = 10, b = 7$, then $5p = 1$ | (B) If $a = 20, b = 5$, then $5p = 3$ |
| (C) If $a = 20, b = 19$, then $13p = 1$ | (D) If $a = 17, b = 7$, then $12p = 5$ |

NUMERICAL ANSWER TYPE QUESTIONS

Q.(14) Let a_k denote the minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least k . Let b_k denote the maximum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at most k . Then value of $a_{0.96} + b_{0.96}$ is equal to

Q.(15) Let C_1 and C_2 be two biased coins such that the probabilities of getting head in a single toss are $\frac{2}{3}$ and $\frac{1}{3}$, respectively. Suppose α is the number of heads that appear when C_1 is tossed twice, independently, and suppose β is the number of heads that appear when C_2 is tossed twice, independently. Then the probability that the roots of the quadratic polynomial $x^2 - \alpha x + \beta$ are real and equal, is p . Then $81p$ is equal to

Q.(16) A dice is randomly rolled 7 times. If the probability that both the outcomes of 4^{th} and 5^{th} throw, is(are) also one of the outcomes of the remaining throws is $\frac{p}{6^6}$, then the remainder obtained when p is divided by 7, is equal to

Q.(17) Eight ants start at different corners of a cube. Suddenly each ant moves to an adjacent corner at random. That is, each ant walks along one of its three adjacent edges with equal chance. If p is the probability that none of the ants collide, then $6561p$ is equal to

Q.(18) There are 5 bags of colors Blue, Green, Pink, Red and Yellow. Each of these bags have 5 balls of same color as that of the bag. A ball is drawn from blue bag and is transferred to one of the other bags. Then, a ball is drawn from green bag and transferred to one of the other bags. Similarly, 1 ball is transferred from the pink bag, then from red bag and then from yellow bag (to one of the other bags). If p denotes the probability that at the end each bag has 5 balls of same color, given that there are 5 balls in each bag, then value of $[(216p)^{-1}]$ is equal to ($[.]$ denotes the greatest integer function)

Q.(19)

Let S be the sample space of all 3×3 matrices with entries from the set $\{0, 1\}$. Let the event E_1 and E_2 be given by

$$E_1 = \{A \in S : \det A = 0\} \text{ and}$$

$$E_2 = \{A \in S : \text{sum of entries of } A \text{ is } 7\}.$$

If a matrix is chosen at random from S , then the conditional probability $P(E_1|E_2)$ equals ____

Q.(20) Players $P_1, P_2, P_3 \dots P_8$ of equal strength play in a knockout tournament, where the players are divided into pairs at random, and winner from each pair moves into the next round. If p denotes the probability that player P_1 defeats players P_2 and P_3 to reach and final, and wins the tournament also, then $42p$ is equal to

PASSAGES

Passage-I:

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls.

Now, answer the following questions:

Q.(21) If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is p . Then $181p$ is equal to

Q.(22) If 1 ball is drawn from each of the boxes, then the probability that all 3 drawn balls are of the same color is p . Then $324p$ is equal to

Passage-II:

Box 1 contains three cards bearing numbers, 1,2,3 ; box 2 contains five cards bearing numbers 1,2,3,4,5; and box 3 contains seven cards bearing numbers 1,2,3,4,5,6,7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the i^{th} box, $i = 1,2,3$.

Now, answer the following questions:

Q.(23) If p denotes the probability that $x_1x_2 + x_2x_3 + x_3x_1$ is odd, then $15p$ is equal to

Q.(24) If p denotes the probability that $x_1 + x_2 + x_3$ is even, then $105p$ is equal to

MATRIX MATCHING

Q.(25)

Consider an experiment of tossing a fair of coin six times. Let A and B be two events defined as follows

A : Number of heads greater than number of tails.

B : The third toss results in head. Now match the columns.

Column-I
Column-II

- | | |
|---|--|
| (A) If $P(A) = \frac{a}{b}$ where a and b are relatively prime then 'a' is | (P) an even number |
| (B) If $P(B) = \frac{p}{q}$, where p and q are relatively prime then 'q' is | (Q) an odd number |
| (C) If $P\left(\frac{A}{B}\right) = \frac{u}{v}$ where u and v are relatively prime then 'u' is | (R) a prime number |
| (D) If $P\left(\frac{B}{A}\right) = \frac{m}{n}$ where m and n are relatively prime then
3(m + n) is | (S) a composite number
(T) neither prime nor composite number |

KEY

1. C	2. A	3. A	4. B	5. B
6. ABD	7. ABD	8. ABC	9. ABD	10. BCD
11. ABC	12. 2,3	13. BD	14. 15	15. 20
16. 6	17. 24	18. 4	19. 0.50	20. 0.25
21. 55	22. 41	23. 10	24. 52	25. A-QR,B-PR, C-QT,D-QS

GR-16 : COMPLEX NUMBERS

SINGLE CORRECT TYPE QUESTIONS

Q.(1)

Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex numbers z

satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(2)

Let z be a complex number such that the imaginary part of z is nonzero and $a = z^2 + z + 1$ is real.

Then a **cannot** take the value -

- (A) -1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

Q.(3)

Let $z = \frac{-1 + \sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is

- (A) 1 (B) 2 (C) 3 (D) 4

Q.(4)

Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$. If the complex number z_0 is

such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set $\left\{ \frac{1}{|z - 1|} : z \in S \right\}$, then the principal argument of $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$

is

(A) $\frac{\pi}{4}$

(B) $-\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$

(D) $\frac{\pi}{2}$

Q.(5) Let z_1, z_2, z_3 be any 3 complex numbers, such that

$$|z_1| = |z_2| = 2; |z_3| = 1 \text{ and } \left| \arg \left(\frac{z_1 - z_3}{z_2 - z_3} \right) \right| = \frac{\pi}{2}.$$

Then, the difference between the maximum and minimum values of $|z_1 + z_2|$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

MULTIPLE CORRECT TYPE QUESTIONS

Q.(6)

Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$.

Then $P^2 \neq 0$, when $n =$

(A) 57

(B) 55

(C) 58

(D) 56

Q.(7)

Let $w = \frac{\sqrt{3} + i}{2}$ and $P = \{w^n : n = 1, 2, 3, \dots\}$. Further $H_1 = \left\{ z \in \mathbb{C} : \operatorname{Re} z > \frac{1}{2} \right\}$ and $H_2 = \left\{ z \in \mathbb{C} : \operatorname{Re} z < \frac{-1}{2} \right\}$, where $\mathbb{C}$ is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$

and O represents the origin, then $\angle z_1 O z_2 =$

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

Q.(8)

Let $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S = \left\{ z \in \mathbb{C} : z = \frac{1}{a + ibt}, t \in \mathbb{R}, t \neq 0 \right\}$, where $i = \sqrt{-1}$.

If $z = x + iy$ and $z \in S$, then (x, y) lies on

- (A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a > 0, b \neq 0$
 (B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a < 0, b \neq 0$
 (C) the x-axis for $a \neq 0, b = 0$
 (D) the y-axis for $a = 0, b \neq 0$

Q.(9)

Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are TRUE ?

- (A) $\left|z + \frac{1}{2}\right| \leq \frac{1}{2}$ for all $z \in S$ (B) $|z| \leq 2$ for all $z \in S$
 (C) $\left|z + \frac{1}{2}\right| \geq \frac{1}{2}$ for all $z \in S$ (D) The set S has exactly four elements

Q.(10) Let z_1, z_2 be two complex numbers satisfying

$$\frac{z_1 - 1}{z_1 + 1} = \left(\frac{\sqrt[3]{5} + \sqrt[3]{7} - \sqrt[3]{3} - \sqrt[3]{9}}{|\sqrt[3]{5} + \sqrt[3]{7} - \sqrt[3]{3} - \sqrt[3]{9}|} \right) \left(\frac{z_2 - 1}{z_2 + 1} \right)^n \quad (n \in \mathbb{N}).$$

If z_2 lies on the circle $|z| = 1$, then the value of $|z_1 - 1| + |z_1 + 1|$ is

- (A) equal to 2, if $n = 2$ (B) greater than 2, if $n = 3$
 (C) greater than 2, if $n = 4$ (D) greater than 2, if $n = 5$

Q.(11) Let α, β, γ be three real numbers such that

$$e^{i3\alpha} + e^{i3\beta} + e^{i3\gamma} + e^{i(\alpha+\beta+\gamma)} = 0, k = |e^{i\alpha} + e^{i\beta} + e^{i\gamma}|$$

Then, k can become equal to which of the following options?

- (A) 1 (B) 2 (C) $\frac{3}{2}$ (D) 4

Q.(12) Let z and ω be complex numbers satisfying

$$\omega = \frac{z^2 - 1}{z^2 - 2}, \arg\left(\frac{z - \sqrt{2}}{z + \sqrt{2}}\right) = \frac{\pi}{2}.$$

If $\arg z$ denotes the principal value of argument, then which of the following options is/are correct?

- (A) $4|\omega - 3 - i|_{\min} = 9$ (B) $4|\omega - 3 + i|_{\min} = 9$
 (C) $2(\arg \omega)_{\max} = \pi$ (D) $(\arg \omega)_{\min} = 0$

NUMERICAL ANSWER TYPE QUESTIONS

Q.(13)

For a complex number z , let $\text{Re}(z)$ denote the real part of z . Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\text{Re}(z_1) > 0$ and $\text{Re}(z_2) < 0$, is _____

Q.(14) Let ω represents a non-real cube root of unity, z_1 and z_2 are two complex numbers such that there exist positive real numbers a, b, c , such that

$$\frac{z_1}{z_2} = \frac{a + b\omega + c\omega^2}{a + b + c}; |z_1 - z_2| = 4; |z_1 - z_2\omega| = 5; |z_1 - z_2\omega^2| = 3.$$

Then, the value of $|z_2|^2 - 4\sqrt{3}$ is equal to (correct to 2-decimal places)

Q.(15)

If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $|2z - 6 + 5i|$ is

Q.(16) Consider the following sets

$$\mathbb{A} = \left\{ (x, y) \in \mathbb{Z}^2 : x^2 - y^2 = \frac{x}{x^2 + y^2} \right\}, \quad \mathbb{B} = \left\{ (x, y) \in \mathbb{Z}^2 : 2xy + \frac{y}{x^2 + y^2} = 2 \right\},$$

$$\mathbb{C} = \left\{ (x, y) \in \mathbb{Z}^2 : \frac{x^3 - 3xy^2 + 2y + 3x^2}{1 + 3x^2} = 1 \right\}, \quad \mathbb{D} = \left\{ (x, y) \in \mathbb{Z}^2 : \frac{4x^2y - 2x}{x^2 + y^2} = y \right\}.$$

If $|A|$ denotes the cardinal number of set A , then $|\mathbb{A} \cap \mathbb{B}| - |\mathbb{C} \cap \mathbb{D}|$ is equal to

Q.(17) Let z_1 and z_2 be two complex numbers satisfying $|z_1 + z_2| = 20$ and $|z_1^2 + z_2^2| = 16$, then $\max|z_1^3 + z_2^3| - \min|z_1^3 + z_2^3|$ is equal to

Q.(18) Let $w_1, w_2, \dots, w_n$ be the distinct roots of $w^{105} + 1 = w^7(w^{91} + 1)$, and z_1, z_2 be two complex numbers such that

$$\lambda = n + \prod_{r=1}^n (2 + 3w_r^2), \arg(z_1 - z_2) = \frac{\pi}{4}, |z_1 - 3| = \operatorname{Re}(z_1), \text{ and } |z_2 - 3| = \operatorname{Re}(z_2)$$

If $\arg(z)$ denotes principal argument of complex number z , $\{.\}$ denotes the fractional part function, then $\left\{\frac{\lambda}{10}\right\} \cdot \operatorname{Im}(z_1 + z_2)$ is equal to

Q.(19) Let z_1, z_2, z_3 be any three complex numbers satisfying

$$\mathbb{A} = \left\{ |z_1 - z_2|^2 + |z_3 - z_2|^2 + |z_1 - z_3|^2 : |z_1| = \frac{5}{2}, |z_2| = \frac{3}{2}, |z_3| = 2 \right\}.$$

Further, let

$$\lambda = \left| (\alpha_3^{1001})^{2021} + \left(\alpha_4^{669 + \frac{1}{3}} \right)^{2021} + \left(\alpha_5^{501 + \frac{3}{4}} \right)^{2021} \right|; \text{ where } \alpha_k = e^{\frac{2\pi i}{5}(k-1)}; k \in \mathbb{N}.$$

Then the value of $\lambda \cdot \max(\mathbb{A})$ is equal to

Q.(20) If z is any complex number lying on the curve $C: |z^2 - 6| = 6$, and

$$a = \min \left(\sum_{r=0}^4 \sum_{k=0}^{2021} \left| z e^{\frac{ik\pi}{3}} - \frac{e^{\frac{ir\pi}{2}}}{1011} \right| \right),$$

then number of positive divisors of 'a' are equal to

PASSAGES

Passage-I:

Let a, b and c be three real numbers satisfying

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad \dots(E)$$

Now, answer the following questions:

Q.(21)

If the point $P(a, b, c)$, with reference to (E), lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is

Q.(22)

Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying (E),

then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to -

(A) -2

(B) 2

(C) 3

(D) -3

Q.(23)

Let $b = 6$, with a and c satisfying (E). If α and β are the roots of the quadratic equation

$$ax^2 + bx + c = 0, \text{ then } \sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n \text{ is -}$$

MATRIX MATCHING

Q.(24)

Match the statements in **Column-I** with those in **Column-II**.

[Note : Here z takes values in the complex plane and $\text{Im } z$ and $\text{Re } z$ denote, respectively, the imaginary part and the real part of z .]

Column I

Column II

(A) The set of points z satisfying

$$|z - i| = |z + i| \text{ is contained in or equal to}$$

(B) The set of points z satisfying

$$|z + 4| + |z - 4| = 10 \text{ is contained in or equal to}$$

(C) If $|w| = 2$, then the set of points

$$z = w - \frac{1}{w} \text{ is contained in or equal to}$$

(D) If $|w| = 1$, then the set of points

$$z = w + \frac{1}{w} \text{ is contained in or equal to}$$

(p) an ellipse with eccentricity $\frac{4}{5}$

(q) the set of points z satisfying $\text{Im } z = 0$

(t) the set of points z satisfying $|\text{Im } z| \leq 1$

(s) the set of points z satisfying $|\text{Re } z| \leq 2$

(t) the set of points z satisfying $|z| \leq 3$

Q.(25)

Match the statements given in **Column I** with the values given in **Column II**

Column I

Column II

(A) If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}$, $\vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle,

then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is

(B) If $\int_a^b (f(x) - 3x)dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is

(C) The value of $\frac{\pi^2}{\ln 3} \int_{7/6}^{5/6} \sec(\pi x)dx$ is

(D) The maximum value of $\left| \text{Arg} \left(\frac{1}{1-z} \right) \right|$ for

$|z| = 1, z \neq 1$ is given by

(p) $\frac{\pi}{6}$

(q) $\frac{2\pi}{3}$

(r) $\frac{\pi}{3}$

(s) π

(t) $\frac{\pi}{2}$

KEY

1. A

2. D

3. A

4. B

5. B

6. BCD	7. CD	8. ACD	9. BC	10. ABD
11. AB	12. AB	13. 8	14. 8.33	15. 5
16. 0	17. 960	18. 1.8	19. 112.5	20. 4
21. 6	22. B	23. 7	24. A-QR,B-P, C-PST,D-QRST	25. A-Q,B-P/PQRST, C-S,D-T

GR-18 : THREE DIMENSIONAL GEOMETRY**SINGLE CORRECT TYPE QUESTIONS**

Q.(1)

The equation of the plane passing through the point $(1,1,1)$ and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is-

(A) $14x + 2y + 15z = 31$

(B) $14x + 2y - 15z = 1$

(C) $-14x + 2y + 15z = 3$

(D) $14x - 2y + 15z = 27$

Q.(2)

Let P_1 be the plane passing through $(1,1,-1)$ and $(2,6,1)$ and perpendicular to the plane $P_2 : x + 5y + 2z = 0$. Line of intersection of P_1 and P_2 is parallel to vector $\hat{i} - \hat{j} + 2\hat{k}$. If distance between P_1 and y-axis is d then $5d^2$ is

(A) 8

(B) 9

(C) 16

(D) 4

Q.(3)

Let A, B, C, D be four non-coplanar points. Then the number of planes which are equidistant from all the four points is equal to

(A) 8

(B) 4

(C) 6

(D) 7

MULTIPLE CORRECT TYPE QUESTIONS

Q.(4)

Let $A(1,0,0)$, $B(1,1,0)$ are fixed points. If OA (where O is the origin) is rotated through O in the xz -plane by a variable angle ' θ ' to reach the new position OP , then which of the following is (are) true-

- (A) maximum volume of tetrahedron $OAPB$ is $\frac{1}{6}$
- (B) maximum volume of tetrahedron $OAPB$ is $\frac{1}{2}$.
- (C) equation of the plane OAP is $\vec{r} \cdot (\cos \theta \hat{i} + \sin \theta \hat{k}) = 0$
- (D) length of projection of $\overline{OP}$ on $\overline{OB}$ is $\frac{|\cos \theta|}{\sqrt{2}}$

Q.(5)

From point $A(-1,1,-1)$, the perpendicular is drawn to a plane P_1 whose foot is $(1,-1,1)$. From point $B(3,-1,-1)$, the perpendicular is drawn to a plane P_2 whose foot is $(1,2,1)$. P_3 is a plane containing the line of intersection of P_1 and P_2 and passing through the mid point of the line joining A and B . Then which of the following statement(s) is/are correct ?

- (A) Direction ratios of line of intersection of P_1 and P_2 are $5,4,-1$
- (B) Equation of plane P_3 is $16x - 19y + 4z = 12$
- (C) Equation of line of intersection of P_1 and P_2 is $\frac{x-15}{5} = \frac{y-12}{4} = \frac{z-0}{-1}$
- (D) The length of perpendicular from $(0,0,0)$ to plane P_3 is $\frac{2}{\sqrt{18}}$

Q.(6)

A line $\frac{2x-1}{1} = \frac{2y-k}{2} = \frac{2z-3}{2}$ cuts the xy - plane and xz - plane at points A and B respectively. If AB subtends a right angle at the origin, then –

(A) the value of k is 2

(B) the value of k is 3

(C) the area of ΔAOB is $\frac{\sqrt{5}}{8}$ (where O is the origin)

(D) the area of ΔAOB is $\frac{\sqrt{5}}{16}$ (where O is the origin)

Q.(7)

Let L_1 and L_2 be the following straight lines.

$$L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3} \text{ and } L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$$

Suppose the straight line

$$L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$$

lies in the plane containing L_1 and L_2 , and passes through the point of intersection of L_1 and L_2 . If the line L bisects the acute angle between the lines L_1 and L_2 , then which of the following statements is/are TRUE?

(A) $\alpha - \gamma = 3$

(B) $\ell + m = 2$

(C) $\alpha - \gamma = 1$

(D) $\ell + m = 0$

Q.(8) In $\mathbb{R}^3$, let L be a line passing through origin. Suppose that all the points on L are at a constant distance from two planes

$$P_1: x + 2y - z + 1 = 0 \quad \text{and} \quad P_2: 2x - y + z = 1.$$

Let M be the locus of feet of perpendiculars drawn from points on L to plane P_1 . Then which of the

following points lie(s) on M ?

- (A) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$ (C) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (D) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

Q.(9)

Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with $OP = 3$. The point S is directly above the mid-point T of diagonal OQ such that $TS = 3$. Then-

- (A) the acute angle between OQ and OS is $\frac{\pi}{3}$.
- (B) the equation of the plane containing the triangle OQS is $x - y = 0$
- (C) the length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$
- (D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

Q.(10) Given a line L and a plane Π :

$$L \equiv \frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}, \quad \Pi \equiv x + y + z = 3.$$

Let lines L_1 and L_2 respectively be projection and image of line L in plane Π . Let planes Π_1, Π_2 respectively be planes containing L and parallel, perpendicular to Π . Then shortest distance of

- (A) Π_1 from origin is $\sqrt{3}$ (B) L_1 from $(-9, 6, -3)$ is $\sqrt{53}$.
- (C) L_2 from origin is $\sqrt{\frac{285}{7}}$ (D) Π_2 from origin is $\sqrt{\frac{49}{26}}$.

NUMERICAL ANSWER TYPE QUESTIONS

Q.(11)

Three lines are given by

$$\vec{r} = \lambda \hat{i}, \lambda \in \mathbb{R}$$

$$\vec{r} = \mu(\hat{i} + \hat{j}), \mu \in \mathbb{R} \text{ and}$$

$$\vec{r} = v(\hat{i} + \hat{j} + \hat{k}), v \in \mathbb{R}.$$

Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the area of the triangle ABC is Δ then the value of $(6\Delta)^2$ equals ____

Q.(12)

Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is ____.

Q.(13)

Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0, 0, 0) is the origin. Let $S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p} = \overrightarrow{SP}$, $\vec{q} = \overrightarrow{SQ}$, $\vec{r} = \overrightarrow{SR}$ and $\vec{t} = \overrightarrow{ST}$, then the value of $|(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})|$ is ____.

Q.(14) Let $P(1, 1, 0)$, $Q(6, 4, 4)$ and $R(a, b, 2 - a - b)$ be any three points in three-dimensional space. Find the value of $|2\ell^2 + ab|$, if

$$\ell = \max\left(\frac{QP + PR}{QR}\right); a, b \in \mathbb{R}.$$

Q.(15) Let us consider three points in three-dimensional space,

$$A(3, 5, 4), B(4, 3, 5) \text{ and } P(\alpha, \beta, 0).$$

If point P is in such a place, that $\angle APB \in [0^\circ, 180^\circ]$ is maximum, then find the value of λ , where

$$\lambda = 10 \left(\frac{\alpha + 1}{13 - \beta} \right) + [\alpha] + [\beta].$$

Q.(16) From point $P(x, y, z)$ three mutually perpendicular lines L, M, N are drawn. Line L intersects x -axis at A , M intersects y -axis at B , and N intersects z -axis at C . If the volume of tetrahedron $PABC$ is $\frac{4}{3}$, then locus of point P is

$$(x^2 + y^2 + z^2)^\lambda = \mu xyz.$$

The value of $3\lambda + \mu$ is equal to

Q.(17) $ABCD$ is a regular tetrahedron of volume 1 *cu. units*. Points A', B', C' and D' are 4 points in space opposite to faces BCD, DCA, CAB and ABC respectively, such that $A'BCD, AB'CD, ABC'D, ABCD'$ are also regular tetrahedrons. If V denotes the volume of tetrahedron $A'B'C'D'$, then the value of $54V$ is equal to

PASSAGES

Passage-I:

Given 4 lines $L_i; i = 1, 2, 3, 4$ as follows:

$$L_1: \vec{r} = (t + 3)\hat{i} + (2t - 1)\hat{j} + (2t + 4)\hat{k}; t \in \mathbb{R}, \quad L_3: x = 5, \frac{y}{3 - \alpha} = \frac{z}{-2},$$

$$L_2: \vec{r} = (2s + 3)\hat{i} + (2s + 3)\hat{j} + (s + 2)\hat{k}; s \in \mathbb{R}, \quad L_4: x = \alpha, \frac{y}{-1} = \frac{z}{2 - \alpha}.$$

Now, answer the following questions:

Q.(18)

A line L passing through origin and perpendicular to L_1 and L_2 intersects L_1 at A . Point B lies on L_2 such that $AB = \sqrt{17}$. Then possible coordinates of B is/are

(A) $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3} \right)$

(B) $(-1, -1, 0)$

(C) $(1, 1, 1)$

(D) $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9} \right)$

Q.(19) If lines L_3 and L_4 are collinear, then which of the following options is/are possible value(s) of α ?

(A) 1

(B) 2

(C) 3

(D) 4

MATRIX MATCHING

Q.(20)

Consider the lines $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$, $L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes $P_1 : 7x+y+2z=3$, $P_2 : 3x+5y-6z=4$. Let $ax+by+cz=d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 and P_2 .

Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

 P. $a =$

 Q. $b =$

 R. $c =$

 S. $d =$

List-II

1. 13

2. -3

3. 1

4. -2

KEY

1. A	2. B	3. D	4. AD	5. ABC
6. AD	7. AB	8. AB	9. BCD	10. ABD
11. 0.75	12. 8	13. 0.5	14. 19	15. 11
16. 73	17. 250	18. BD	19. AD	20. P-3, Q-2, R-4, S-1

GR-19 : MATRICES AND DETERMINANTS**SINGLE CORRECT TYPE QUESTIONS**

Q.(1): Let M denotes the set of all 2×2 matrices. Define the relation

$$R = \{(A, B) \in M \times M \mid A + B^T \text{ is a symmetric matrix}\}.$$

Then R is

- (A) symmetric, transitive but not reflexive relation
- (B) reflexive, symmetric but not transitive relation
- (C) reflexive, transitive but not symmetric relation
- (D) an equivalence relation

Q.(2): If A, B, C are three matrices of order 2, such that

$$|A + B| - |C| = -10,$$

$$|B + C| - |A| = 25,$$

$$|C + A| - |B| = 15,$$

then the value of $|A + B + C|$ is equal to ($|A|$ denotes the determinant of matrix A)

- (A) 15 (B) 30 (C) 45 (D) 60

Q.(3): If $|A|$ denotes the determinant of matrix A , where A is a square matrix of order 3, satisfying

$$\text{tr}(A) = 4, \text{tr}(A^2) = 10, \text{and } \text{tr}(A^3) = 28,$$

then find the value of $|A|$.

- (A) 1 (B) 2 (C) 3 (D) 0

MULTIPLE CORRECT TYPE QUESTIONS

Q.(4):

Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$,
 $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$ [JEE(Advanced)-2019]

where P_k^T denotes the transpose of the matrix P_k . Then which of the following options is/are correct?

- (1) $X - 30I$ is an invertible matrix (2) The sum of diagonal entries of X is 18

(3) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$

- (4) X is a symmetric matrix **MT0144**

Q.(5):

Let $x \in \mathbb{R}$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$.

Then which of the following options is/are correct ?

(1) For $x = 1$, there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

- (2) There exists a real number x such that $PQ = QP$

(3) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

(4) For $x = 0$, if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then $a + b = 5$

Q.(6): Which of the following option(s) represents a possible value k , satisfying

$$\begin{bmatrix} 1 + \frac{1}{\sqrt{3}} & 1 - \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} - 1 & 1 + \frac{1}{\sqrt{3}} \end{bmatrix}^{2021} = \left(\frac{2}{\sqrt{3}}\right)^{\frac{6063-3k}{2}} \begin{bmatrix} 1 - \frac{1}{\sqrt{3}} & 1 + \frac{1}{\sqrt{3}} \\ -1 - \frac{1}{\sqrt{3}} & -1 + \frac{1}{\sqrt{3}} \end{bmatrix}^k.$$

(A) 12173

(B) 48413

(C) 12217

(D) 24265

Q.(7): Given that a, b, c are non-zero real and distinct constants, such that

$$\begin{vmatrix} x^2 - a^2 & x^2 - b^2 & x^2 - c^2 \\ (x - a)^3 & (x - b)^3 & (x - c)^3 \\ (x + a)^3 & (x + b)^3 & (x + c)^3 \end{vmatrix} = 0.$$

If $a = 11, b = 12, c = 13$, then the given equation has m solutions.

If $a = 10, b = -12, c = 3$, then the given equation has n solutions.

If $a = 60, b = -12, c = 15$ then the given equation has p solutions.

Which of the following options is/are correct?

- (A) $2m + 3n = 9$ (B) $3n + 4p = 7$ (C) $5m - 6p = 4$ (D) $3n + 4p = 4$

Q.(8): Let $a, b, c \in \mathbb{R}, b \neq c \neq 0$, such that

$$\begin{vmatrix} bc - 4a^2 & 2ac - b^2 & 2ab - c^2 \\ 2ac - b^2 & 2ab - c^2 & bc - 4a^2 \\ 2ab - c^2 & bc - 4a^2 & 2ac - b^2 \end{vmatrix} = 0.$$

Then, which of the following options is/are correct for all possible values of a, b, c ?

- (A) $24ax^2 + 4bx + c = 0$ has at least one root in $\left(0, \frac{1}{2}\right)$
 (B) $ax + by + c = 0$ always passes through a fixed point
 (C) $ax^2 + bx + c = 0$ has at least one root in $(0, 2)$
 (D) $6ax^2 + 2bx + c = 0$ has at least one root in $(0, 1)$

Q.(9):

Let A and B are any two square matrices of order 2×2 , with real entries, such that

$$AB = A^2 B^2 - (AB)^2 \text{ and } |B| = 3.$$

If $|A|$ denotes the determinant of matrix A , then which of the following options is/are correct?

- (A) $|A + I| = 1$ (B) $|A + 3B| - |3A + B| = 24$
 (C) $|2A + B| - |A + 2B| = 9$ (D) $\text{adj}(AB) = O$.

Q.(10): Let A be a square matrix of order 3, satisfying $A^{19} = A + I$, then which of the following options is/are correct ($|A|$ denotes the determinant of matrix A)?

- (A) $|A^{38} - I| > 0$ (B) $|A - I| < 0$ (C) $|A + I| < 0$ (D) $|A| > 0$

Q.(11): For 3×3 matrices M and N , which of the following statement(s) is(are) NOT correct?

- (A) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric.
 (B) $MN - NM$ is skew symmetric for all symmetric matrices M and N .

- (C) MN is symmetric for all symmetric matrices M and N .
 (D) $\text{adj } M \text{ adj } N = \text{adj } MN$ for all square matrices M and N of same order.

NUMERICAL ANSWER TYPE QUESTIONS

Q.(12): The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is -18 , then the value of the determinant of A is

Q.(13): Let $M = \begin{bmatrix} \sin^4 x & -1 - \sin^2 x \\ 1 + \cos^2 x & \cos^4 x \end{bmatrix} = f(x)I_2 + g(x)M^{-1}$, where I_2 is an identity matrix of order 2,

$$f_m = \min\{f(x) | x \in [0, 2\pi)\}, \quad \text{and} \quad f_M = \max\{f(x) | x \in [0, 2\pi)\}.$$

Then the value of $1 - 16(f_m + f_M)$ is equal to

Q.(14): How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5?

Q.(15): For a real number 'a', if the system

$$\begin{bmatrix} 1 & a & a^2 \\ a & 1 & a \\ a^2 & a & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

of linear equations, has infinitely many solutions, then $1 + a + a^2$ equals

Q.(16):

Let β be a real number. Consider the matrix

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$$

If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____.

Q.(17): If a, b, c, x_i, y_i, z_i are real numbers ($i = 1, 2, 3$) satisfying

$$\begin{pmatrix} ax_1^2 + by_1^2 + cz_1^2 = 36, \\ ax_2^2 + by_2^2 + cz_2^2 = 25, \\ ax_3^2 + by_3^2 + cz_3^2 = 20, \end{pmatrix}, abc = 8, \begin{pmatrix} ax_2x_3 + by_2y_3 + cz_2z_3 = 21, \\ ax_1x_3 + by_1y_3 + cz_1z_3 = 20, \\ ax_2x_1 + by_2y_1 + cz_2z_1 = 23 \end{pmatrix},$$

then find the value of $\begin{vmatrix} y_2z_3 - y_3z_2 & x_3z_2 - x_2z_3 & y_3x_2 - y_2x_3 \\ y_3z_1 - y_1z_3 & x_1z_3 - x_3z_1 & y_1x_3 - y_3x_1 \\ y_1z_2 - y_2z_1 & x_2z_1 - x_1z_2 & y_2x_1 - y_1x_2 \end{vmatrix}$.

Q.(18): Let $A = [a_{ij}]_{3 \times 3}$, and $adj A = [c_{ij}]_{3 \times 3}$ be two matrices satisfying

$$AA^T = 4I_3, 2a_{ij} + c_{ij} = 0 \forall i, j \text{ and } \lambda = 3|A + I_3| + |A + 2I_3| + |A + 3I_3|.$$

Then the value of $|\lambda|$ is equal to

PASSAGES

Passage-I:

Let

$$X = \left\{ [a_{ij}]_{3 \times 3} : a_{ij} = a_{ji}, a_{ij} \in \{0,1\}, \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} = 5 \right\},$$

$$Y = \left\{ [b_{ij}]_{3 \times 3} : B = [b_{ij}]_{3 \times 3}, B \in X, B \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has unique solution} \right\},$$

$$Z = \left\{ [c_{ij}]_{3 \times 3} : C = [c_{ij}]_{3 \times 3}, C \in X, C \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has NO solution} \right\},$$

$$T = \left\{ [d_{ij}]_{3 \times 3} : D = [d_{ij}]_{3 \times 3}, D \in X, D \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has infinite solutions} \right\}.$$

Now, answer the following questions:

Q.(19): Which of the following options is/are correct?

- (A) $n(X) = 12$ (B) $n(Y) = 6$ (C) $n(Z) = 5$ (D) $n(T) = 3$

Q.(20): Which of the following options is/are correct?

- (A) $n(X) = 13$ (B) $n(Y) = 7$ (C) $n(Z) = 4$ (D) $n(T) = 2$

Passage-II:

If A and B are two square matrices of order n, then answer the following questions:

Q.(21): If A, B and A+B are idempotent matrices, then

- (A) there exists A, B such that $AB \neq O$. (B) $BA = O$
 (C) there exists A, B such that $BA \neq O$. (D) $AB = O$

Q.(22): Which of the following options is/are correct?

- (A) If A and B are orthogonal matrices, then BA is also orthogonal.
 (B) If A and B are orthogonal matrices, then AB is also orthogonal.
 (C) If $A + B = I$, and A is an idempotent matrix, then $AB = BA = O$.
 (D) If $A^2 + B^2 = AB$, then $\text{tr}(ABAB) = \text{tr}(A^2B^2)$.

MATRIX MATCHING

Q.(23): If matrices A and B are defined as

$$A_{n,m} = [a_{ij}]_{n \times n}, a_{ij} = \begin{cases} n, & i = j \\ m, & i \neq j \end{cases} \text{ and } B = \begin{bmatrix} -bc & b^2 + bc & b^2 + bc \\ c^2 + ca & -ca & c^2 + ca \\ a^2 + ab & a^2 + ab & -ab \end{bmatrix},$$

where a, b, c are roots of $3x^3 + 11x^2 - 21x - 29 = 0$. Then, match the following columns

Column-I

- (A) $\gcd(|\det A_{6,3}|, |\det B|)$
 (B) $\gcd(|\det A_{5,15}|, |8 + \det B|)$
 (C) $\gcd(|\det A_{11,7}|, |2 - \det B|)$
 (D) $\gcd(|\det A_{8,11}|, |2 + \det B|)$

Column-II

- (P) 1
 (Q) 3
 (R) 5
 (S) 7

KEY

1. D	2. B	3. D	4. 234	5. 3,4
6. CD	7. AB	8. ABCD	9. AB	10. AD
11. CD	12. 7	13. 30	14. 198	15. 1
16. 3	17. 108	18. 2	19. AB	20. CD
21. BD	22. ABCD	23. A-S, B-R, C-Q, D-P	24.	25.

GR-20 : BINOMIAL THEOREM, STATISTICS, SETS AND RELATIONS

SINGLE ANSWER TYPE QUESTIONS

Q.(1)

For the frequency distribution :

Variate (x) : $x_1 \quad x_2 \quad x_3 \dots x_{15}$

Frequency (f) : $f_1 \quad f_2 \quad f_3 \dots f_{15}$

where $0 < x_1 < x_2 < x_3 < \dots < x_{15} = 10$ and $\sum_{t=1}^{15} f_t > 0$, the standard deviation cannot be

(1) 1

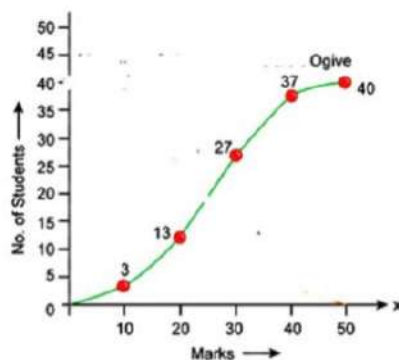
(2) 4

(3) 2

(4) 6

Q.(2)

Consider the Continuous frequency distribution of marks of 40 students plotted in Ogive curve where Cumulative frequency (Number of students obtained less than the required marks) is plotted along y axis and Marks (in class size of 10) are plotted along x axis.



Find median of the distribution.

(A) 28

(B) 26

(C) 25

(D) 27.5

Q.(3) Which of the following option is equal to $a_{11} + b_9$, where

$$(1+x^2)^4 \cdot (1+x^3)^7 \cdot (1+x^4)^{12} = \sum_{r=0}^{77} a_r x^r; a_r \in \mathbb{R}, \text{ and}$$

$$(1+x)(1+x^2)(1+x^3) \dots (1+x^{100}) = \sum_{r=0}^{5050} b_r x^r; b_r \in \mathbb{R}.$$

(A) 1059

(B) 1114

(C) 1121

(D) 1128

MULTIPLE ANSWER TYPE QUESTIONS

Q.(4) Let $\{.\}$ denotes the fractional part function,

$$S_n = {}^nC_0^2 + 1 \cdot \frac{{}^nC_1^2}{{}^nC_0} + 2 \cdot \frac{{}^nC_2^2}{{}^nC_1} + 3 \cdot \frac{{}^nC_3^2}{{}^nC_2} + \dots + 1 \quad \text{and} \quad R(n, m) = m \left\{ \frac{n}{m} \right\}.$$

Then which of the following options is/are correct?

(A) $R(S_{10}, 7) = 2$

(B) $R(S_{102}, 13) = 11$

(C) $R(S_{13}, 13) = 2$

(D) $R(S_{100}, 17) = 15$

Q.(5) Let m be the median of the data set ${}^{10}C_0, {}^{10}C_1, {}^{10}C_2, \dots, {}^{10}C_{10}$.

If $x_i \in \mathbb{R}; i = 1, 2, 3, \dots, n$,

$$\sum_{r=1}^n x_i = 80 \quad \text{and} \quad \sum_{r=1}^n x_i^2 = 400.$$

then n iCANNOT be equal to

(A) $m - 30$

(B) $m - 33$

(C) $m - 27$

(D) $m - 35$

Q.(6) Let $a_i \in \mathbb{R} \forall i = 0, 1, 2, \dots, 100$, such that

$$(1 + 2x + 3x^2)^{50} = a_0 + a_1x + a_2x^2 + \dots + a_{100}x^{100}.$$

Then, which of the following options is/are correct?

(A) $a_{51} : a_{49} = 153 : 51$

(B) $a_{53} : a_{47} = 153 : 51$

(C) $11a_{11} = 80a_{10} + 273a_9$

(D) $27a_{27} = 48a_{26} + 225a_{25}$

Q.(7)

Consider a discrete frequency distribution.

Variable	X_1	X_2	X_3		X_n
Frequency	f_1	f_2	f_3		f_n

 Where $X_i, f_i \in \mathbb{R}^+ \forall i = 1, 2, 3, \dots, n$

 If deviation of X_i from mean M say ' x_i ' ($x_i = X_i - M$) is small compared to mean such

 that $\left(\frac{x_i}{M}\right)^3$ and higher powers of $\frac{x_i}{M}$ can be neglected $\forall i = 1, 2, 3, \dots, n$.

 If G is geometric mean and σ is standard deviation of the distribution, then

A) $G = M \left(1 - \frac{\sigma^2}{M^2}\right)$

B) $G = M \left(1 - \frac{\sigma^2}{2M^2}\right)$

C) $M^2 - \sigma^2 = G^2$

D) $\sigma^2 - M^2 = G^2$

Q.(8) Let $a \in \mathbb{Z}$ such that

$$\frac{2x^2 + ax + 2}{x^2 + x + 1} = \frac{e^y}{[y + 1]}$$
 is true for all $x \in \mathbb{R}$ and $y \geq 0$.

 If $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$, then which of the following options is/are correct?

 (A) A possible value of a is 3.

 (B) A.M. of possible values of a equals 1

 (C) Variance of possible values of a equals 2

 (D) A.M. of possible values of a equals 1

Q.(9) Consider the following statements

 S-1: The mean of 5 distinct integers is also an integer. Let their variance lie in the interval $(5, 6)$ and μ denote their mean-deviation about the mean.

S-2: The standard deviation of a set of data representing temperatures in °C is 5. Let σ denote the corresponding standard deviation when the data were converted into °F.

Then which of the following options is/are correct?

- (A) $\mu + \sigma = 11$ (B) $\sigma - \mu = 7$ (C) $2\sigma + 3\mu = 25$ (D) $3\sigma + 4\mu = 25$

Q.(10)

95% of the students in a class have taken Marketing, 80% have chosen Finance, 84% have chosen operations (ops), and 90% have chosen Human Resources (HR). Then, which of the following options is/are correct?

- (A) Minimum percentage of people who have chosen all of the four is 49%
 (B) Minimum percentage of people who have chosen all of the four is 50%
 (C) Maximum percentage of people who have chosen all of the four is 84%
 (D) Maximum percentage of people who have chosen all of the four is 80%

Q.(11)

Of 60 students in a class, anyone who has chosen to study maths elects to do physics as well. But no one does maths and chemistry, 16 do physics and chemistry. All the students do at least one of the three subjects and the number of people who do exactly one of the three is less than or equal to the number who do more than one of the three. Then which of the following options is/are correct?

- (A) Maximum number of people doing Chemistry only is 30.
 (B) Minimum number of people doing Physics is 30.
 (C) Minimum number of people doing Physics and Maths is 14.
 (D) Maximum number of people not doing Maths is 44.

Q.(12) Let $A = \{2,3,4\}$ and $B = \{8,9,12\}$. The relation R is defined as

$$R = \left\{ ((a_1, b_1), (a_2, b_2)) \in (A \times B, A \times B) \mid \frac{b_2}{a_1} \in \mathbb{N}, \frac{b_1}{a_2} \in \mathbb{N} \right\}.$$

Further let $C = \{1,2,3, \dots, 10\}$ and $D = \{0,1,2,3,4\}$. The relation S is defined as

$$S = \left\{ (a, b) \in A \times A \mid 2(a - b)^2 + 3(a - b) \in B \right\}.$$

If $|X|$ denotes the cardinal number of set X , then

- (A) $|R| = 2|S|$ (B) $|R| + 2|S| = 54$ (C) $|R| - |S| = 18$ (D) $2|R| + |S| = 72$

Q.(13) If $[.]$ denotes the greatest integer function, $\binom{m}{n} = \frac{m!}{n!(m-n)!}$,

$$\mathbb{A} = \left\{ n \in \mathbb{N} : 2^n \left[2^{-n} \cdot \sum_{r=0}^9 (-1)^r \cdot 3^{9-r} \cdot \binom{20}{k} \cdot \binom{20-k}{9-k} \right] = \sum_{r=0}^9 (-1)^r \cdot 3^{9-r} \cdot \binom{20}{k} \cdot \binom{20-k}{9-k} \right\},$$

$$(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r \text{ and } (3x^2 + 4x + 9)^{10} = \sum_{r=0}^{20} b_r x^r,$$

then which of the following options is/are correct?

(A) $a_6 \max(\mathbb{A}) = 192a_{14}$

(B) $a_7 \max(\mathbb{A}) = 96a_{13}$

(C) $27b_{13} \max(\mathbb{A}) = 4b_7$

(D) $b_6 \max(\mathbb{A}) = 972b_{14}$

Q.(14)

Which of the following is (are) correct

A) The Mean of the numbers a,b,8,5,10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the Mean, then 25M is equal to 60

B) The number of values of $a \in N$ such that the variance of 3,7,12,a,43-a is a natural number is 0

C) The Mean and standard deviation of 15 observations are found to be 8 and 3 respectively on rechecking it was found that, in the observations 20 was misread as 5 then the correct variance is equal to 17

D) Let the Mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$

respectively. If the mean and variance of the first 4 observations are $\frac{7}{2}$ and a

respectively, then $(4a + x_5)$ is equal to 18

NUMERICAL ANSWER TYPE QUESTIONS

Q.(15)

A club has 280 members of whom 10 can play all of football, tennis and cricket. Moreover, 90 members can play exactly one game, 180 can play exactly 2 games and 90 can play **football** or tennis but not cricket. If 120 members cannot play football and 100 members can play cricket but not football, then number of members who can play football is equal to

Q.(16)

Students in a class have to choose at least one from chemistry, mathematics and physics. The number of students choosing all three subjects is 18, choosing mathematics as one of their subjects is 23, choosing mathematics as one of their subjects is 27 and choosing physics as one of their subjects is 25. Also, 10 students chose exactly two subjects. Let m be maximum possible number of students who could choose either chemistry or physics as one of their subjects, and n be maximum possible number of students who could choose either chemistry or mathematics as one of their subjects, then $2m + 3n$ is equal to

Q.(17)

If $|A \cup B \cup C| = 250, |A \cap B \cap C| = 15 = |\bar{A} \cap B \cap C| - 5 = |A \cap \bar{B} \cap C|, |(A \cup B) \cap \bar{C}| = 170, |A \cap \bar{B} \cap \bar{C}| < 75, |B \cap \bar{A} \cap \bar{C}| < 75, 2|A| = |B| + |C|$, then find the maximum possible value of $|A \cup B \cup C| - |A \cap B| - |C \cap B| - |A \cap C| + 2|A \cap B \cap C|$.

Q.(18) Let

$$(1+x)^2 + (1+x)^3 + (1+x)^4 + \dots + (1+x)^{49} + (1+mx)^{50} = \sum_{r=0}^{50} a_r x^r; a_r \in \mathbb{R}.$$

Also, let m be the smallest possible positive integer such that

$$\frac{a_2}{20825} = (3n+1) \text{ for some } n \in \mathbb{N}.$$

Then the value of n is equal to

Q.(19) The marks obtained by 40 students are grouped in a frequency table in class intervals of 10 marks each. The mean and variance obtained from this distribution are found to be 40 and 49 respectively. It was later discovered that the two observations belonging to class interval (21-30)

were included in the the class interval (31-40) by mistake. Then the mean after correcting the error is equal to

Q.(20)

A construction company has three different sites located in the city. Analysis of the daily wages paid to the labourers in the three sites are as follows.

	Site 1	Site 2	Site 3
No. of labourers	20	25	40
Average daily wages (Rs)	305	300	340
Standard deviation (Rs)	50	40	45

If the standard deviation of the wages of all the 85 labourers in the city is given by $\text{Rs } \sqrt{\frac{p}{q}}$,

where $p, q \in \mathbb{N}, H.C.F(p, q) = 1$, then find the value of $\left[\frac{p}{q} - 2362 \right]$

(where $[.]$ denotes the greatest integer function)

Q.(21) Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression with $x_3 = \frac{3}{4}$ and $x_1 = 3$. A new data set is constructed by replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ denotes the mean of new data then $\lambda = \max\{a \in \mathbb{Z} : a \leq \bar{x}\}$ is equal to

Q.(22) Let α and β are distinct reals satisfying

$$\sum_{i=1}^{18} (x_i - \alpha) = 36 \quad \text{and} \quad \sum_{i=1}^{18} (x_i - \beta)^2 = 90.$$

If the standard deviation of $x_1, x_2, x_3, \dots, x_{18}$ is 1, then the value of $|\alpha - \beta|$ is equal to

Q.(23) Let σ_{a_i} denote the standard deviation of observations $a_1, a_2, a_3, \dots, a_n$ and $\bar{a}$ denote their arithmetic mean. Given that $n = 20$,

$$\sigma_{x_i} = 4, \quad \sigma_{y_i} = 3 \quad \text{and} \quad \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 90.$$

Then $\sigma_{x_i - y_i}$ is equal to

Q.(24) Suppose for some positive integer n , we have

$$\det \begin{bmatrix} \sum_{k=0}^n k & \sum_{k=0}^n ({}^nC_k \cdot k^2) \\ \sum_{k=0}^n ({}^nC_k \cdot k) & \sum_{k=0}^n ({}^nC_k \cdot 3^k) \end{bmatrix} = 0 \quad \text{and} \quad \lambda = \sum_{k=0}^n \frac{{}^nC_k}{k+1}.$$

Then the value of $\left(\frac{\lambda}{31} + \frac{1}{2860} \left(({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + \dots + 10({}^{10}C_{10})^2 \right) \right)^{\frac{1}{3}}$ is equal to

Q.(25) The value of

$$\lambda = \log_8 \left(2^{50} \binom{50}{0} + 2^{49} \binom{51}{1} + 2^{48} \binom{52}{2} + \dots + 2^0 \binom{100}{50} \right),$$

is equal to (correct to 2-decimal places) Note: $\binom{n}{r}$ represents the value of nC_r

PASSAGES

Passage-I:

The first of the two samples has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$

Q.(26)

The mean of the second group is

A) 16

B) 4

C) 8

D) 2

Q.(27)

The standard deviation of the second group is

- A) 16 B) 8 C) 4 D) 2

Passage-II:

A shop sells three type of products i.e. Pen, Pencil & Notebook. On a survey for checking the sales of each product of the shop he found that the number of people who bought only pen, only pencil, & only notebook are such that an A.P. can be formed from them. Similarly, an A.P. can be formed from the numbers corresponding to exactly two of the three products. It was also found that the number of people who bought all the products is $\left(\frac{1}{20}\right)^{th}$ of the number of people who bought pencil only, which in turn is equal to half of the number of people who bought notebook only. The number of people that bought both pen & pencil is 15, whereas that of those who bought pencil & notebook is 19. The number of people who bought notebook are 120, which is more than the number of people who bought pen (which is a 2 digit number above 50).

Q.(28)

What is the total no people that visits the shop?

Q.(29)

How many people bought both pen and notebook?

MATCHINGS

Q.(30) Match the digit in units place of the answers in Column-I with correct option in Column-II.

Column-I

 (A) Let $\mathbb{A} = \{1,2,3,4\}$ and $\mathbb{R}$ be a relation on the set

 $\mathbb{A} \times \mathbb{A}$ defined by $\mathbb{R} = \{(a, b), (c, d) \mid 2a + 3b = 4c + 5d\}$.

 Then number of elements in $\mathbb{R}$ is equal to

 (B) Let $\mathbb{A} = \{0,3,4,6,7,8,9,10\}$ and $\mathbb{R}$ be a relation defined on $\mathbb{A}$
Column-II

(P) 9

(Q) 6

such that $\mathbb{R} = \{(x, y) | x - y \text{ is odd positive or } 2\}$. Then minimum number of elements that must be added to $\mathbb{R}$ so that it is a symmetric relation is equal to

(C) Let $\mathbb{A} = \{1, 3, 4, 6, 9\}$ and $\mathbb{B} = \{2, 4, 5, 8, 10\}$. Let $\mathbb{R}$ be a relation (R) 7

such that $\mathbb{R} = \{(a, b), (c, d) | a \leq d, b \leq c\}$. Then the number of elements in $\mathbb{R}$ is equal to

(D) Let $\mathbb{A} = \{-4, -3, -2, 0, 1, 3, 4\}$ and relation $\mathbb{R}$ be defined as (S) 2

$\mathbb{R} = \{(a, b) \in \mathbb{A} \times \mathbb{A} | b = |a| \text{ or } b^2 = a + 1\}$. Then the minimum number of elements that must be added to relation $\mathbb{R}$ such that it becomes reflexive and symmetric is equal to

KEY

1. D	2. A	3. C	4. AC	5. ABD
6. ACD	7. BC	8. ABC	9. AB	10. AD
11. ABC	12. ABC	13. ABD	14. ABC	15. 160
16. 145	17. 137	18. 5	19. 49.25	20. 8
21. 142	22. 4	23. 4	24. 7	25. 33.33
26. A	27. C	28. 233	29. 23	30. A-Q, B-P, C-S, D-R

GR-21 : TRIGONOMETRIC IDENTITIES AND EQUATIONS

SINGLE CORRECT

Q.(1)

Number of integral values of x which satisfies equation $\frac{\sin^3 x - \cos^3 x}{\sin x + \cos x} = \frac{\operatorname{cosec}^2 x - \cot x - 2 \cos^2 x}{1 - \cot^2 x}$, $x \in [0, 2\pi]$, is-

- (A) 0 (B) 3 (C) 6 (D) 7

Q.(2)

If $4\cos\theta\cos2\theta\cos3\theta = 1$, then number of values of $\theta \in [0, \pi]$ are -

- (A) 9 (B) 5 (C) 6 (D) 8

Q.(3) $\left\{x \in \left[-\frac{3\pi}{14}, \frac{45\pi}{14}\right] : \sin^2 x + \sin x \cos x = \frac{1}{\sqrt{2}}(\sin 2x + 1)\right\}$.

Then which of the following options is/are correct?

- (A) $|A \cap \overline{B}| = |A|$ (B) $|A \cap \overline{B}| = |B|$ (C) $|A - B| > |B - A|$ (D) $A \cap B = A \cup B$

Q.(4) Let $[x]$ denotes the greatest integer $\leq x$. If

$$2 \sin(x + y) = \frac{2}{3} = \cos(x - y); -\frac{\pi}{4} < y < 0 < x < \frac{\pi}{4},$$

$$\lambda = \frac{\sin x}{\cos y} + \frac{\sin y}{\cos x} + \frac{\cos x}{\sin y} + \frac{\cos y}{\sin x},$$

Then $[\lambda^2]$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

MULTIPLE CORRECT

Q.(5)

If solution of the equation $\frac{\sqrt{6}-\sqrt{2}}{\sin x} + \frac{\sqrt{6}+\sqrt{2}}{\cos x} = 8$ in $x \in \left(0, \frac{\pi}{2}\right)$ is $\frac{m\pi}{n}$ (where m & n are coprime

numbers), then (m + n) can be

- (A) 23 (B) 39 (C) 47 (D) 13

Q.(6)

Let $-\frac{\pi}{2} < \alpha_1 < \alpha_2 < \alpha_3 < \dots < \alpha_n < 2\pi$ and $0 < \beta_1 < \beta_2 < \dots < \beta_m < 2\pi$ are values of α and β respectively such that $2\cos 2\alpha - 3 = 2\cos(2\alpha + \beta) - 2\cos \beta$, then which of the following is(are) true ?

- (A) $2n > 3m$ (B) $n > m$ (C) $\sum_{i=1}^m \beta_i \leq 2\pi$ (D) $\sum_{i=1}^m \beta_i \leq \sum_{i=1}^n \alpha_i$

Q.(7)

Consider the following statements. Then which of the following is/are correct -

- (A) If A, B, C are in A.P. & $B = \frac{\pi}{4}$ then $\tan A \cdot \tan B \cdot \tan C = 1$ (where $A, B, C \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{I}$)
 (B) $\sin 55^\circ + \sin 5^\circ = \sin 65^\circ$
 (C) If $\cos \theta + \sec \theta = 2$ then $\sin \theta = 0$
 (D) The minimum value of the expression $(3\sin x - 4\cos x - 10)(3\sin x + 4\cos x - 10)$ is 49.

Q.(8)

For a positive integer n, let $f_n(\theta) = \left(\tan \frac{\theta}{2}\right)(1 + \sec \theta)(1 + \sec 2\theta)(1 + \sec 4\theta) \dots (1 + \sec 2^{n-1}\theta)$

- (A) $f_2\left(\frac{\pi}{16}\right) = 1$ (B) $f_3\left(\frac{\pi}{32}\right) = 1$ (C) $f_4\left(\frac{\pi}{64}\right) = 1$ (D) $f_5\left(\frac{\pi}{128}\right) = 1$

Q.(9)

If A,B,C are the angles of a triangle such that $\cos^2 A + \cos^2 B + \sin^2 C - \sqrt{2} \cos A - \sqrt{2} \sin C + 1 = 0$, then

(A) $\sin^2 A + \sin^2 B + \cos^2 C = 1$

(B) $\sin^2 A + \sin^2 B + \cos^2 C = 2$

(C) ΔABC is a right angled Δ

(D) ΔABC is an isosceles triangle.

Q.(10) Consider the following sets

$$A(\alpha) = \left\{ 60 \cos x \left(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha} \right) : x \in \mathbb{R} \right\}; \alpha \in [0, \pi].$$

Then, which of the following options is/are correct?

(A) $A\left(\sin^{-1} \frac{3}{4}\right) = [-75, 75]$

(B) $A\left(\sin^{-1} \frac{4}{5}\right) = [-100, 100]$

(C) $A\left(\sin^{-1} \frac{5}{12}\right) = [-65, 65]$

(D) $A\left(\sin^{-1} \frac{12}{13}\right) = [-156, 156]$

Q.(11) Consider the following matrix formed by functions f and g .

$$[p_{ij}]_{1 \leq i, j \leq 2} = \begin{bmatrix} (f(x))^{f(x)} & (f(x))^{g(x)} \\ (g(x))^{f(x)} & (g(x))^{g(x)} \end{bmatrix}.$$

Then which of the following options is/are correct?

(A) If $f(x) = \tan x, g(x) = \cot x$, then

$$p_{22} > p_{21} > p_{11} > p_{12} \quad \forall x \in \left(0, \frac{\pi}{4}\right).$$

(B) If $f(x) = \tan x, g(x) = \cot x$, then

$$p_{22} < p_{21} < p_{11} < p_{12} \quad \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right).$$

(C) If $f(x) = \sin x, g(x) = \cos x$, then

$$p_{21} > p_{22} > p_{11} > p_{12} \quad \forall x \in \left(0, \frac{\pi}{4}\right).$$

(D) If $f(x) = \sin x, g(x) = \cos x$, then

$$p_{21} < p_{22} < p_{11} < p_{12} \quad \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right).$$

Q.(12) Let sets $\mathbb{A}$ and $\mathbb{B}$ are defined as

$$\mathbb{A} = \left\{ \left(\frac{1 + \sin x}{\cos x} + \frac{\cos y}{1 - \sin y} \right) \in \mathbb{R} : x, y \in (0, \pi) \right\},$$

$$\mathbb{B} = \left\{ \frac{2(\sin x - \sin y)}{\sin(x - y) + \cos x - \cos y} \in \mathbb{R} : x, y \in (0, \pi) \right\}.$$

Then which of the following options is/are correct?

(A) $\mathbb{A} \subseteq \mathbb{B}$

(B) $\mathbb{B} \subseteq \mathbb{A}$

(C) $\mathbb{A} \neq \mathbb{B}$

(D) $\mathbb{A} \cap \mathbb{B} = \emptyset$

NUMERICAL TYPE

Q.(13)

Determine the smallest positive value of x (in degrees) for which

$$\tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ).$$

Q.(14)

Number of solution of equation, $9\cos^{12}x + \cos^{22}x + 1 = 6 \cos^6x \cos 2x + 6 \cos^6x - 2\cos 2x$ in $[0, 4\pi]$ are

Q.(15)

Number of solution(s) of the equation $1 + (\sec^2 2x) + \cot^2 x - 2(\sin x + \cos x \cdot \cot x) = 0$ in $(0, 2\pi)$ is/are

Q.(16) Let

$$f(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{\sin^4 2x}{4} + \frac{\sin^4 4x}{16} + \frac{\sin^4 8x}{64} + \dots + \frac{\sin^4 2^n x}{4^n} \right).$$

Then $125f\left(\sin^{-1}\frac{3}{5}\right)$ is equal to

Q.(17) Let f and g be functions defined as

$$f(x) = \sec^2 x + \tan x \text{ and } g(x) = \sec^2 x - \tan x; x \in \mathbb{R} - \left\{ \frac{(2n+1)\pi}{2} : n \in \mathbb{Z} \right\}.$$

Then the integer nearest to the value of

$$\lambda = f\left(\frac{\pi}{18}\right)g\left(\frac{5\pi}{18}\right)f\left(\frac{7\pi}{18}\right)$$

is equal to

Q.(18) Find λ , satisfying

$$\lambda = \frac{(a + \sqrt{10})(\sqrt{5 + \sqrt{5}} - \sqrt{3 - \sqrt{5}})}{\sin \frac{\pi}{20} \sin \frac{69\pi}{180} \sin \frac{51\pi}{180}};$$

$$a \in \left\{ (1 - \sin x)(1 - \cos x) : (1 + \sin x)(1 + \cos x) = \frac{5}{4}, x \in \mathbb{R} \right\}.$$

Q.(19) Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \left\{ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] : \sqrt{3}a \cos x + 2b \sin x = c, a, b, c \in \mathbb{R} - \{0\} \right\},$$

$$\mathbb{B} = \left\{ \frac{b}{a} : |\mathbb{A}| = 2, \mathbb{A} = \{\alpha, \beta\}, \alpha + \beta = \frac{\pi}{3} \right\}.$$

If $x \in \mathbb{B}$, then x is equal to

Q.(20)

If $A + B + C = \pi$ and $L = \sum \left(\frac{\operatorname{cosec} B \operatorname{cosec} C}{\sec A} \right)$ (wherever defined), $M = \frac{\sin 50^\circ + \sin 70^\circ + \sin 20^\circ}{\cos 25^\circ \sin 35^\circ \cos 10^\circ}$ and

$N = \left(1 - \cos \frac{2\pi}{9} \right) \left(1 - \cos \frac{4\pi}{9} \right) \left(1 - \cos \frac{6\pi}{9} \right) \left(1 - \cos \frac{8\pi}{9} \right)$ then the value of $(L^2 MN)$ is

PASSAGES

Passage-I

Let S_1 be the set of all those solutions of the equation $(1 + a) \cos \theta \cos (2\theta - b) = (1 + a \cos 2\theta) \cos (\theta - b)$, which are independent of a & b and S_2 be the set of all such solutions which are dependent on a & b then

Q.(21)

 Conditions that should be imposed on a & b such that S_2 is non-empty :

- (A) $\left| \frac{a}{2} \sin b \right| < 1$ (B) $\left| \frac{a}{2} \sin b \right| \leq 1$ (C) $|a \sin b| \leq 1$ (D) none of these

Q.(22)

 All the permissible values of b , if $a = 0$ and S_2 is a subset of $(0, \pi)$:

- (A) $b \in (-n\pi, 2n\pi); n \in \mathbb{Z}$ (B) $b \in (-n\pi, 2\pi - n\pi); n \in \mathbb{Z}$
 (C) $b \in (-n\pi, n\pi); n \in \mathbb{Z}$ (D) None of these

Passage-II:

Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Suppose sets $\mathbb{A}$ and $\mathbb{B}$ whose elements are written in then increasing order are defined as

$$\mathbb{X} = \{x : f(x) = 0\}, \mathbb{Y} = \{x : f'(x) = 0\},$$

$$\mathbb{Z} = \{x : g(x) = 0\}, \mathbb{W} = \{x : g'(x) = 0\}.$$

Now, based on above information, answer the following two questions

Q.(23) Which of the following options is/are correct?

- (A) Elements of set $\mathbb{X}$ are in A.P.(Arithmetic Progression).
 (B) Elements of set $\mathbb{Y}$ are in A.P.(Arithmetic Progression).
 (C) Elements of set $\mathbb{Z}$ are in A.P.(Arithmetic Progression).
 (D) Elements of set $\mathbb{W}$ are in A.P.(Arithmetic Progression).

Q.(24) Which of the following options is/are correct?

- (A) $\left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, \frac{7\pi}{3} \right\} \subseteq \mathbb{X}$ (B) $\left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\} \subseteq \mathbb{Y}$
 (C) $\left\{ \frac{\pi}{2}, 4\pi, \frac{\pi}{6}, \frac{7\pi}{6} \right\} \subseteq \mathbb{Z}$ (D) $\left\{ \frac{\pi}{2}, 4\pi, \frac{\pi}{6}, \frac{7\pi}{6} \right\} \subseteq \mathbb{W}$

MATRIX MATCHING TYPE

Q.(25) Consider the following sets

$$\mathbb{A} = \left\{ x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3} \right] : \cos^2 x + \sin x \cos x + \cos x + 2 \sin x = 2 \right\},$$

$$\mathbb{B} = \left\{ x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18} \right] : \sqrt{3} \tan x = \frac{1 - 4 \sin^2 x}{4 \cos^2 x - 1} \right\},$$

$$\mathbb{C} = \left\{ x \in [-6\pi, 6\pi] : 8 \left(2 \sin^2 \frac{x}{5} - 1 \right) \left(1 - 2 \cos^2 \frac{x}{5} \right) \left(1 - \tan^2 \frac{x}{5} \right) = 3\sqrt{3} \left(1 + \tan^2 \frac{x}{5} \right) \right\},$$

$$\mathbb{D} = \left\{ x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4} \right] : \sin^2 x + 2 \sin x = 3 + \sin x \cos x + 3 \cos x \right\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then match the following columns

Column-I
Column-II

 (A) $|\mathbb{A}|$

(P) 6

 (B) $|\mathbb{B}|$

(Q) 8

 (C) $|\mathbb{C}|$

(R) 4

 (D) $|\mathbb{D}|$

(S) 2

KEY

1. C	2. C	3. D	4. A	5. CD
6. BC	7. ABCD	8. ABCD	9. BCD	10. AC
11. ACD	12. AB	13. 30	14. 12	15. 1
16. 27	17. 17	18. 52	19. 0.5	20. 9
21. C	22. B	23. ABC	24. BD	25. A-S, B-S, C-P, D-R

GR-22 : INVERSE TRIGONOMETRIC FUNCTIONS

SINGLE CORRECT

Q.(1) Which of the following option(s) is/are equal to

$$\frac{48 \cos \left(2 \sin^{-1} \frac{3}{5} \right)}{\sin \left(2 \cos^{-1} \frac{4}{5} \right)} + \tan \left(\tan^{-1} \tan \frac{91\pi}{9} + \tan^{-1} (\tan 65^\circ - 2 \tan 40^\circ) \right).$$

- (A) 15 (B) 8 (C) $14 + \sqrt{3}$ (D) $7 - \sqrt{3}$

Q.(2) Which of the following option(s) is/are equal to $[\lambda + \mu]$, where

$$\lambda = \cos^{-1} \cos 2 + \csc^{-1} \csc 2 + \cot^{-1} \cot 2, [x] = \max\{a \in \mathbb{Z} : a \leq x\} \text{ and}$$

$$\mu = \cos^{-1} \cos f(x) + f(x); \text{ where } f(x) = \pi + |\sqrt{3} \sin x + \cos x|.$$

- (A) 11 (B) 12 (C) 10 (D) 9

Q.(3) Which of the following option(s) is/are equal to $\frac{\alpha}{\beta - \delta} + 4\gamma^2$, where

$$\alpha = 2 \tan^{-1} \sqrt{3 - 2\sqrt{2}} + \sin^{-1} \sqrt{\frac{1}{8 - 4\sqrt{3}}},$$

$$\beta = \cot^{-1}(\sqrt{3} - 2) + \frac{\sec^{-1}(-2)}{8},$$

$$\gamma = \sin \left(\frac{\pi}{\sqrt{3}} \tan \left(\tan^{-1}(\sqrt{2} - 1) + \cos^{-1} \frac{1}{2} \sqrt{2 + \sqrt{2 + \sqrt{3}}} \right) \right),$$

$$\delta = \tan^{-1} \frac{1}{\sqrt{2}} + \cos^{-1} \frac{1}{\sqrt{3}}.$$

- (A) 4 (B) 5 (C) 6 (D) 7

MULTIPLE CORRECT

Q.(4) Which of the following option(s) lie in the range of

$$f(x) = \frac{818(\cot^{-1} x + \sec^{-1} x + \csc^{-1} x)}{\sin \left(\frac{1}{2} \cdot \cos^{-1} \frac{1}{2} \right) + \sin \left(2 \cdot \cos^{-1} \frac{5}{13} \right)}.$$

(A) 339π

 (B) 507π

 (C) 845π

 (D) 1014π
Q.(5) Which of the following options is/are correct for $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \cos^{-1}(-\{x\}); \{x\} = x - \max\{a \in \mathbb{Z} : a \leq x\} \text{ and}$$

$$g(x) = \cos^{-1}(\cot^{-1} \ln x).$$

 (A) Domain of g is $[e^{\cot^{-1} 1}, \infty)$ and range is $\left[0, \frac{\pi}{2}\right]$

 (B) f is neither an Injective function nor an Even function

 (C) Range of f contains two prime numbers

 (D) f is periodic function with period 1.

Q.(6) Which of the following options is/are correct, if set $\mathbb{A}$ defined below contains 4 elements

$$\mathbb{A} = \left\{x \in \mathbb{R} : |\tan^{-1} 3x| = \frac{3}{4}|x| + \frac{\pi}{3} - \frac{\sqrt{3}}{k}\right\}.$$

 (A) $k = 1$

 (B) $k = 2$

 (C) $k = 3$

 (D) $k = 4$
Q.(7) If $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}$, then which of the following option(s) lie in set

$$\left\{\left(4\tan^{-1} \sqrt{\lfloor x \rfloor + \lfloor -x \rfloor} + 4\sqrt{2 - |x|} + \frac{4}{x^2}, 2\sin^{-1}[2y] + 3\cos^{-1}(\lfloor y \rfloor - 1)\right) \in \mathbb{R}^2 : (x, y) \in \mathbb{R}^2\right\}.$$

 (A) $(1, 3\pi)$

 (B) $(8, 4\pi)$

 (C) $(4, 4\pi)$

 (D) $(12, 5\pi)$
Q.(8) Let functions f and g be defined as

$$f: \left[-\frac{\pi}{3}, \frac{\pi}{6}\right] \rightarrow [0, 4]; f(x) = 2\cos^2 x + \sqrt{3}\sin 2x + 1 \text{ and}$$

$$g(x) = \min(\tan^{-1}(x+2), \cot^{-1}(x+2)) - \frac{\pi}{3}.$$

 If $\alpha < \beta$ are the integers lying in the range of g , then

 (A) $4f^{-1}(1) > -\pi$

 (B) $\cos^{-1} \cos \alpha + \sin^{-1} \sin \beta = 1$

 (C) $f^{-1}(\sqrt{1.25} + 2.5) = \frac{\pi}{15}$

 (D) $\cos^{-1} \cos \beta + \sin^{-1} \sin \alpha = 3 - \pi$

Q.(9) Which of the following options is/are correct, if

$$x_1 = \sin\left(\frac{1}{3}\sin^{-1}\frac{1}{\sqrt{3}}\right), x_2 = \cos\left(\cos^{-1}\frac{1}{\sqrt{5}} - \sin^{-1}\frac{2}{\sqrt{5}}\right),$$

$$\{\sin \cot^{-1} \cos \tan^{-1} x \in \mathbb{R} : x \in \mathbb{R}\} = [p, q]; p, q \in \mathbb{R}.$$

Here $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}$.

(A) $x_2^2 = 3(3x_1 - 4x_1^3)^2$

(B) $2x_2^2 = 3(3x_1 - 4x_1^3)^2$

(C) $\left\lfloor q + \frac{1}{p} \right\rfloor = 2$

(D) $\left\lfloor q + \frac{1}{p} \right\rfloor = 3$

Q.(10) Let $|\mathbb{X}|$ denote the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \left\{x \in \mathbb{R} : 2 \tan^{-1}|x| = \frac{1}{\ln|x|}\right\},$$

$$\sec^{-1}\sqrt{1+x^2} + \csc^{-1}\frac{\sqrt{1+y^2}}{y} + \cot^{-1}\frac{1}{z} = \pi; x, y, z > 0 \text{ and}$$

$$\tan^{-1}(a+1) + \cot^{-1}(a-1) = \sin^{-1}\frac{4}{5} + \cos^{-1}\frac{3}{5}.$$

Then, which of the following options is/are correct?

(A) $|\mathbb{A}| = 2$

(B) $x + y + z = xyz$

(C) $7a^2 = 48$

(D) $|\mathbb{A}| > 2$

NUMERICAL TYPE

Q.(11) If a, b are the roots of $x^2 = 4x - 1$ ($a > b$), then find

$$\lambda = \frac{b^3}{2} \csc^2\left(\frac{1}{2}\tan^{-1}\frac{b}{a}\right) + \frac{a^3}{2} \sec^2\left(\frac{1}{2}\tan^{-1}\frac{a}{b}\right).$$

Q.(12)

If $\alpha = \tan\left(\frac{1}{3}\tan^{-1}\frac{4}{3}\right)$ & $\beta = \sin\left(\frac{1}{2}\sin^{-1}\frac{4\sqrt{2}}{9}\right)$, then the value of $\frac{1}{|\alpha^3 - 4\alpha^2 - 3\alpha + \beta|}$ is

Q.(13)

If $\cos^{-1} \frac{p}{a} + \cos^{-1} \frac{q}{b} = \alpha$ and $\frac{p^2}{a^2} - \frac{2pq}{ab} \cos \alpha + \frac{q^2}{b^2} = f(\alpha)$, then number of integers in the range of $f(\alpha)$ is

Q.(14)

Let $x = \sin^{-1}(\sin 8) + \cos^{-1}(\cos 11) + \tan^{-1}(\tan 7)$, then the least value of the integer which is greater than x is equal to

Q.(15)

If $\sum_{r=1}^{\infty} \cot^{-1} \left(\frac{(r+1)^2}{2} \right) = \tan^{-1} x$, then x is equal to -

Q.(16)

Let x_1, x_2, x_3 be the solution of $\tan^{-1} \left(\frac{2x+1}{x+1} \right) + \tan^{-1} \left(\frac{2x-1}{x-1} \right) = 2\tan^{-1}(x+1)$ where $x_1 < x_2 < x_3$ then $2x_1 + x_2 + x_3^2$ is equal to

PASSAGES

Passage-I

Let roots of equation $x^3 - ax^2 + \left(1 + \left(\frac{\pi}{4} + \frac{4}{\pi} \right) \tan^{-1} 2 \right) x - \tan^{-1} 2 = 0$ are $\frac{4}{\pi}$, $\tan^{-1} \alpha$ & $\tan^{-1} \beta$ where α & β are positive integers.

Q.(17)

$\tan^{-1}(\alpha) + \tan^{-1}(\beta) + \tan^{-1}(\alpha + \beta)$ is equal to-

Q.(18)

If $\alpha > \beta$, then number of solutions of equation $\sin^{-1}(x + \beta) + \sin^{-1}(\alpha x) = \frac{\pi}{2}$, is -

Passage-II

Let sets $\mathbb{A}$ and $\mathbb{B}$ are defined as

$$\mathbb{A} = \left\{ k \in \mathbb{R} \mid \text{Function } f: \mathbb{R} \rightarrow \left(0, \frac{\pi}{3}\right]; f(x) = \cos^{-1} \frac{x^2 - k}{x^2 + 1} \text{ is Surjective} \right\} \text{ and}$$

$$\mathbb{B} = \left\{ x \in \mathbb{Z} \mid x = \csc^{-1} \left(a + \frac{1}{a} \right) + \sec^{-1} \left(b + \frac{1}{b} \right); a, b \in \mathbb{R}, ab < 0 \right\}.$$

Now, based on above information, answer the following questions

Q.(19) The number of elements in set $\mathbb{A}$ is equal to

Q.(20) The number of elements in set $\mathbb{B}$ is equal to

KEY

1. A	2. A	3. D	4. ABC	5. BCD
6. BC	7. AB	8. ABCD	9. AC	10. ABC
11. 56	12. 1	13. 2	14. 4	15. 3
16. 1	17. 3.14	18. 1	19. 1	20. 2

GUIDED REVISION – ERRATA

GR-01	9: ABCD
GR-02	2: Replace k by r 5: ACD 7: AC 14: 55 19: replace $\frac{2}{\pi}$ in question by $\frac{-2}{\pi}$
GR-03	2: Replace θ by ϕ 3: Use colon notation in the set involving mod 6: ACD 11: Replace AB by BC 12: ABCD
GR-04	-
GR-05	1: in option D $m_A \neq m_B = m_C$ 11: ABC 13: BD 15: Replace word ‘used’ by ‘discarded’ 16: 9.6 25: In option P, replace -2 by 2 B→PQ, C→P
GR-06	3: In option A) $\frac{3\pi^2}{2}$ 10: dt is missing
GR-07	-
GR-08	4: ABD 5: Needs language change Q.(5) Let M and N be two lines. Line L_1 is reflected in M and becomes line L_2 . Then line L_2 is reflected in N and becomes line L_3 . Finally line L_3 is reflected in M and becomes line L_4 . If 12: 3 16: D
GR-09	8: replace 16 by -16 and k by -k(see below). Also in options A and C replace 8 by -8. Q.(8) Circle $x^2 + y^2 - 8x - 6y - 16 = 0$ and line $y = mx + c$ intersect at two distinct points A and B, such that circle with AB as diameter always passes through origin. If $x^2 + y^2 + 2gx + 2fy - k = 0$ represents the locus of center of circle with AB as diameter, then (A) $c^2 + (4m - 3)c - 8(m^2 + 1) = 0$ (B) $k + g = 6$ (C) $4c^2 + (4m - 3)c - 8(m^2 + 1) = 0$ (D) $k - 2f = 11$ 9: passing through every point (see below) Q.(9) Let $x^2 + y^2 + 2gx + 2fy + c = 0$ represent the equation of circle passing through every point lying in $A \cup B$, given that it exists and $ A = 2 = B $, where $ X $ denotes the cardinal number of set X, $A = \{(x, y) \in \mathbb{R}^2 x^2 + y^2 = 4, x + 2y = a\},$ $B = \{(x, y) \in \mathbb{R}^2 x^2 + y^2 - 4x - 2y + 1 = 0, 12x - 6y = 41\}.$ Then which of the following options is/are correct? (A) $a + 2 = 0$ (B) $5g - 5f = 4$ (C) $c = -36$ (D) $a - 5c = 34$

GR-10

16:

Q.(16)→ If $t = 4a = 4$, then $r \cdot AB$ is equal to